

SmartRecruit: AI-Driven Automated Recruitment System

A project report submitted as a partial fulfilment of the degree of
Master of Computer Applications (MCA)

Submitted by

Rajesh Girishkumar Lalwani

Seat No.: 75

of

Post Graduate Department of Computer Science and Technology

Sardar Patel University

Vallabh Vidyanagar

Submitted to

SARDAR PATEL UNIVERSITY

Vallabh Vidyanagar

April 2026

**Post Graduate Department of Computer Science and Technology Sardar Patel
University
Vallabh Vidyanagar**

CERTIFICATE

This is to certify that **Rajesh Girishkumar Lalwani** of Master of Computer Applications (MCA) IV semester has worked on the project entitled **SmartRecruit: AI-Driven Automated Recruitment System** satisfactory towards the partial fulfilment of the degree of Master of Computer Applications (MCA) during the final semester at the Post Graduate Department of Computer Science and Technology, Sardar Patel University, Vallabh Vidyanagar, Gujarat, India.

Date of Submission

18th April, 2026

Dr. Vinod Desai
Internal Project Guide

Dr. Priti. S. Sajja
Head of the Department

TO WHOM IT MAY CONCERN

This is to certify that Rajesh Girishkumar Lalwani, a student of Post Graduate Department Of Computer Science & Technology has successfully completed his internship in the field of Data Science from 15th December to 15th April (16 weeks) under the guidance of Mr. Satyam Raval at TEPL. His internship activities include successful completion of the assigned project at the given period of time along with abiding by company's rules and regulations. During the period of his internship program with us, he had been exposed to different processes and was found diligent, hardworking, and inquisitive. We wish him every success in his life and career.

Tech Elecon Pvt. Ltd.,

ABSTRACT

The Smart Recruit System is an advanced, web-based platform designed to streamline and automate every aspect of the recruitment lifecycle, from job postings to candidate onboarding. It enhances operational efficiency, minimizes manual intervention, and provides a seamless, user-friendly experience for both candidates and recruiters. Through a secure, scalable, and highly functional interface, the system facilitates job searches, application screening, interview scheduling, assessments, and reporting - offering a complete talent acquisition solution within one integrated platform.

For Candidates, the system delivers an intuitive and efficient browsing experience. Users can easily register and log in, explore real-time job openings, and filter roles based on various criteria such as location, job type (full-time, remote, internship), salary expectations, and required skills. Each job listing includes comprehensive details such as role responsibilities, company culture insights, required qualifications, and application instructions. Users can apply for jobs, track their application status through various pipeline stages, and complete practical coding or technical assessments directly through the platform. Integrated notification mechanisms ensure candidates are kept informed via instant alerts regarding their application progress.

The Recruiter Dashboard is a powerful backend module that empowers HR professionals and hiring managers to manage job listings, candidate profiles, interview pipelines, and evaluations efficiently. Recruiters can add, update, or archive job postings, manage hiring teams, set application deadlines, and dynamically define pipeline stages. Job applications, assessment scores, and resume submissions can be tracked, progressed, or rejected in real time. Recruiters can also manage candidate communications, technical evaluations, and AI-generated interview questions, ensuring timely feedback and an enhanced hiring velocity.

A critical component of the Smart Recruit System is its robust assessment and documentation tracking. The integration of specialized coding challenges and an AI-driven question generator ensures accurate evaluation of a candidate's technical capabilities. Once a hiring stage is complete, the system automatically generates and sends status updates and interview confirmations via email. Additionally, the platform supports digital document uploads and validation for resumes, portfolios, and employment verification, streamlining traditionally paperwork-heavy HR processes.

The system also supports proactive talent acquisition tools such as candidate pooling, pipeline tracking, and job referral mechanisms, which can be utilized by recruiters to increase job visibility, retain top talent, and boost application conversions. These tools are particularly effective for managing campus recruitment drives, mass hiring campaigns, or sourcing passive candidates for highly specialized roles.

Analytics and Reporting are key features that provide administrators with valuable insights into hiring velocity, candidate drop-off rates, recruiter performance, and role fulfillment timelines. The system can generate detailed reports on job listing views, application conversion rates, pipeline bottlenecks, popular skillsets, and location-based talent supply, enabling data-driven hiring decisions. These insights assist organizations in optimizing sourcing strategies, refining job requirements, and identifying high-potential talent pools for future expansion.

Security and privacy are foundational pillars of the system. The Smart Recruit System uses industry-grade encryption to store sensitive data, such as candidate resumes, personal contact details, and evaluation scores. Role-based access control ensures that only authorized personnel can access or modify specific applicant data, protecting both candidates and company HR operations from unauthorized access.

The system is designed with scalability in mind, capable of supporting a growing number of applicants, job postings, and hiring departments as the organization expands. Its modular architecture enables the easy integration of future enhancements, such as AI-driven candidate ranking, automated resume parsing, integrated video interviews, and mobile apps, keeping the platform competitive in a rapidly evolving digital HR landscape.

Moreover, the system's responsive design ensures a seamless experience across all devices, whether accessed via desktops, tablets, or smartphones. This cross-platform compatibility widens accessibility, increases candidate engagement, and enhances the overall application experience.

In conclusion, the Smart Recruit System stands as a comprehensive, future-ready solution for modern talent acquisition. By automating and centralizing critical operations, from job posting and pipeline management to assessments and analytics, the system reduces operational HR burdens, elevates the candidate experience, and drives organizational growth by securing the right talent faster. With its secure, scalable, and customizable framework, the platform is well-equipped to evolve with market demands and emerging HR technologies.

ACKNOWLEDGMENT

Acknowledging everyone is difficult, but I will still try to express our gratitude to each and every one who has helped me in the completion of this project.

I am indebted to **Mr. Satyam Raval** who devoted his precious time to help me understand the application that I have part of these without whose assistance I would not have been able to build up a good Application.

I am grateful to **Dr. Vinod Desai** for their prolonged interest in my work and excellent guidance. They have been a constant source of motivation for me. Bly their uncommon promising demand for quality and their insistence on meeting the deadlines. I am able to accomplish such an excellent work.

I express my sincere gratitude to the Head of the Department **Dr. Priti Sri Sajja** They have been very good mentors who were always available for helping me with any kind of queries that i would have regarding the topic and anything that would be a hurdle in the development of my application.

And last but not least I would like to acknowledge the cooperation extended by all those persons who have directly or indirectly helped me during the course of this project.

Thanking you all in anticipation.

With sincere Regards,

Rajesh Lalwani

INTERNSHIP LETTER



HR/TEPL/325

16.01.2026

To,
Post Graduate Department of Computer Science and Technology,
V V Nagar.

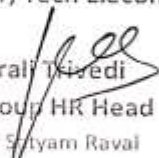
Permission for Project

With reference to your letter dated 16.11.2025, we are pleased to grant permission to *Mr.Harshil Faldu/Mr.Rajesh Lalwani/Ms.Dhvani Rana & Ms.Sanjana Soni* student of MCA for Project in our organization from **15.12.2025 to 15.04.2026**.

The following points are to be adhered by the student.

- Laptops, Pen-Drives & other hard drives are not permitted inside the company premises.
- Appropriate dressing & grooming will be appreciated. Jeans & T Shirts are not allowed.
- Female students have to wear dress and Kurtis only.
- Photography & Videography is strictly prohibited in the premises.
- Trainee/s has to strictly comply with the Company's timings & schedule.
- In case of any casualty the Company will not be held responsible.
- On Completion of training the trainee/s has to submit a copy of Training Report duly signed by concern guide to obtain the Training Certificate.
- In case of any ambiguity / difficulty, the trainee is required to approach HR Department. The detailed guideline are attached overleaf.

For, Tech Elecon Pvt.Ltd.


Nirali Khedri
Group HR Head
Cc. Satyam Raval

TECH ELECON PVT. LTD.

Anand Sojitra Road, Vallabh Vidyanagar - 388 120, Gujarat, India | Email: inquiry@techelecon.com | Web: www.techelecon.com
Phone: +91 2692 227292/91. CIN: U72900GJ2012PTC070532

Branch Office: Vadodra: S1/19-21, Panorama Complex, R. C. Dutt Road, Akapuri, Vadodra - 390007. Mobile: +91 90990 36045 E-mail: inquiry@techelecon.com
Kolkata: 68, Sarat Bose Road, Kolkata - 700025. Ph. No. 033 24761861 / 24760904

INDEX

Chapter No.	Chapter Title	Subsections	Page No.
1	Introduction	1.1 Department Profile 1.2 Company Profile 1.3 Project Overview 1.4 Purpose & Objective 1.5 Scope of the System	1
2	System Analysis & Requirements	2.1 Existing System 2.2 Proposed System 2.3 Module Description	14
3	Feasibility Study	—	20
4	System Features	—	22
5	Hardware and Software Requirements	—	23
6	System Design	6.1 System Flow Diagram 6.2 Data Flow Diagrams	24
7	Data Dictionary	7.1 Data Dictionary	48
8	Design of Input-Output Screen & Reports	—	60
9	Testing	9.1 Testing Plan 9.2 Testing Strategy 9.3 Types of Testing 9.4 Test Cases & Results	97
10	Future Enhancement	—	108
11	Conclusion	—	110
12	Bibliography	—	111

INTRODUCTION

1.1 DEPARTMENT PROFILE



GH Patel PG Department of Computer Science & Technology,

Sardar Patel University, Vallabh Vidyanagar

The Department of Computer Science was established in year 1986 as an institution offering postgraduate in Computer Science. The computing facility, however, was introduced at the Sardar Patel University somewhere in 1973-74 at Computer centre of the University with an IBM 1620, a second-generation mainframe computer. The University started one-year Post Graduate Diploma in Computer Applications (PGDCA) course in 1986, M.Sc. (Computer Science) in 2008, Ph.D. (Computer Science) in 1990 and Ph.D. (Bioinformatics) in 2011.

Besides these programs, the University also started computer related short-term programs from time to time. The computing resources were also used by different postgraduate departments of the University for their Research Work. The institute took lead in Establishment started a CSI (Computer Society of India is a national professional body having links with various professional bodies in India and abroad) in 1985-86. Various workshops, seminars and symposia have been organized under the CSI banner so far, which have strengthened both the teaching and research activities of the Department. In the year 1988, Late Shree Goradhanbhai Hathibhai Patel donated money to the University for a Separate building of the Post Graduate Department of Computer Science. The Department also recognized by UGC to provide technical training to personnel at various levels of academia and administration. Several PC based systems had been Installed and major research work carried out in the areas of CAI (Computer Aided Instruction), KBS (Knowledge Based Systems) and DSS (Decision Support System). In 1992. The M.Sc. In Computer Science Program was terminated and intake of the MCA Program was Increased from 30 to 40. Three faculty members were selected on the basis of merit for the CICC-Japan training in the years 1992, 1994 and 1995.

Early 2000 was the time when vast technological changes took place in the IT industry all over the world. The department could meet the technological requirements through support from the UGC, AICTE and various bodles of the state. A networked laboratory was established at the department, providing state-of-the-art facilities. The intake to MCA was increased to 100 by introducing self-financed seats in the year 2000, and a separate self-financed batch of MCA was started. The department was recognized and offered 'Refresher Courses in Computer Science' for the Computer Science faculty members in India by the University Grants Commission. In the year 2002, a special MCA program was introduced for providing lateral entry Into the Second year to computer science graduates. In a short span, the department grew to the extent that a new building was required. With one more generous donation from Late Shree Goradhanbhai Hathibhai Patel, grants from the UGC and he state government and the department with modern infrastructure was constructed. The building was inaugurated on July 22nd, was then changed to G H Patel Post Graduate Department of Computer Science and Technology in the year 2002. Subsequently, new powerful servers and several workstations were added to the laboratory and electives for bioinformatics and wireless networks were added to the syllabus. The M.Sc. (Bioinformatics) program under the UGC Innovative Program scheme was introduced in the year 2005.

1.2 COMPANY PROFILE



Tech Elecon Private Limited

Tech Elecon Pvt. Ltd. is a trusted and prominent IT services and software development company based in Vallabh Vidyanagar, Anand, Gujarat. As an integral part of the prestigious Elecon Group, we are dedicated to delivering high-quality, innovative, and result-driven IT solutions. With a passion for technology and a strong commitment to excellence, we serve clients across the globe with customized digital solutions that empower businesses to stay ahead in a competitive digital landscape.

We specialize in web and mobile app development, custom software solutions, and creative digital services that are tailored to meet the specific needs of businesses across various sectors, including manufacturing, retail, healthcare, logistics, education, hospitality, and corporate enterprises. Our solutions are designed to not only solve real-world problems but also to deliver measurable business outcomes.

At Tech Elecon, our expert team of developers, designers, and strategists work collaboratively to transform ideas into functional, scalable, and visually compelling digital products. We believe that great technology starts with understanding the client's vision and ends with a solution that exceeds expectations. Our development process is agile, transparent, and client-centric. From discovery and planning to design, development, testing, and deployment, we ensure every stage is executed with precision and professionalism. Our focus remains on:

- **Enhancing brand visibility** through smart digital strategies
- **Improving customer engagement** with intuitive UI/UX design
- **Driving business growth** through scalable technology solutions

Beyond core development, we offer comprehensive IT consulting and innovation services. We help businesses adopt the latest technologies, such as cloud computing, artificial intelligence, machine learning, IoT, and automation, to increase operational efficiency, reduce costs, and unlock new opportunities. Our team ensures that clients are well-prepared to adapt to the ever-evolving digital ecosystem.

Our graphic designing services are a perfect blend of creativity and strategy. We craft visually engaging and brand-consistent designs that communicate your brand story effectively, whether it's for digital marketing campaigns, branding materials, or user interfaces.

In addition, we provide website designing services, focusing on creating responsive, SEO-friendly, and high-performance websites that deliver seamless user experiences. Our websites are not just beautiful, they are built to convert visitors into loyal customers and improve your digital footprint.

We take pride in delivering:

- **End-to-end project support**, from ideation to deployment and maintenance
- **Timely delivery** without compromising on quality
- **Customized solutions** that align with your business goals
- **Long-term partnerships** built on trust, transparency, and continuous improvement

Whether you're a startup looking to build your first product or an established business in need of digital transformation, Tech Elecon Pvt. Ltd. is your reliable technology partner, committed to building your success, one solution at a time.

Address: Anand - Sojitra Road, Vallabh Vidyanagar, Anand, Gujarat 388120.

Email: info@techelecon.com

Website: www.techelecon.com

Phone: +91 2692 236 513

1.3 PROJECT OVERVIEW

Project Name	SmartRecruit: AI Driven Automated Recruitment System
Organization	Tech Elecon Pvt. Ltd.
Address	Anand - Sojitra Road, Vithal Udyognagar, Vallabh Vidyanagar, Anand, Gujarat - 388120.
Project Guide	Dr. Vinod Desai
Duration	4 Months
Developed By	Rajesh Girishkumar Lalwani
Tools	Vs code, Git, GitHub
Front End	HTML5, CSS3, JavaScript, Bootstrap 5
Back End	Python (Django Web Framework)
Database	SQLite
Submitted to	PG Department of Computer Science & Technology

➤ TEAM INFORMATION:

Name of Member	Exam No.	Email Address	Name of Institute
Rajesh Girishkumar Lalwani	75	rajlalwani511@gmail.com	GDCST
Raghunandannath Ramjanamnath Tiwari	74	raghunandantiwari30@gmail.com	GDCST
Dhruv Ravindrabhai Sonvane	13	dhruvsonvane8401@gmail.com	GDCST

➤ DIVISION OF WORK:

NAME	PRIMARY ROLE & MODULE	ANALYSIS	DESIGN	CODING	TESTING
Rajesh Lalwani	AI Orchestrator & System Lead (Agentic AI, RAG Logic)	✓	✓	✓	✓
Dhruv Sonvane	Full-Stack Recruiter Architect (Analytics, Backend APIs)	✓	✓	✓	✓
Raghunandannath Tiwari	Candidate Experience Head (UI/UX, Glassmorphism Portal)	✓	✓	✓	✓

1.4 PROJECT DEFINITION

The Smart Recruit System is a comprehensive and robust web-based Applicant Tracking System (ATS) developed to streamline, automate, and simplify the complete recruitment lifecycle. This system serves as a unified digital platform where both candidates (job seekers) and administrators (recruiters or HR managers) can interact seamlessly to perform various hiring operations. The primary goal of this project is to eliminate traditional manual resume screening and spreadsheet tracking, offer a faster and more secure application process, and provide a user-centric digital alternative that enhances operational efficiency and the overall hiring experience.

From the candidate's perspective, the system provides a smooth and intuitive interface where users can browse a catalog of active job openings, filter listings based on preferences (e.g., job role, location, required skills, and experience level), and view comprehensive role details including responsibilities, prerequisites, and company insights. Users can submit applications, attempt practical coding assessments, or engage with interview schedules, all from the comfort of their homes. Registered candidates also get access to their personal application pipeline and status updates, improving transparency and building trust in the recruitment service.

For recruiters and administrators, the system offers a powerful backend dashboard to manage the entire hiring workflow. Admins can create new job postings, manage automated job expiry logic, set skill requirements, manage candidate pipelines, move applicants through various stages (Screening, Interview, Assessment, Offered), and generate performance reports. The interface allows real-time tracking of application statuses, simplifies talent pipeline management, and utilizes AI to automatically generate role-specific technical questions for interviews. A built-in evaluation module enables the easy grading of candidate assessments and feedback management.

Overall, this project is particularly valuable for corporate HR departments, recruitment agencies, and independent talent acquisition teams looking to digitize their operations. It significantly boosts recruiter productivity, reduces human errors in candidate tracking, and ensures faster, more reliable hiring decisions. By centralizing the complete job posting and assessment lifecycle, the Smart Recruit System enhances the transparency, scalability, and overall effectiveness of recruitment operations in the modern digital landscape.

In addition to core functionalities, the system includes several user-centric features that elevate the digital experience. A responsive, modern design (featuring a Glassmorphism UI layout) ensures accessibility across all devices—desktop, tablet, and mobile, enabling candidates and recruiters to stay connected on the go. The integrated notification and tracking mechanisms dynamically highlight pending candidate actions and upcoming evaluation assessments, helping prevent applicant drop-offs and easing the administrative workload.

The system also features an advanced AI Generation and Assessment module, which automatically compiles technical interview questions and effectively tracks candidate programming solutions. This module strengthens evaluation accuracy, ensures unbiased hiring metrics, and lends a highly professional, modern edge to the screening service.

From a technical standpoint, the system is built using widely adopted, secure, and highly scalable web technologies such as Python and the Django Web Framework for backend functionality, coupled with SQLite/MySQL for database management. The frontend is elegantly powered by HTML5, CSS3, JavaScript, and Bootstrap 5. The database schema is thoughtfully designed to

support relational integrity, accurately linking users, job postings, applications, AI modules, and assessments.

Scalability is a core strength of the platform. Its modular architecture allows it to grow seamlessly with the organization, whether through the integration of advanced AI resume parsers, the expansion into multi-department or global location hiring, or the future implementation of live video interviews and mobile applications. This forward-compatible design ensures the system remains flexible in response to rapidly evolving digital HR trends.

The admin data analytics board also provides valuable performance summaries, empowering HR directors and hiring managers with real-time insights. Metrics such as the number of active applicants, job-wise application pipelines, recruiter efficiency tracking, and platform activity logs are visually processed and presented for quick decision-making and strategic hiring planning.

In conclusion, the Smart Recruit System is more than just a digital upgrade, it's a strategic transformation aimed at modernizing the talent acquisition sector. By offering a feature-rich, scalable, and user-friendly platform, it empowers recruitment professionals to secure top-tier talent while maintaining complete operational control, data transparency, and long-term organizational growth potential.

1.5 PURPOSE & OBJECTIVE OF THE PROJECT

➤ PURPOSE

The purpose of the Smart Recruit System is to provide a comprehensive digital solution to automate and streamline the management of recruitment pipelines, candidate applications, and technical assessments. This system is designed to enhance operational efficiency for HR professionals and deliver a seamless, transparent, and highly engaging experience for job applicants. It aims to eliminate manual resume tracking via spreadsheets, accelerate hiring timelines, and support organizational scalability in the highly competitive talent acquisition sector.

- **Simplify Operations:** Automate job postings, candidate pipeline tracking, interview scheduling, and AI-driven technical question generation for hiring teams.
- **Enhance Candidate Experience:** Offer a secure, fast, and intuitive platform for applicants to discover active roles, submit applications, and complete coding assessments effortlessly.
- **Improve Efficiency:** Reduce manual screening errors, duplicate candidate files, and evaluation biases through structured pipelines, automated job expiries, and systematic assessment gradings.
- **Increase Transparency:** Provide real-time application status updates, clear pipeline visibility, and reliable communication to both candidates and recruitment administrators.
- **Scalable Solution:** Enable seamless expansion to accommodate multiple hiring departments, global job roles, and future technological integrations such as automated resume parsing, video interviews, or external HRMS syncing.

➤ OBJECTIVES

- **Digitize the Recruitment Process:** Replace traditional spreadsheet tracking and scattered email threads with a fully automated, centralized system to manage job postings, candidate applications, and hiring pipelines.
- **Enable Real-Time Application Tracking:** Allow candidates to view the up-to-date status of their applications (e.g., Screening, Assessment, Interview) and access their assessments instantly without waiting for manual HR intervention.
- **Enhance User Experience:** Design an easy-to-use, responsive interface (featuring modern glassmorphism elements) for both candidates and recruiters to navigate active jobs, submit data, and perform pipeline actions effortlessly across all devices.
- **Manage Job Postings Inventory:** Provide recruiters with comprehensive tools to create, edit, or pulldown job listings, define skill prerequisites, attach specific job requirements, and manage automated posting expiries.

- **Simplify Pipeline & Assessment Management:** Allow recruiters to easily transition candidates between hiring stages, trigger AI-powered technical question generation, seamlessly evaluate practical coding submissions, and maintain a complete historical log of all applicant interactions.
- **Ensure Secure Role-Based Data Isolation:** Implement strict access control frameworks to unconditionally protect candidate data privacy, ensuring candidates only see their own progress while allowing HR structured access exclusively to their assigned hiring campaigns.
- **Provide Reporting and Enterprise Analytics:** Offer insightful, real-time reports and visual tables detailing recruiter productivity, time-wise and role-wise application metrics, pipeline bottlenecks, and overall recruitment velocity to support data-driven hiring decisions.
- **Ensure Security and Data Privacy:** Implement robust security mechanisms, including secure authentication and data encryption, to strictly protect sensitive candidate information, uploaded resumes, and proprietary evaluation parameters.
- **Support Business Growth:** Help corporate HR departments and recruitment agencies effortlessly scale and manage a vastly larger volume of job listings and applicants at once through deep automation and improved communication delivery.
- **Role-Based Access Control:** Assign distinct user roles (System Admin, HR/Recruiter, Candidate) with highly specific system permissions to ensure impenetrable governance and secure access management within the portal.
- **Recruiter Feedback and Assessment System:** Allow hiring managers and technical recruiters to leave structured feedback, interview grades, and assessment ratings for applicants, building a transparent and highly trackable evaluation timeline.
- **Calendar Integration for Interview Schedules:** Provide structured mechanisms for recruiters to visualize, coordinate, and manage scheduled technical rounds and final HR interviews to prevent overlaps and endless email tracking.
- **Document Upload & Validation:** Enable candidates to easily upload necessary professional documents (e.g., PDF Resumes, Portfolios, Cover Letters, Identity Verification) for instant review and secure storage by the recruitment team prior to screening.
- **Talent Pooling and Job Marketing Algorithms:** Allow admins to track proactive sourcing efforts, maintain historical talent pools, or flag highly qualified 'silver-medalist' candidates to boost hiring engagement whenever similar roles open up in the future.
- **Multi-Location & Remote Hiring Support:** Seamlessly manage active job designations across different global office branches, remote designations, or multiple sister companies entirely from a single, unified administrative dashboard.

1.6 SCOPE OF THE SYSTEM

The scope of the Smart Recruit System is to provide a centralized, web-based platform that simplifies, automates, and digitizes talent acquisition and recruitment operations for administrators, HR recruiters, and candidates. It enables seamless cross-user interaction and offers real-time accessibility to hiring-related information.

The system encompasses key operational modules, including:

1. User Management:

- Secure login & registration system with strict role-based access for Admins, Recruiters, and Candidates.
- Individual, specialized dashboards for different user types to comfortably manage their distinct activities.
- Profile management features for users to securely update personal details, upload resumes, and track interaction histories.

2. Job Listings & Management:

- Admins and recruiters can instantly add, edit, or delete active job postings.
- Categorization based on job role, required skills, geographical location, experience level, and status.
- Recruiters can set automated posting expiry dates, required qualifications, and detailed operational responsibilities.

3. Advanced Search & Filtering:

- Candidates can utilize multiple filters such as job title, location, salary parameters, work-type, and required skills.
- Skill-based tracking to enhance the discovery of relevant opportunities for applicants.

4. Application & Assessment Management:

- Candidates can seamlessly submit job applications and engage directly with practical coding assessments from the job detail pages.
- Recruiters receive updates and can transition applicants through pipeline stages (e.g., Screening, Assessment, Interview), or approve/reject candidates.

5. Communication & Notification System:

- Built-in application tracking status updates eliminating the need for disjointed candidate outreach.
- Automated notifications via Email (SMTP) for application confirmations, interview schedules, and recruiter status changes.

6. Admin Dashboard:

- Complete enterprise control over the entire system: manage recruiter accounts, oversee system-wide job postings, access logs, and enforce security policies.
- Ability to view deep recruiter analytics, track time-to-hire metrics, and generate tabular reports on organization-wide recruitment performance.

7. Security & Compliance:

- Strict role-based access control (RBAC) ensures security across users, completely isolating candidate data.
- Data encryption, secure login sessions (Django session framework), CSRF protection, and protection against unauthorized data access.

8. Future Enhancements:

- Advanced AI Resume Parsing (Semantic skills extraction)
- In-App Live Video Interviewing (WebRTC integration)
- Predictive Hiring Analytics & Machine Learning Reports
- Dedicated Mobile App Integration
- Multi-language & global corporate branch compliance support
- Automated Offer Letter Digital Signatures (e-Sign APIs)
- Outbound talent sourcing and bulk email marketing integrations

➤ KEY FEATURES:

- Role-based login architecture for Admin, Recruiter, and Candidate
- Dynamic job listing management with automated expiry logic
- Advanced job filtering & attribute sorting
- Visual, drag-and-drop or status-based application pipeline tracking
- Automated AI-driven technical question generation and assessment modules
- Automated transactional email alerts (via Django SMTP)
- Admin panel loaded with real-time data analytics, tables & platform controls
- Secure, modern, and highly responsive frontend (Bootstrap 5 & Glassmorphism)
- Structured recruiter feedback and technical grading system
- Candidate personal dashboard featuring application history and status milestones
- Future-ready API hooks for external HRMS system syncing
- Enterprise-level backend security and CSRF token validations

➤ SYSTEM DESIGN TECHNOLOGY STACK:

- **Frontend:** HTML, CSS, JavaScript, Bootstrap 5
- **Backend:** Python (Django Web Framework)
- **Database:** SQLite
- **Notification System:** Django Native Email Backend (SMTP)
- **AI Integrations:** Google Gemini API / Hugging Face API
- **Security:** PBKDF2 Password hashing, Secure Session handling, CSRF Tokens, Role-based access control (RBAC)

LIMITATIONS

While the SmartRecruit System offers a comprehensive set of features for streamlining talent acquisition, applicant tracking, and user interaction, there are several areas where future enhancements can further improve functionality and the recruitment experience:

- **Live Video Interviewing (Future Scope):** Integration of in-browser WebRTC video conferencing to allow recruiters to conduct, manage, and record remote interviews directly within the platform, improving convenience and centralizing evaluation data.
- **Third-Party Background Verification:** Integration with external APIs for verifying candidate documents such as educational certificates, previous employment records, and identity proofs, ensuring applicant authenticity and preventing fraudulent hiring.
- **Mobile Application (Future Scope):** A dedicated mobile app for Android and iOS platforms can be developed to provide more accessible, real-time job browsing and application tracking for candidates, along with on-the-go pipeline management for recruiters.
- **AI-Powered Candidate Matching (Resume Parsing):** An advanced intelligent recommendation system could be introduced to semantically parse uploaded resumes and automatically rank or suggest candidates for specific job postings based on technical skills, experience, and system-wide interactions.
- **Automated Offer Generation & e-Signatures:** Implementing a dynamic contract management engine that automatically generates personalized offer letters based on negotiated salaries and securely integrates with e-signature APIs, allowing candidates to sign their contracts digitally without leaving the platform.

SYSTEM ANALYSIS & REQUIREMENTS

2.1 EXISTING SYSTEM

Despite the massive growth of the corporate sector, many organizations still rely on outdated, fragmented, and inefficient manual processes for recruitment. The following limitations highlight the severe drawbacks of the existing traditional hiring system:

1. Manual Candidate Management & Tracking:

- Job postings and applicant pipelines are managed through offline methods like disparate email threads or printed resumes.
- Candidate progress is tracked using Excel spreadsheets or notebooks, significantly increasing the chances of lost profiles, missed follow-ups, and structural human errors.
- Pipeline stage updates are often heavily delayed, leading to outdated candidate statuses and a prolonged time-to-hire.

2. Fragmented Application Channels & Candidate Interaction:

- No centralized, unified online portal exists for candidates to comfortably browse active job roles, submit applications, or track their progress.
- Candidates must rely on back-and-forth emails or phone calls to check their application status, causing severe delays and miscommunication.
- Lack of a dedicated, professional digital career portal severely limits employer marketing, candidate reach, and brand credibility.

3. Inefficient Technical Evaluations & Assessments:

- Technical screening and practical coding assessments are evaluated manually or via unmonitored offline assignments, increasing the risk of evaluation bias.
- Interview questions are generally researched and created from scratch by hand every time, leading to a lack of standardization across technical roles.
- No centralized system exists for automatically tracking assessment submissions or instantly unifying technical feedback from multiple interviewers.

4. Candidate Communication Limitations:

- Candidate queries, feedback sharing, and interview scheduling are handled manually, resulting in frustratingly slow response times.
- No structural database exists to track historical candidate interactions, previous interview feedback, or applicant preferences.
- The absence of automated status updates and continuous communication loops damages the candidate experience and significantly increases applicant drop-off rates.

5. Absence of Reporting & Hiring Insights:

- Recruiter performance reports, time-to-hire logs, and talent pool trends are generated completely manually, often lacking critical accuracy and real-time timeliness.
- Administrators have limited access to key performance indicators (KPIs) such as application conversion rates, job viewing metrics, or pipeline bottlenecks.
- Total inability to track sourcing behavior or recruiter velocity makes it incredibly difficult to optimize workforce planning or specific hiring strategies.

6. Security & Candidate Data Privacy Concerns:

- Sensitive candidate information (PDF resumes, contact details, identification documents) is stored loosely in local hard drives or insecure email inboxes, increasing the risk of data mismanagement.
- No encryption or secure administrative protocols are in place for storing private applicant data or evaluation records.
- Lack of role-based access control (RBAC), audit logs, or tracking of administrative activities opens the HR system to unauthorized access and violates modern data privacy compliance (such as GDPR).

2.2 PROPOSED SYSTEM

The Smart Recruit System is engineered to overcome the severe limitations of existing manual recruitment processes by providing a dynamic, automated, and interactive platform for both candidates and recruiters. The system aggressively aims to ensure hiring transparency, operational efficiency, and a significantly superior job application lifecycle.

❖ FOR CANDIDATES (JOB SEEKERS):

- **Registration/Login:** Candidates can securely create an account and log in to access personalized platform features such as job applications, assessment tracking, and saved career opportunities.
- **Job Search & Applications:** Candidates can search for active job postings using advanced filters such as location, salary expectations, employment type (Remote/On-site), and required skills. Each listing contains detailed information on role responsibilities, prerequisites, and company insights.
- **Assessments & Interview Scheduling:** Users can seamlessly engage with practical coding challenges or technical AI assessments linked to job postings, and confirm interview times directly with recruiters to eliminate email friction.
- **Application Pipeline History:** Candidates can cleanly track their previous applications, evaluation responses, and current pipeline statuses (Screening, Interview, Assessment, Offered). This dramatically improves transparency and eases candidate anxiety.
- **Profile & Resume Management:** Users can update personal details, contact information, portfolio links, and securely upload/manage their PDF resumes directly from their unified candidate dashboard.
- **Results & Feedback Insights:** Instead of being left in the dark, candidates can receive structured feedback or view basic assessment analytics if granted access by the hiring manager, elevating the overall employer brand.
- **Queries & Communication:** Candidates receive automated, real-time status updates via email and can stay completely informed regarding their application progression through internal notification systems.

❖ FOR ADMINISTRATORS & RECRUITERS:

- **Admin Analytics Dashboard:** A commanding, centralized dashboard displaying system-wide data such as total registered candidates, active job postings, application pipelines, and recruiter efficiency tracking. Admins can apply graphical filters to view advanced analytics based on specific timeframes or specific job roles.
- **User & Recruiter Management:** Administrators can view, edit, or deactivate candidate and recruiter accounts. Admins hold the authority to reset passwords, track platform usage, and monitor specific recruitment team behaviors.
- **Job Listings Management:** Recruiters can dynamically add, update, archive, or delete job listings. They can set explicit skill requirements, customize descriptions, configure job constraints, and intelligently manage automated job unpublishing/expiry dates.
- **Application & Pipeline Management:** Recruiters can view all submitted applications and instantly evaluate them. They can dynamically move candidates across structured pipeline stages, evaluate coding assessment submissions, and easily approve or reject candidate profiles.
- **AI Question Generator & Tools:** A dedicated module allowing technical recruiters to instantly generate completely custom, role-specific interview questions securely through integrated Artificial Intelligence (e.g., Gemini API), heavily cutting down preparation time.
- **Security & Access Control:** Administrators can actively manage system roles, track activity via logging functionality, and ensure that highly sensitive operations (and candidate data) are tightly restricted to authorized HR personnel only.

2.3 MODULES

1. User Management Module

Handles registration, login, authentication, and comprehensive profile management for Candidates and Recruiters.

Features:

- Secure registration and login system for Candidates, Recruiters, and Admins
- Forgot password functionality and recovery mechanisms
- Edit profile details (name, email, phone, password, portfolio URLs, etc.)
- Candidates can intimately view personal application history and assessment milestones
- Strict Role-based access control (Admin / Recruiter / Candidate)

2. Job Listings Management Module

Allows Admins and Recruiters to digitally add, update, or seamlessly manage active job postings.

Features:

- Add new job postings detailing title, job description, required technical skills, salary brackets, and location
- Set explicit job availability statuses alongside intelligent automated job posting expiry dates
- Admin functionality to review, approve, or reject job postings created by recruitment staff
- Job structural categorization (Full-time / Part-time / Remote / Internship)

3. Application & Pipeline Management Module

Aggressively manages candidate submissions, practical assessments, and linear hiring pipelines.

Features:

- Candidates can effortlessly submit job applications and attempt attached technical coding challenges
- Recruiters can cleanly view, manually grade, transition, approve, or reject candidate applications
- Pipeline mechanics to visually manage specific candidate stages (e.g., Applied -> Screening -> Assessment -> Interview -> Offered)
- Unique Application ID generation for error-free internal tracking
- Categorized filtering view: Pending Review, Under Assessment, Rejected, and Hired

4. Admin Dashboard Module

Provides complete, top-tier administrative control over all system data and HR activity.

Features:

- High-level overview of total active jobs, registered candidates, applications, and system status
- Graphical data reports detailing recruitment velocities and individual recruiter efficiency
- Advanced filtering of hiring data by timestamp, department, pipeline status, or location
- Quick accessibility to manage user restrictions, live jobs, and platform configurations
- Security notifications and strict auditing logs for critical admin activities

5. AI Features & Assessment Module

Empowers technical recruiters to intensely automate technical screening protocols using modern Artificial Intelligence.

Features:

- Dynamic AI Question Generator capable of compiling custom, role-specific technical questions instantly via integrated AI models
- Dedicated Code snippet environment for candidates to submit solutions to generated technical challenges
- Side-by-side 'Expected Input vs. Expected Output' mapping for incredibly fast recruiter validation
- Internal grading allocation functionality for practical technical submissions

6. Reports & Analytics Module

Helps HR Leads and Business Owners deeply track enterprise-level metrics and generate extremely useful hiring reports.

Features:

- Application trends and recruitment trajectory visualization (daily/weekly/monthly tracking charts)
- Advanced monitoring of Candidate Drop-off rates and structural pipeline bottlenecks
- Track the most-viewed job listings and actively identify highly popular technical skillsets in demand
- Custom range filters specifically built to analyze precise recruiter productivity patterns
- Dashboard tables detailing multi-role hiring closure analytics

7. Communication & Alert Module

Manages real-time notification dispatching and timeline communication between candidates and the internal ATS.

Features:

- Automated email construction for application receipt confirmations and status shifts (via SMTP)
- Trackable dashboard timeline panels reflecting historical application statuses for transparency
- Internal contact queries allowing candidates to report errors or seek support
- Real-time alerts keeping candidates constantly updated on their hiring flow presence

8. Job Search & Filter Module

Allows candidates to precisely discover their most suitable career opportunities using advanced interface filters.

Features:

- Global search bar combined with deep filter options (Job Title, Geographical Location, Salary brackets, Technical Skills)
- Highly responsive frontend architecture delivering dynamic result updates
- Feature to filter roles exclusively by their active/remote working status
- Sorting capabilities to order job postings cleanly based on newest listings, absolute relevance, or compensation

FEASIBILITY STUDY

- Feasibility Study can be considered an investigation that helps the management take a decision about whether the study of the system should be feasible for development or not.
- It is used to obtain a clear picture of the problem and decide whether a feasible or appropriate solution exists or not.
- It identifies the possibility of improving an existing system, developing a new system, and producing refined estimates for the further development of the system.
- The main objective of a feasibility study is to acquire the problem scope instead of immediately solving the problem.
- The output of a feasibility study is a formal system proposal that acts as a decision document, which includes the complete nature and scope of the proposed system.

➤ TYPES OF FEASIBILITY

1. Technical Feasibility:

- **Technology Requirements:**

The system will be developed using modern web technologies such as HTML5, CSS3, JavaScript, and Bootstrap for the frontend. Python and the Django Web Framework will be used for robust backend processing, and an SQLite/MySQL database will be utilized for reliable data storage.

- **Hardware/Software Requirements:**

The system can run efficiently on standard cloud hosting or web servers. The necessary software includes any modern web browser for users, and a Python environment (like VS Code or a virtual environment) for local development and server maintenance.

- **Technical Expertise:**

The development team is highly capable of working with the necessary Python frameworks, REST APIs, and integrated AI models, ensuring the complete technical feasibility of the project.

2. Operational Feasibility:

- **User Experience:**

The system is exclusively designed to be intuitive and user-friendly via a Glassmorphism interface. Both candidates and recruiters will have easy access to their respective pipelines and functionalities with minimal to no training required.

- **User Support:**

The platform will automatically provide support features like real-time application status updates, clear timeline visibility, and automated email alerts, ensuring smooth HR operations and high candidate satisfaction.

- **Maintenance and Updates:**

The modular architecture provided by Django ensures the system will be easy to maintain and update over time. Regular software updates can be implemented to safely add new features and ensure strict data security.

3. Economic Feasibility:

- **Cost Analysis:**

The initial cost of development will primarily include costs for software hosting, AI API usage (like Gemini), and personnel effort. The budget will be heavily allocated for secure web development, thorough testing, and server deployment.

- **Return on Investment (ROI):**

The system is expected to directly bring much higher corporate efficiency, sharply reduce manual resume screening hours, and increase hiring velocity by automating repetitive HR tasks. This can significantly reduce administrative costs and increase overall business productivity.

- **Scalability:**

As the organization grows, the system is designed to be easily scaled to support a massive volume of candidate profiles, new hiring departments, and advanced AI features without requiring a complete redesign, offering exceptional value for money in the long run.

SYSTEM FEATURES

- **Role-Based Login Architecture:** Secure and distinct access portals specifically designed for Administrators, HR Recruiters, and Job Candidates.
- **Automated Job Listing Management:** Recruiters can easily create, publish, edit, and define automatic expiry dates for complete control over active job postings.
- **Advanced Search & Filtering:** Candidates can dynamically filter active jobs by role, location, salary brackets, and required technical skills.
- **Visual Application Pipeline:** Structured, visually grouped tracking stages (e.g., Applied, Screening, Assessment, Interview, Offered) allowing recruiters to move candidates seamlessly across the hiring flow.
- **AI-Driven Assessment Generation:** An integrated AI module (using Gemini API) that empowers recruiters to instantly generate completely custom, role-specific technical interview questions.
- **Practical Coding Environment:** A dedicated technical submission module allowing candidates to input code or practical solutions explicitly mapped against expected outcomes for fast recruiter evaluation.
- **Automated Communication & Alerts:** Pre-configured emails (via SMTP) and dashboard notifications triggered automatically whenever a candidate changes stages or receives an update.
- **Enterprise Analytics Dashboard:** Fully equipped Admin control panel displaying critical hiring metrics, recruiter efficiency velocity, and application drop-off rates through informative charts.
- **Responsive Glassmorphism UI:** A sleek, modern frontend built using Bootstrap 5 that adapts flawlessly across desktops, tablets, and smartphones.
- **Structured Interview Feedback:** Recruiter grading tools that allow hiring managers to leave uniform evaluations and scores attached strictly to a candidate's profile.
- **Candidate Career Dashboard:** A personal profile area where candidates can track their entire application history, upcoming assessment dates, and easily swap out PDF resumes.
- **Top-Tier Security Protocols:** Implementation of powerful Django backend security, including CSRF token protection, strict password hashing, secure session handling, and impenetrable data isolation between candidates.

Hardware and Software Requirements

➤ 5.1 Hardware Requirements:

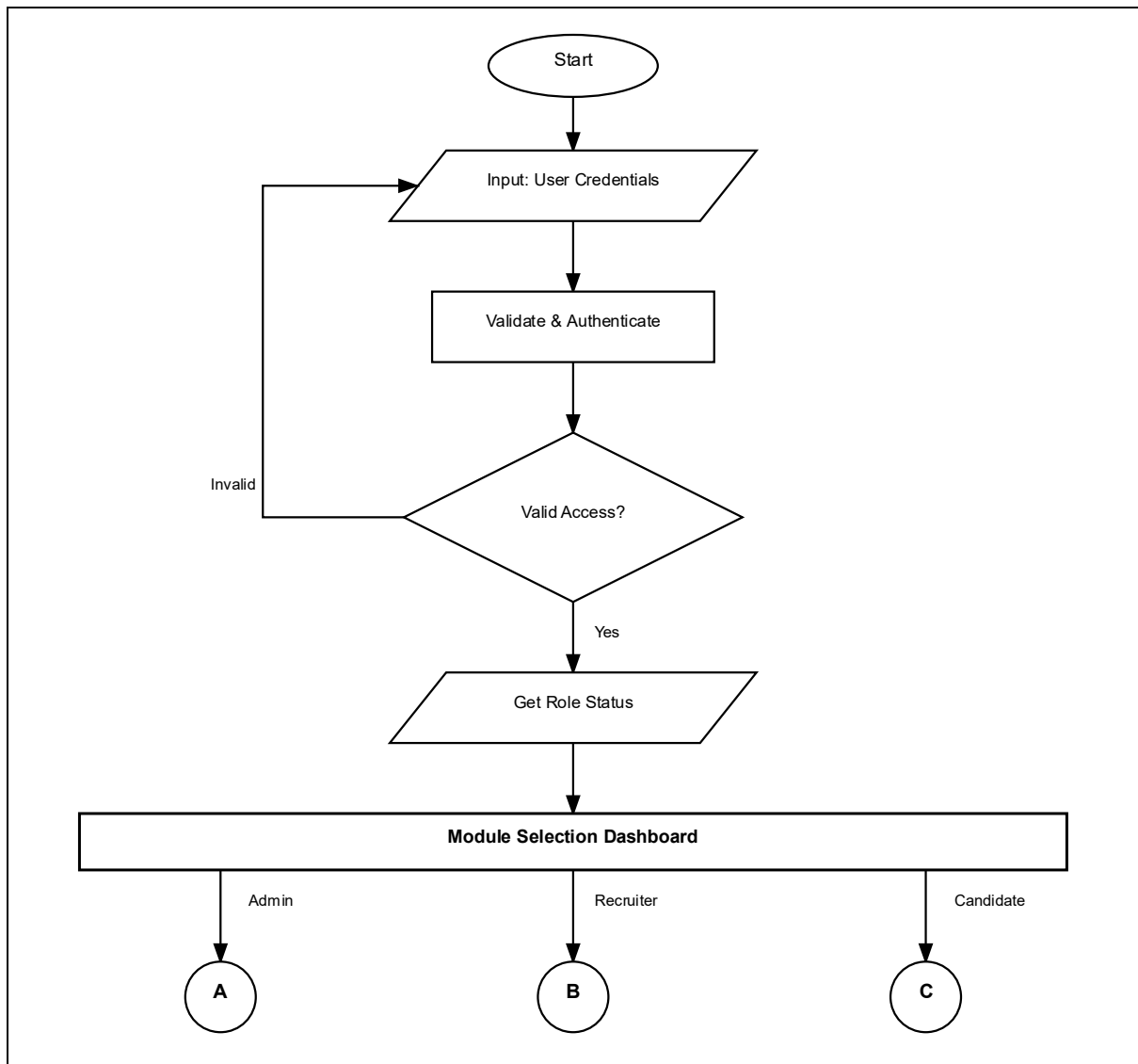
Requirements	Minimum Values
Processor	Intel(R) Core (TM) i5-6400 or AMD Ryzen 3 or Higher
CPU	2.70 GHz or Higher
RAM	8 GB or More
Hard Disk	256 GB or More (SSD Highly Recommended)

➤ 5.2 Software Requirements:

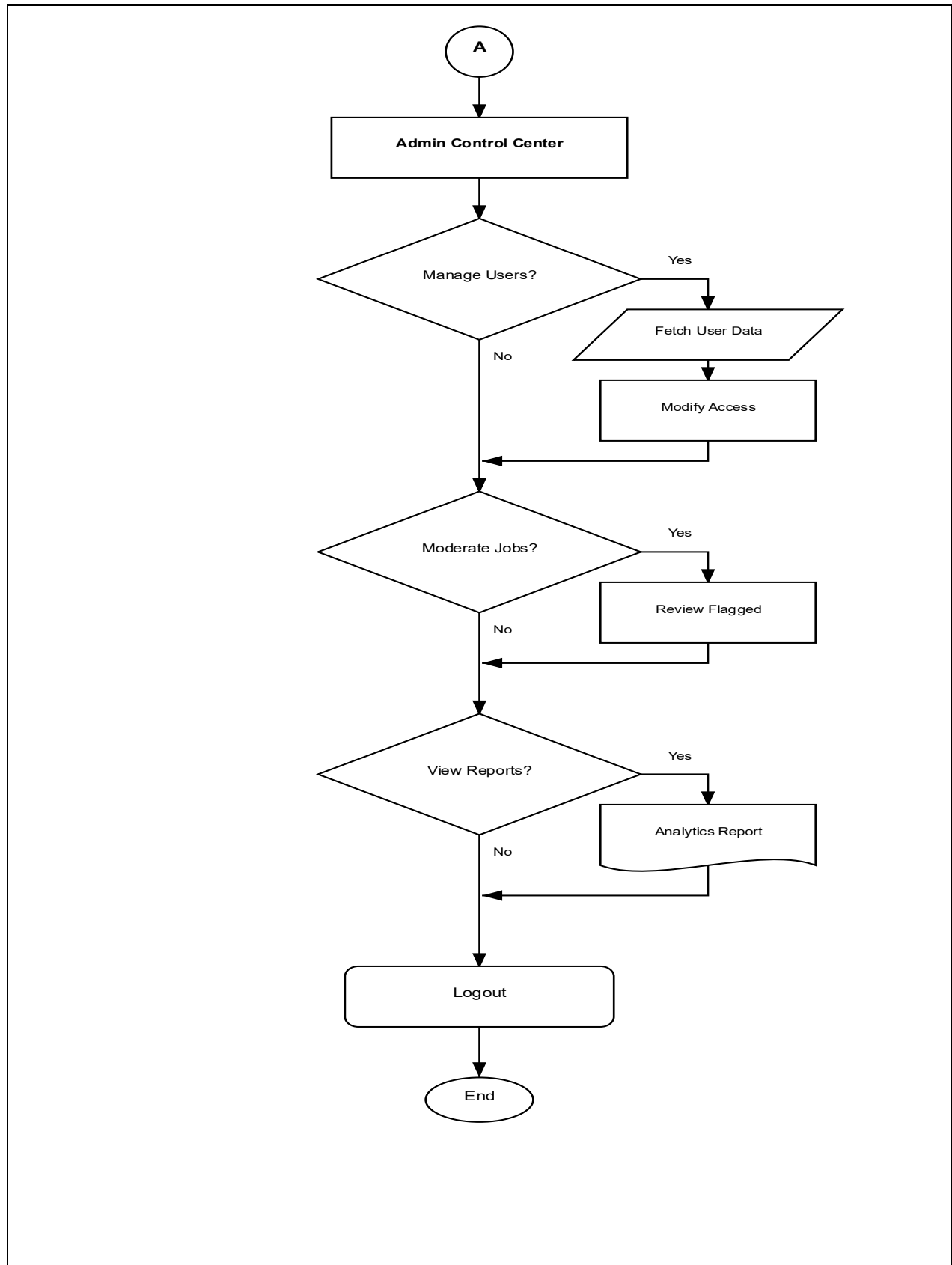
Requirements	Minimum Values
Front-End	HTML5, CSS3, JavaScript, Bootstrap 5
Back-End	Python, Django Web Framework
Server	Django Development Server (WSGI/ASGI), Redis
Database	SQLite3
Documentation	MS Office 2016 or higher (MS Word, MS PowerPoint)
Software (IDE)	Visual Studio Code (VS Code)
Operating System	Windows 10/11, macOS, or Linux

SYSTEM DESIGN

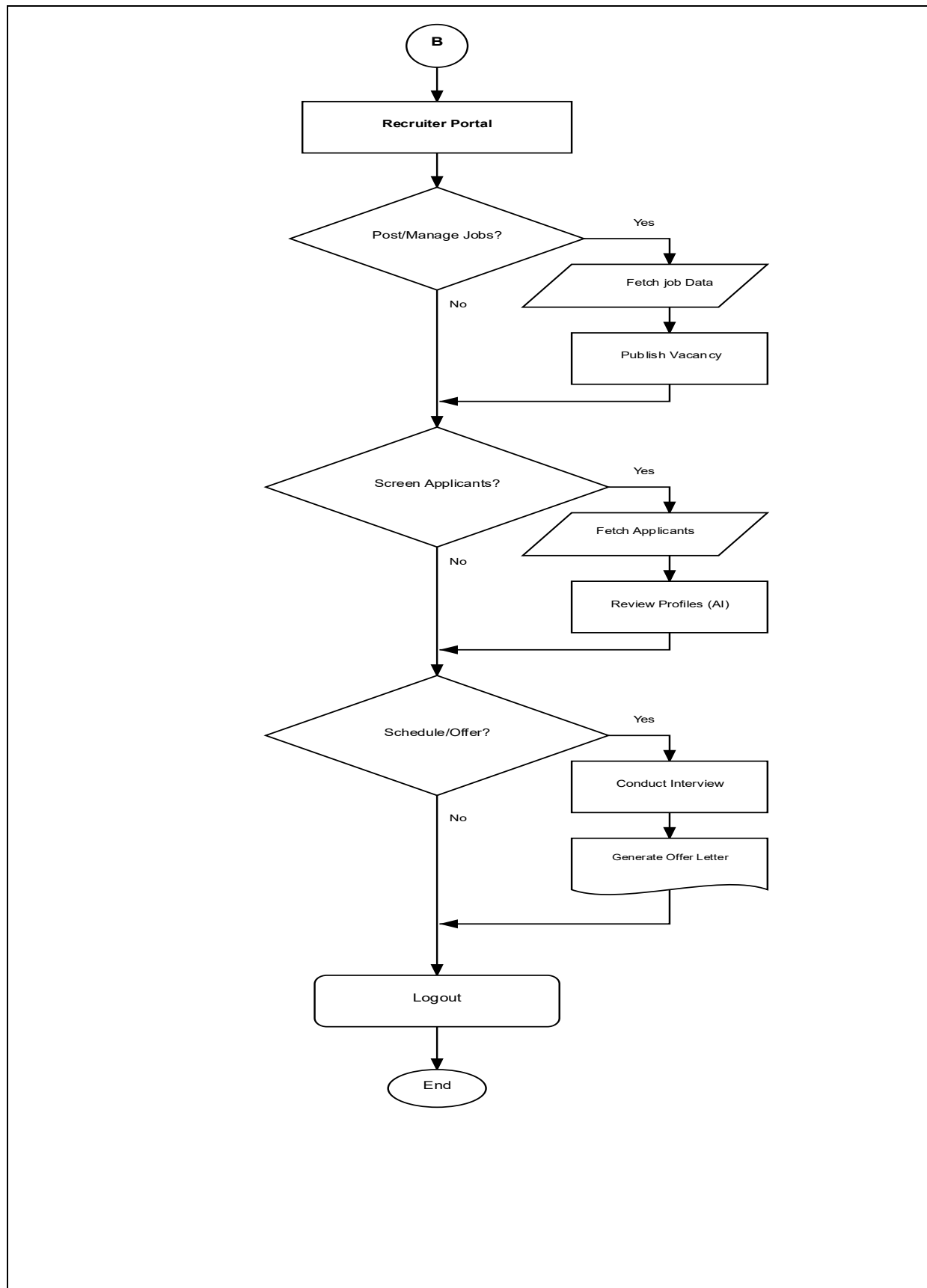
❖ System Flow Diagram (Log in):



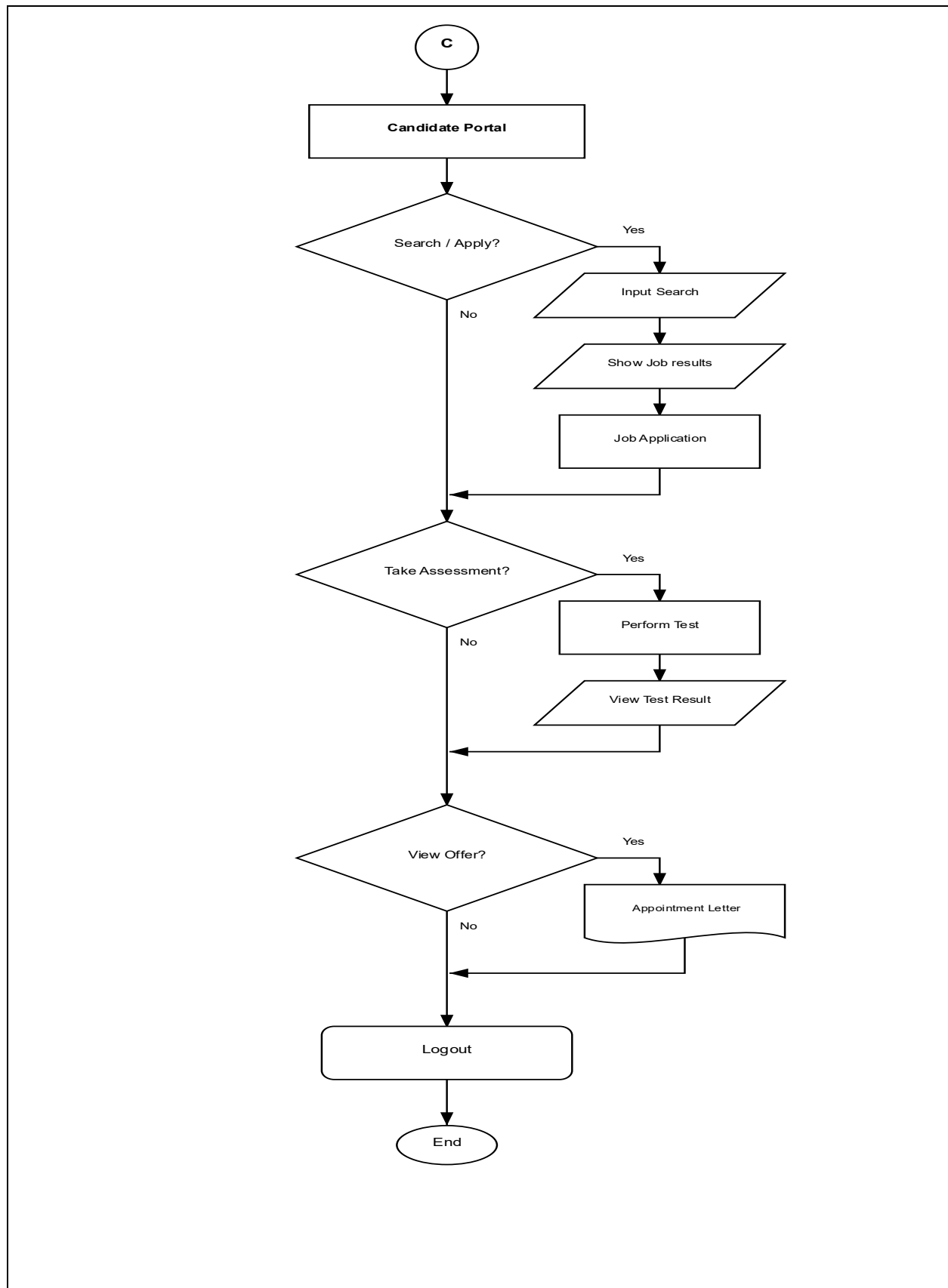
❖ System Flow Diagram (Admin):



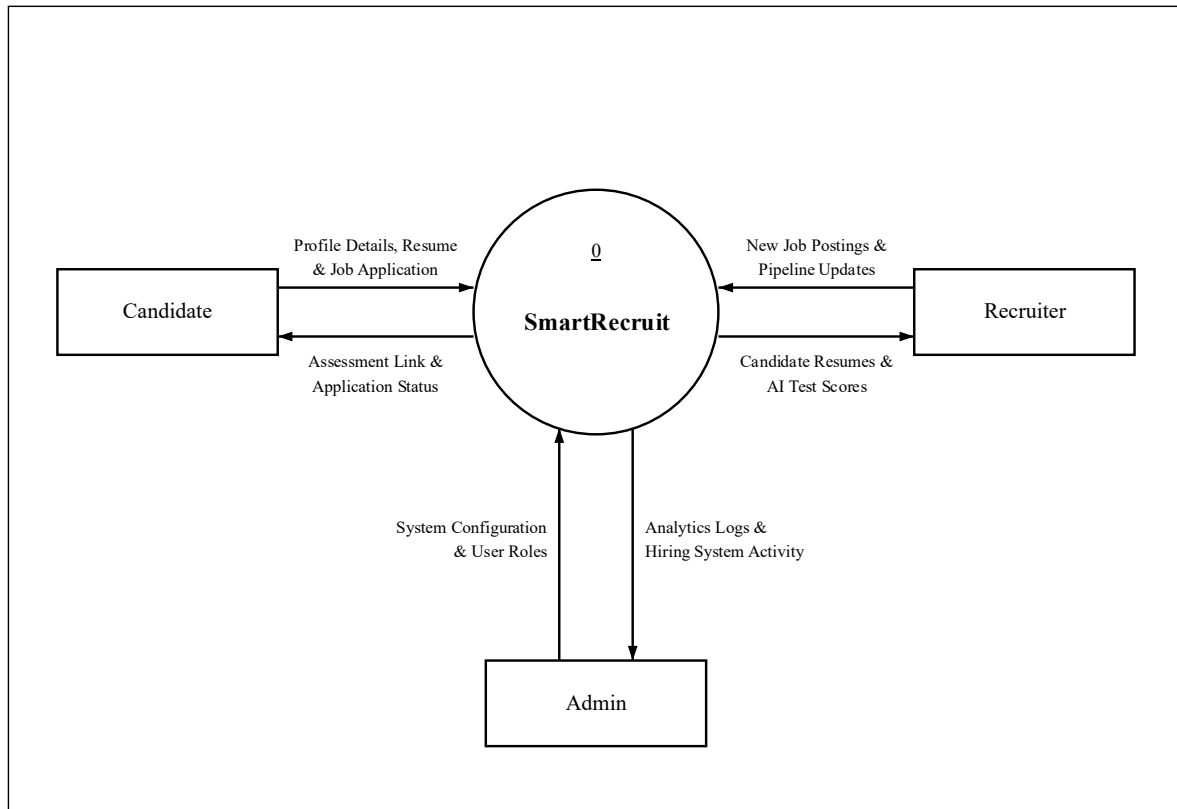
❖ System Flow Diagram (Recruiter):



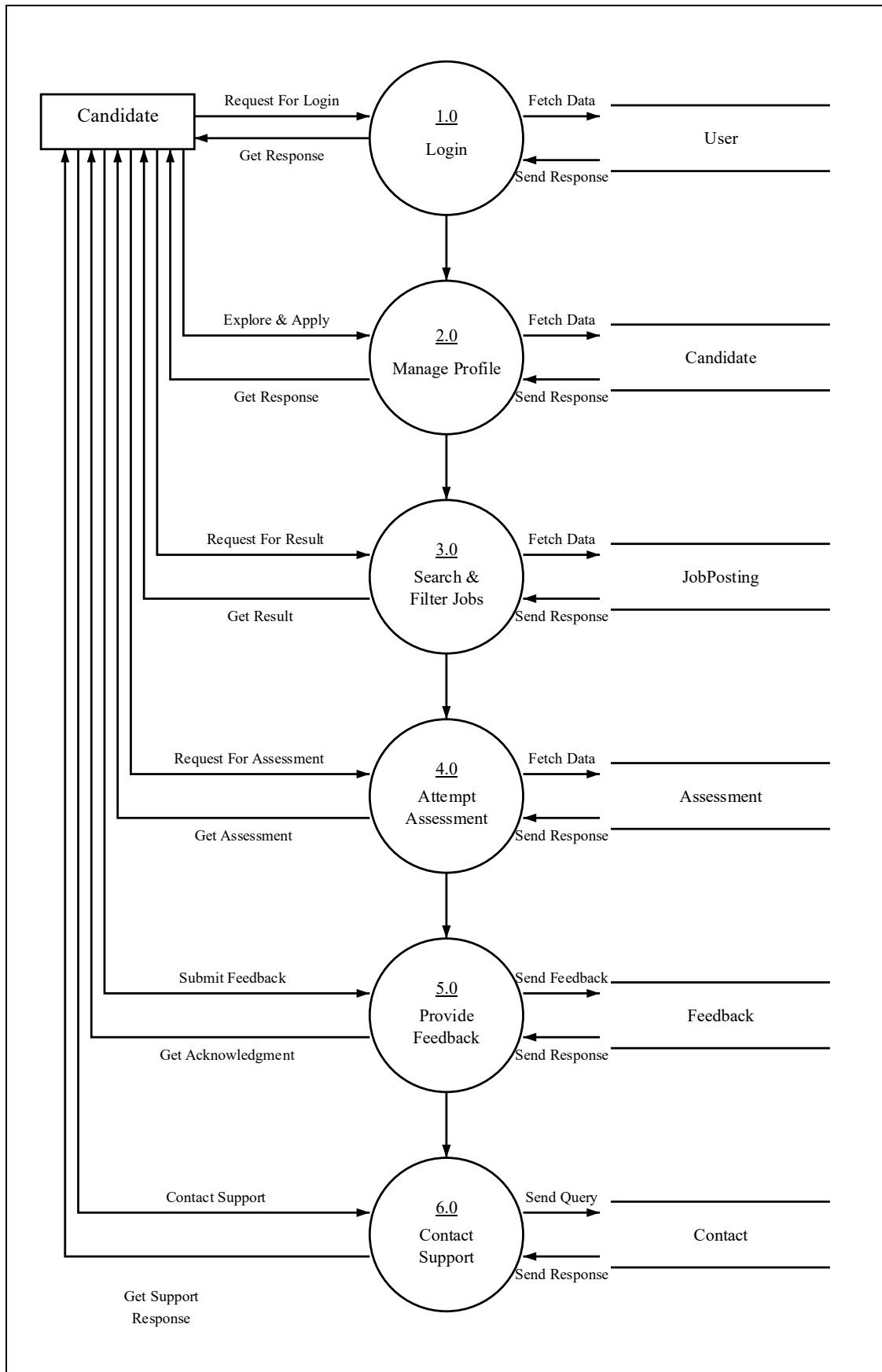
❖ System Flow Diagram (Candidate):



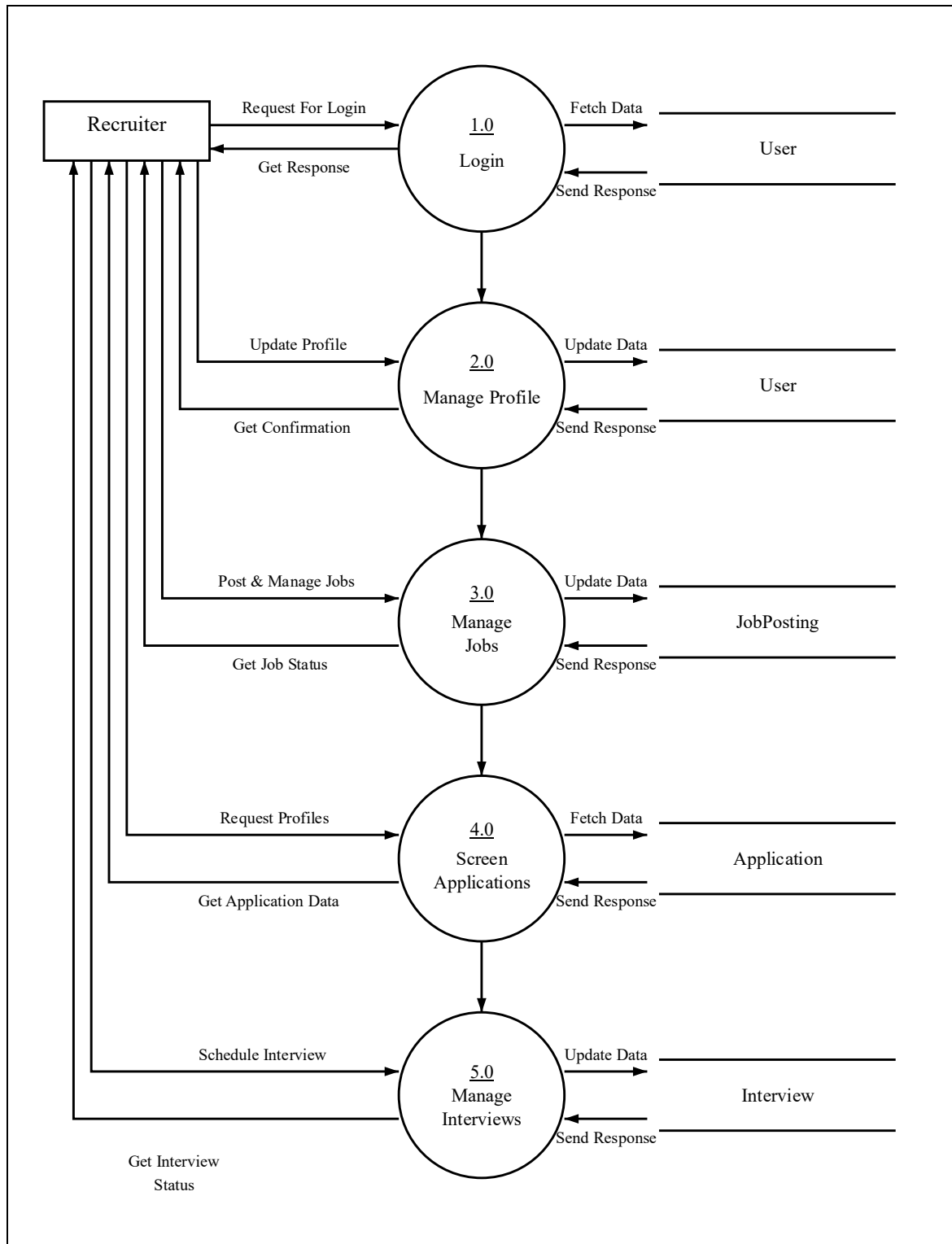
❖ Context Level DFD:



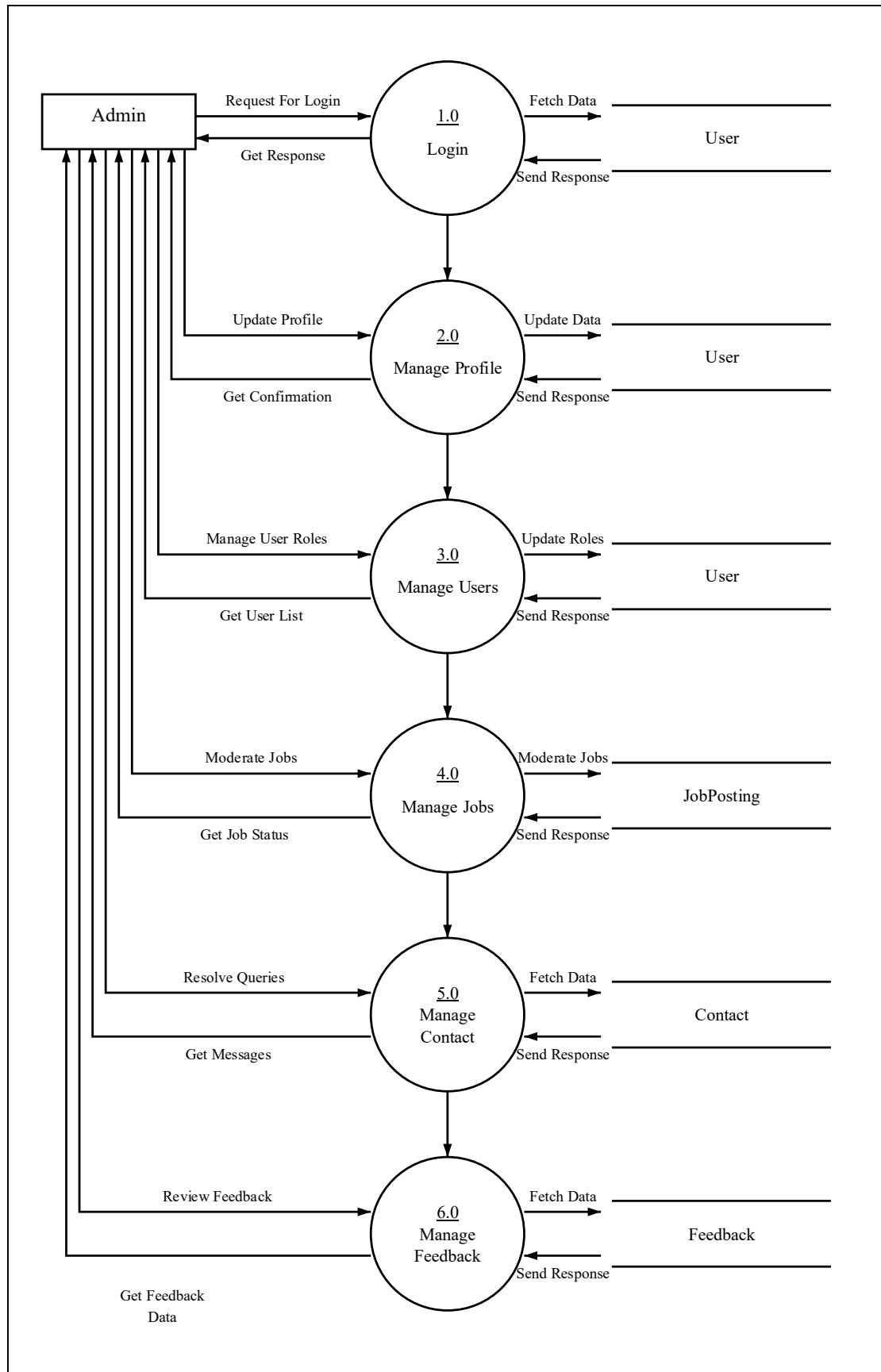
❖ Zero Level DFD (Candidate):



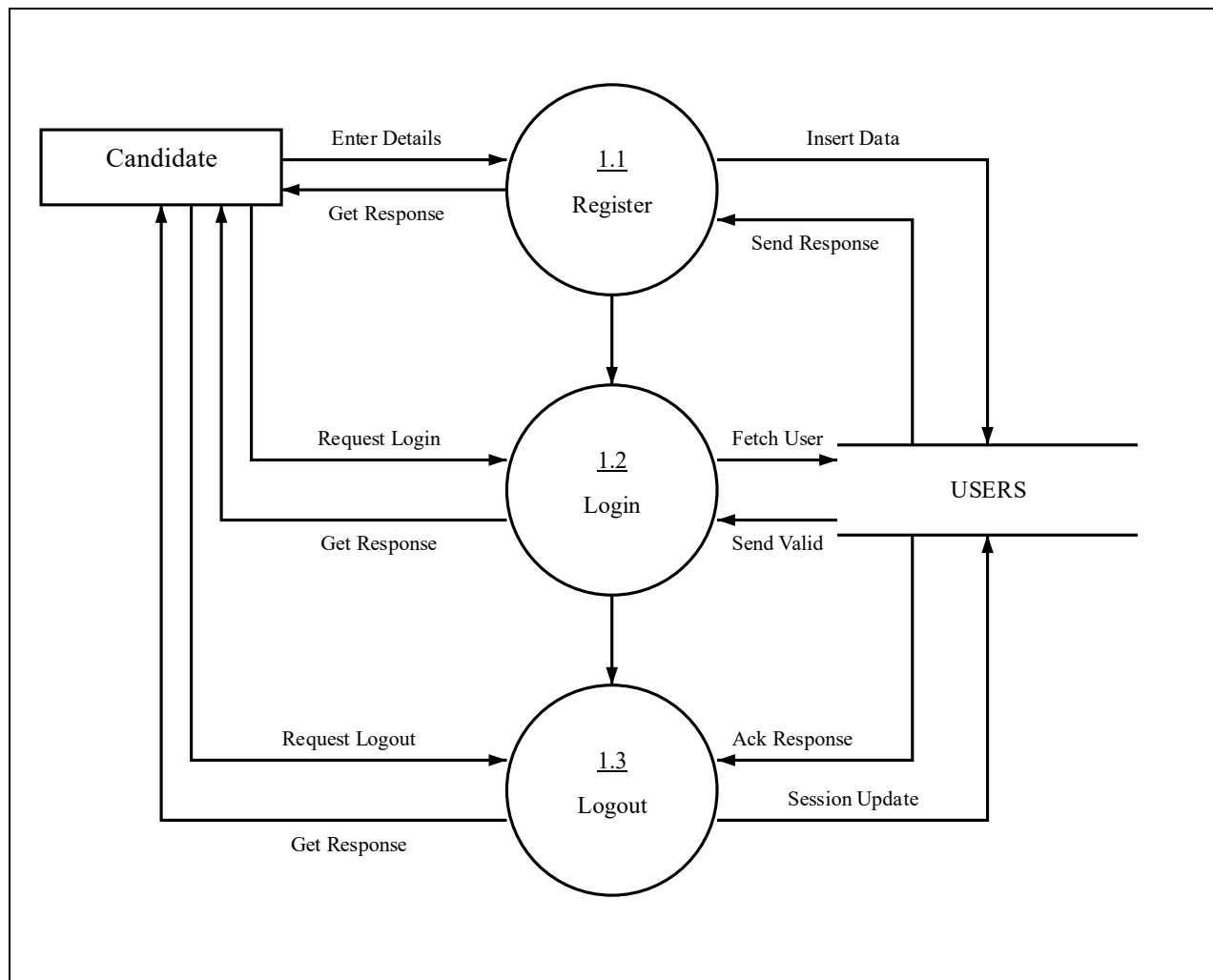
❖ Zero Level DFD (Recruiter):



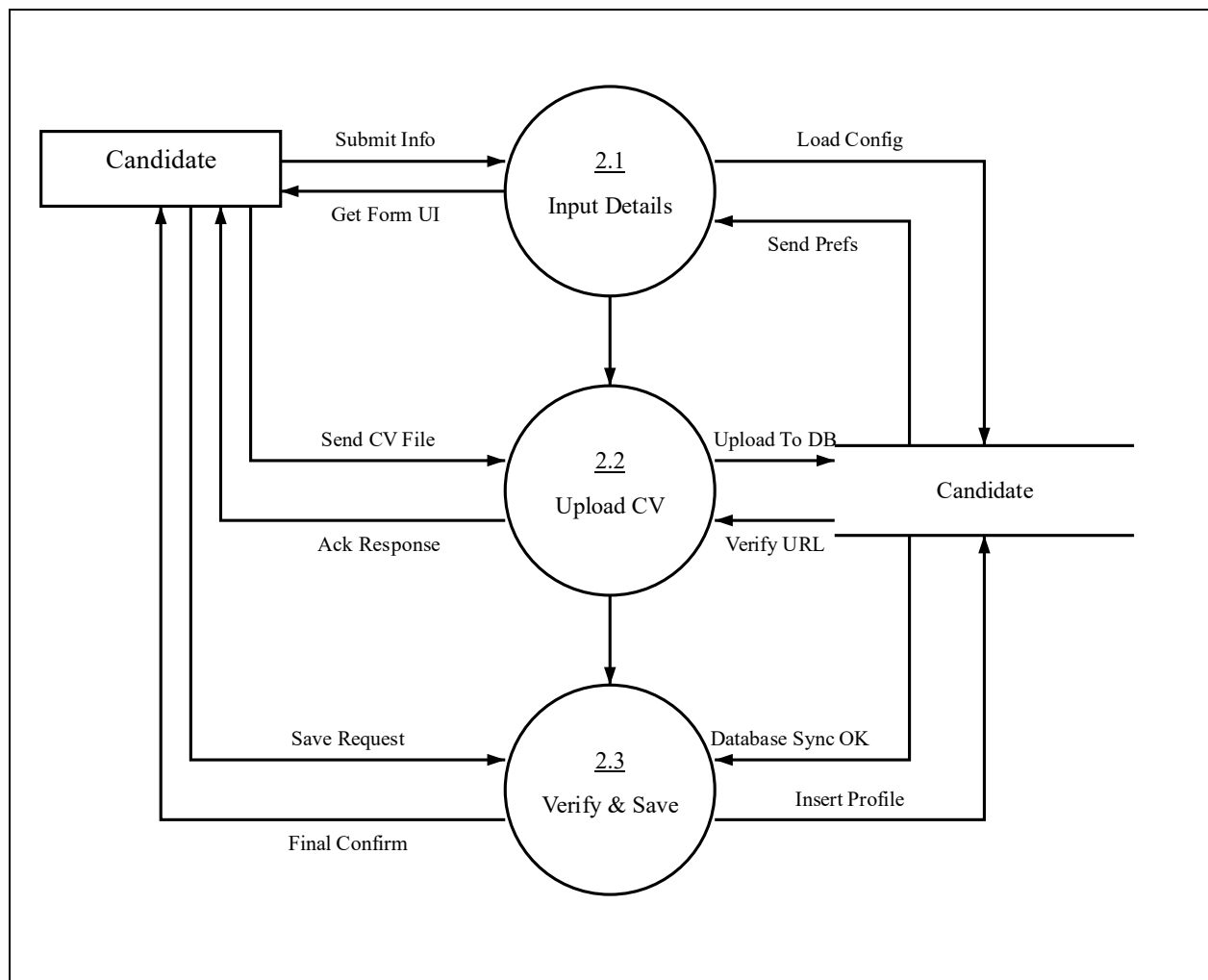
❖ Zero Level DFD (Admin):



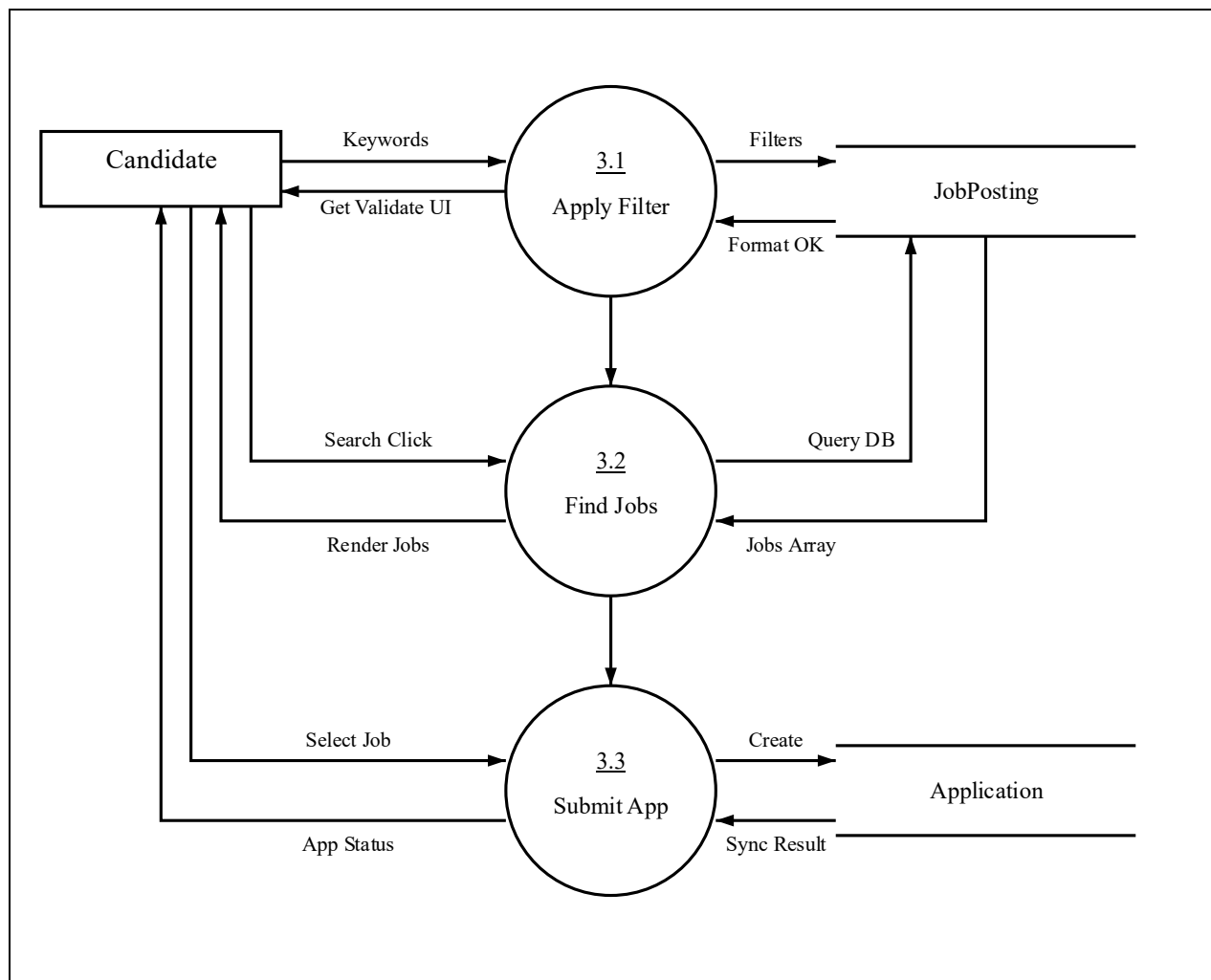
❖ **First Level DFD 1.0 Login (Candidate):**



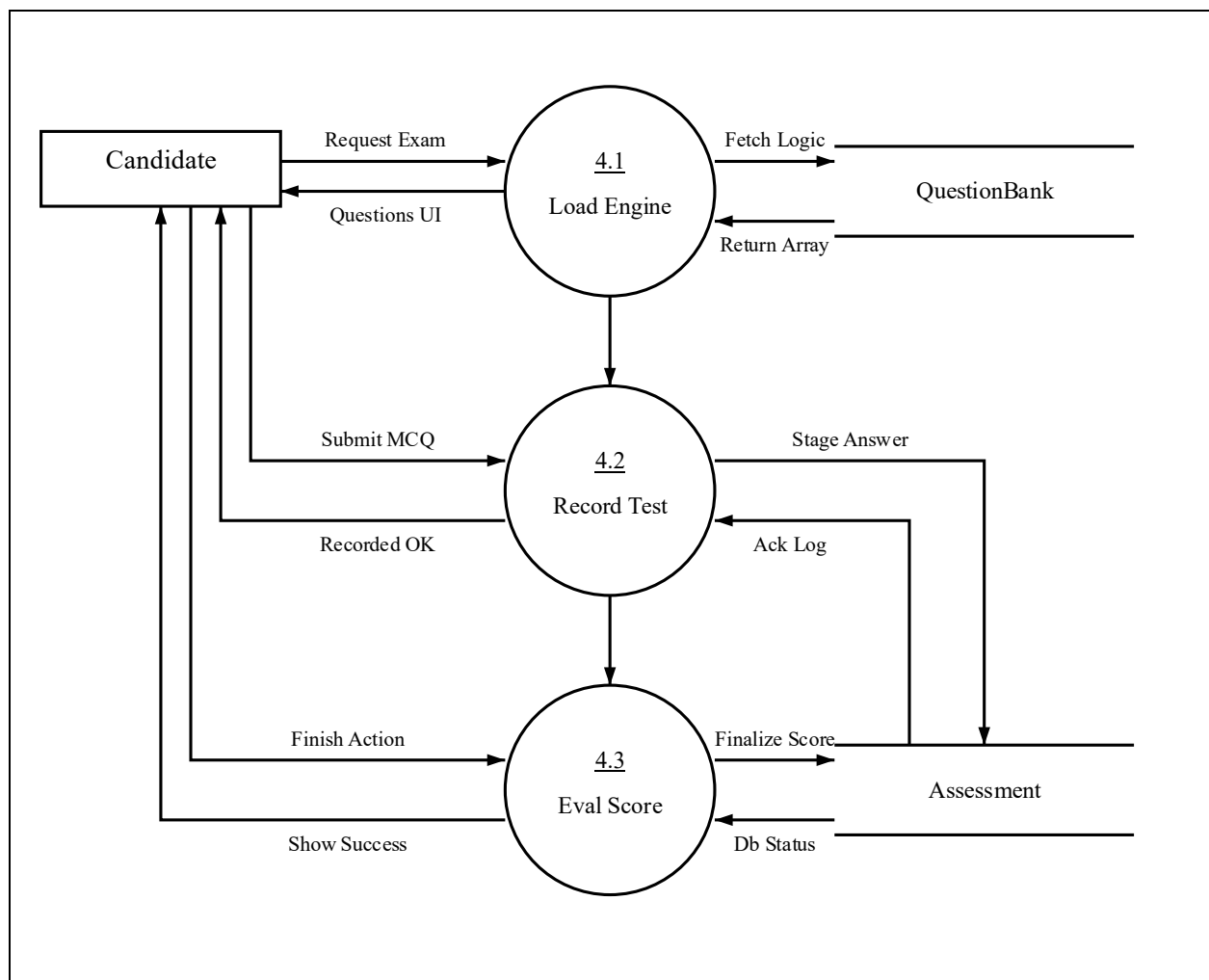
❖ **First Level DFD 2.0 Manage Profile (Candidate):**



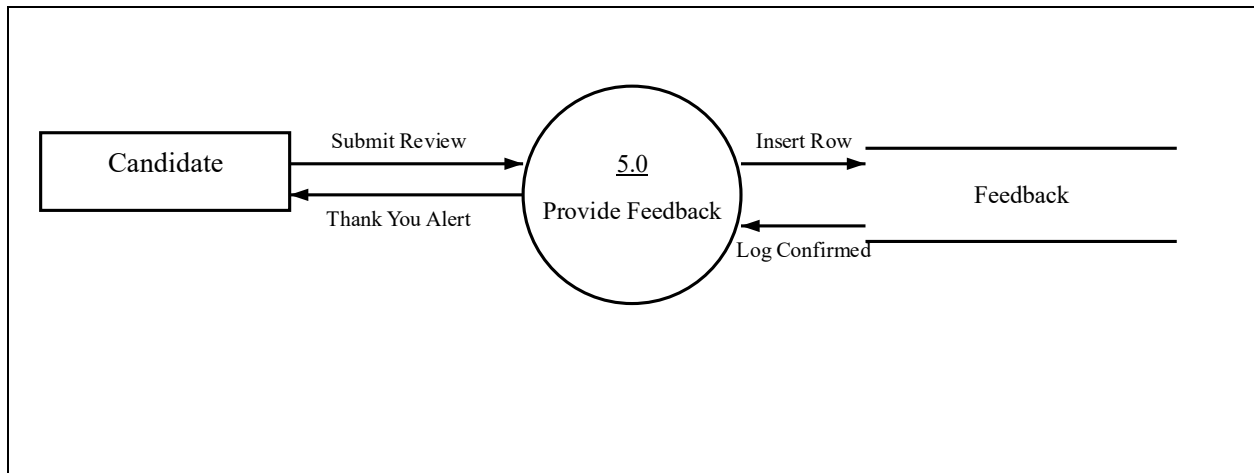
❖ **First Level DFD 3.0 Search & Filter Jobs (Candidate):**



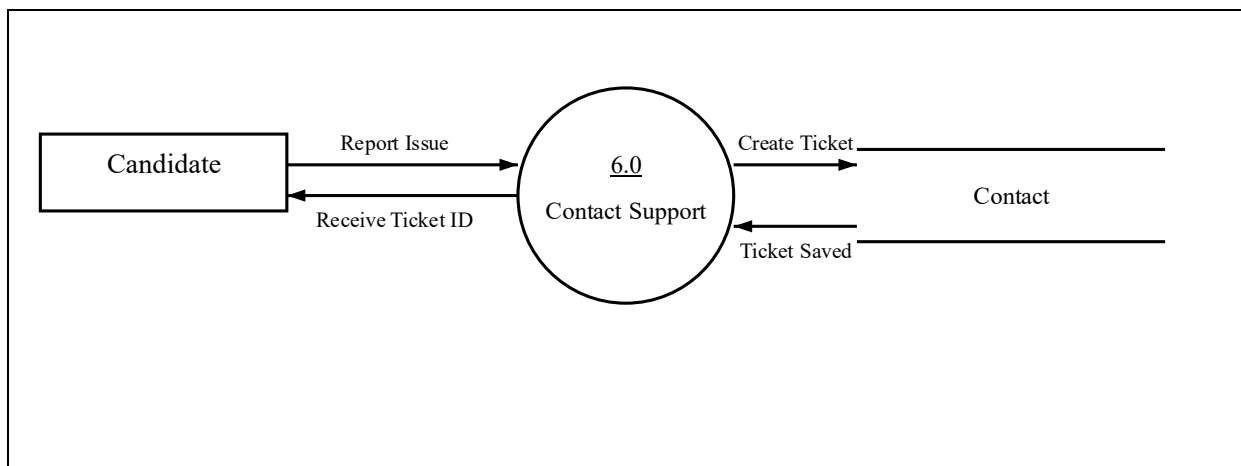
❖ **First Level DFD 4.0 Attempt Assessment (Candidate):**



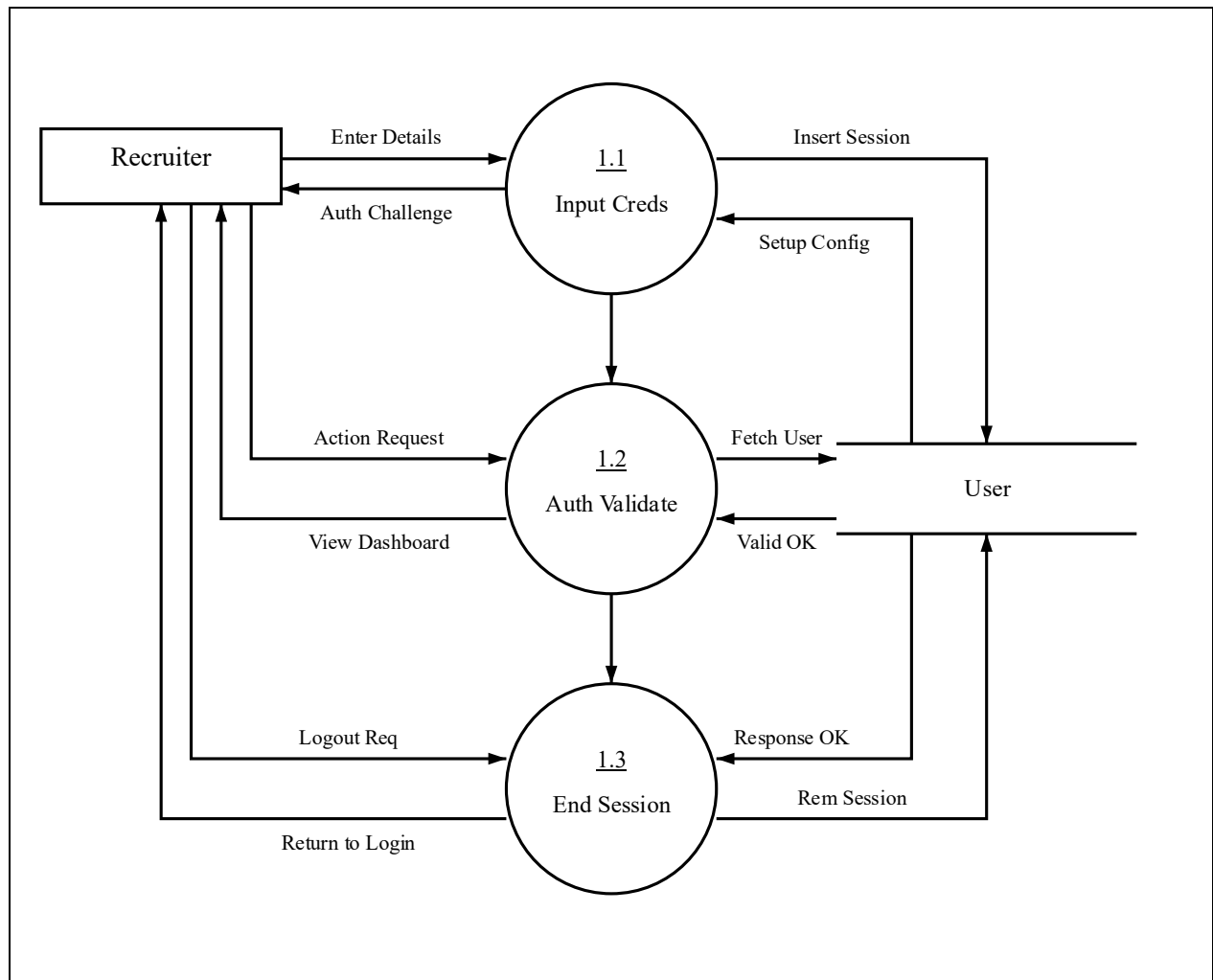
❖ **First Level DFD 5.0 Provide Feed Back (Candidate):**



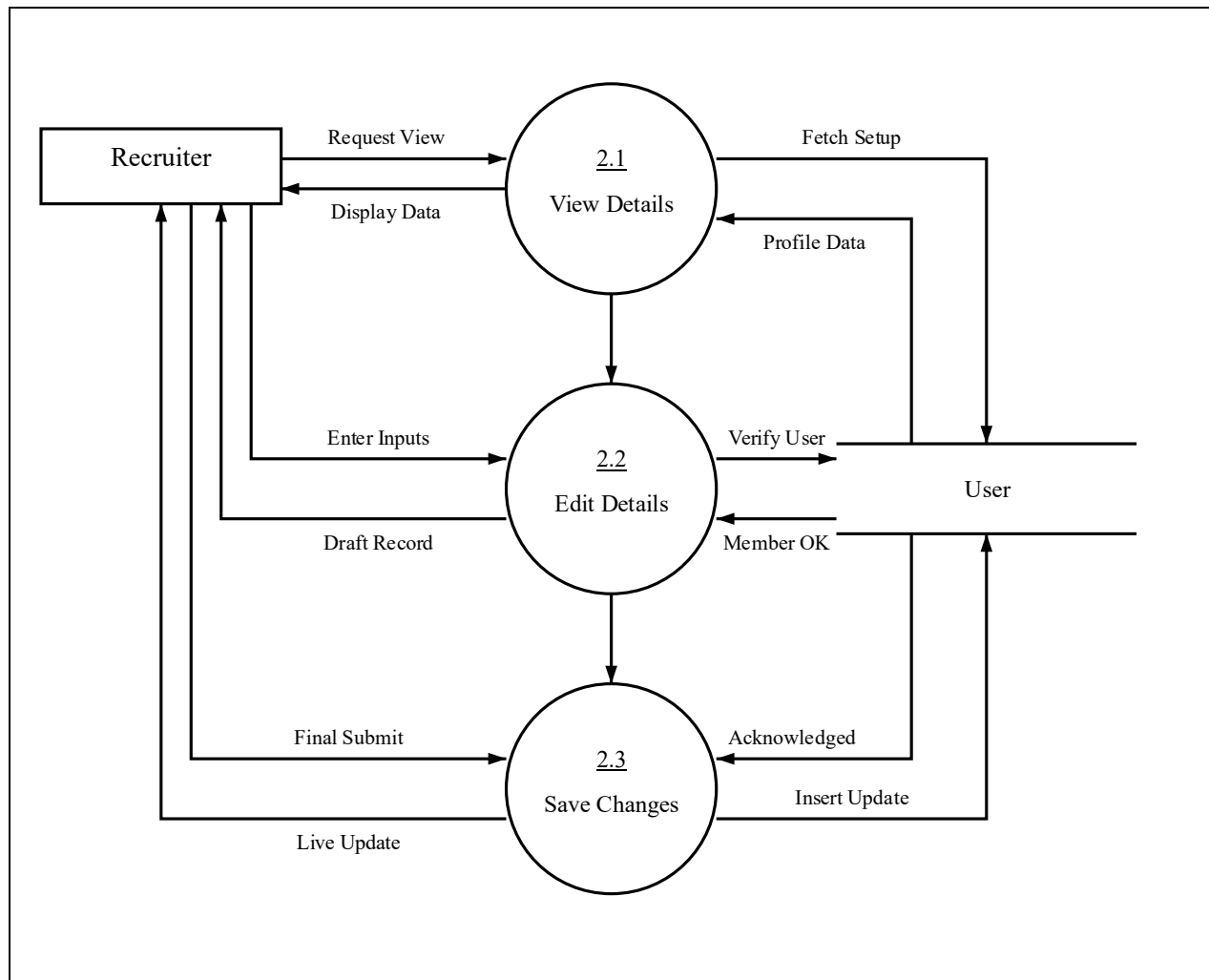
❖ **First Level DFD 6.0 Contact Support (Candidate):**



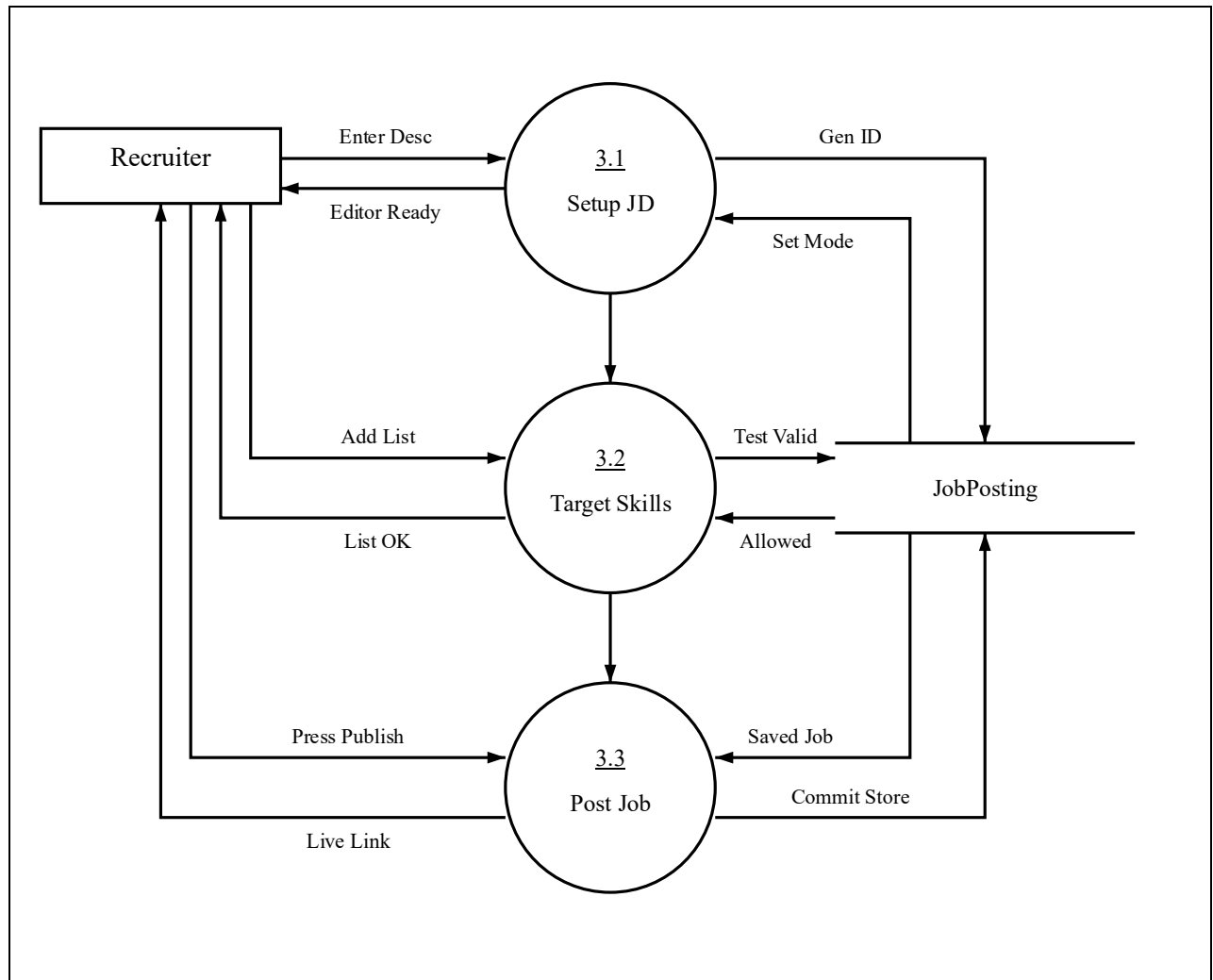
❖ **First Level DFD 1.0 Login (Recruiter):**



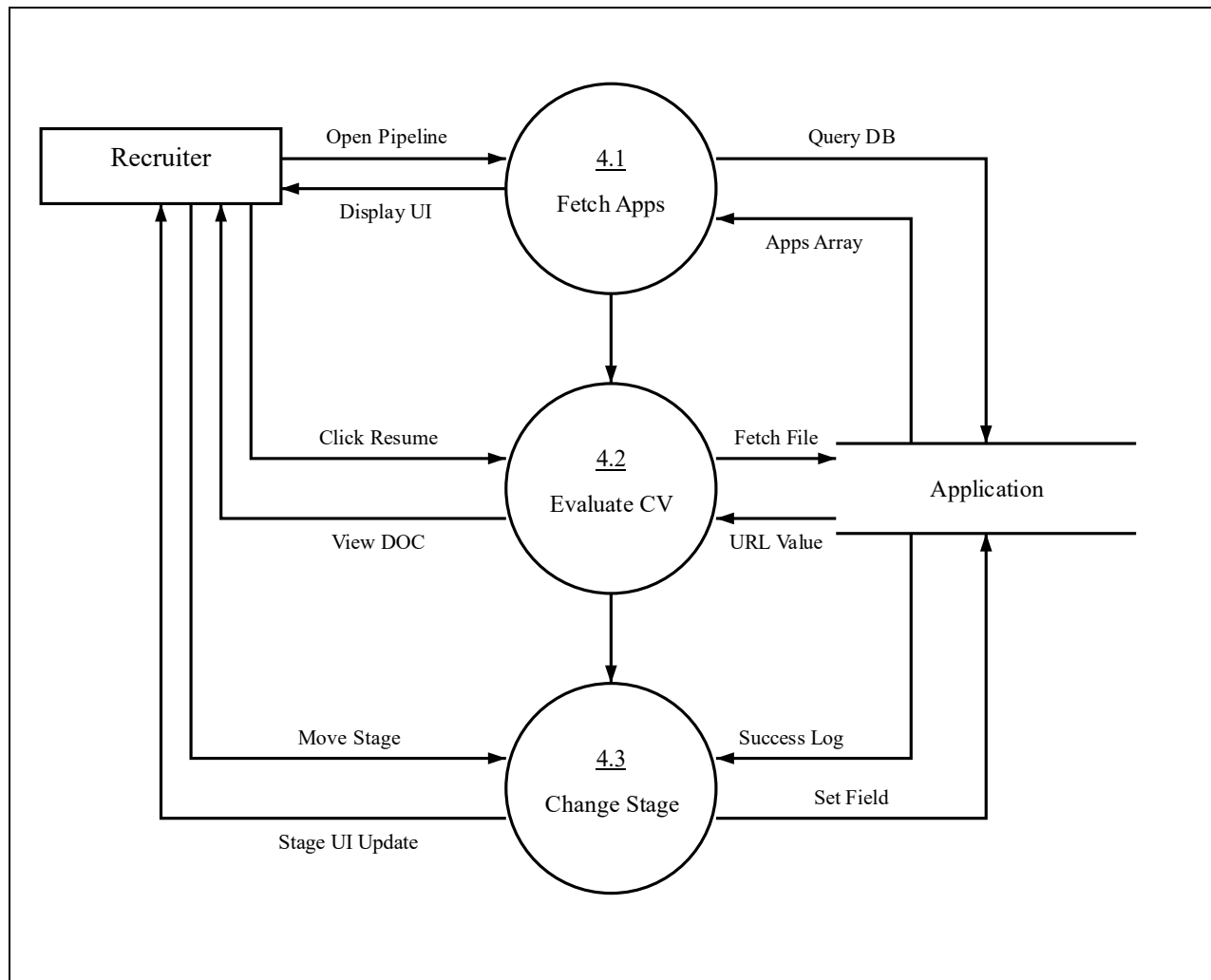
❖ **First Level DFD 2.0 Manage Profile (Recruiter):**



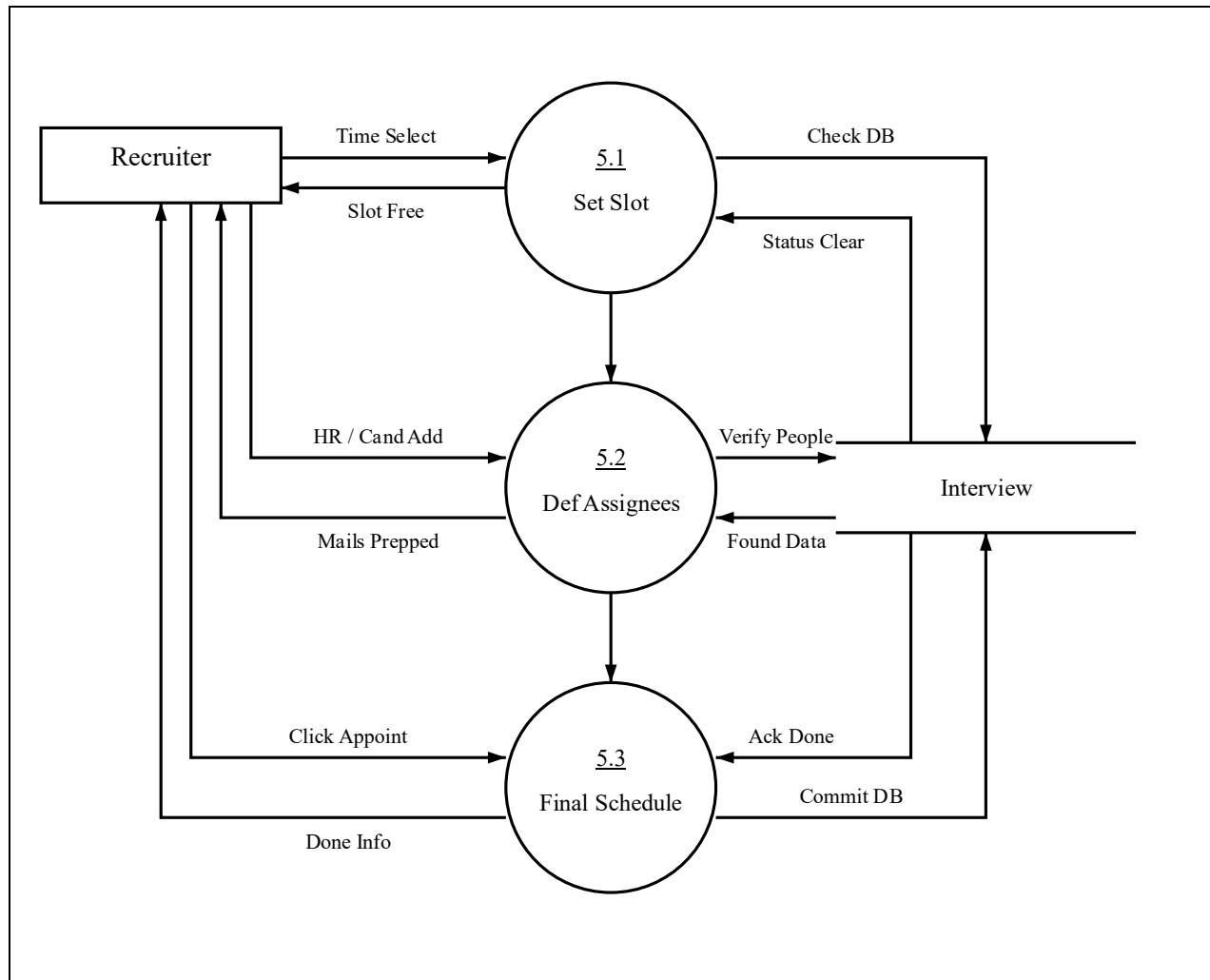
❖ **First Level DFD 3.0 Manage Jobs (Recruiter):**



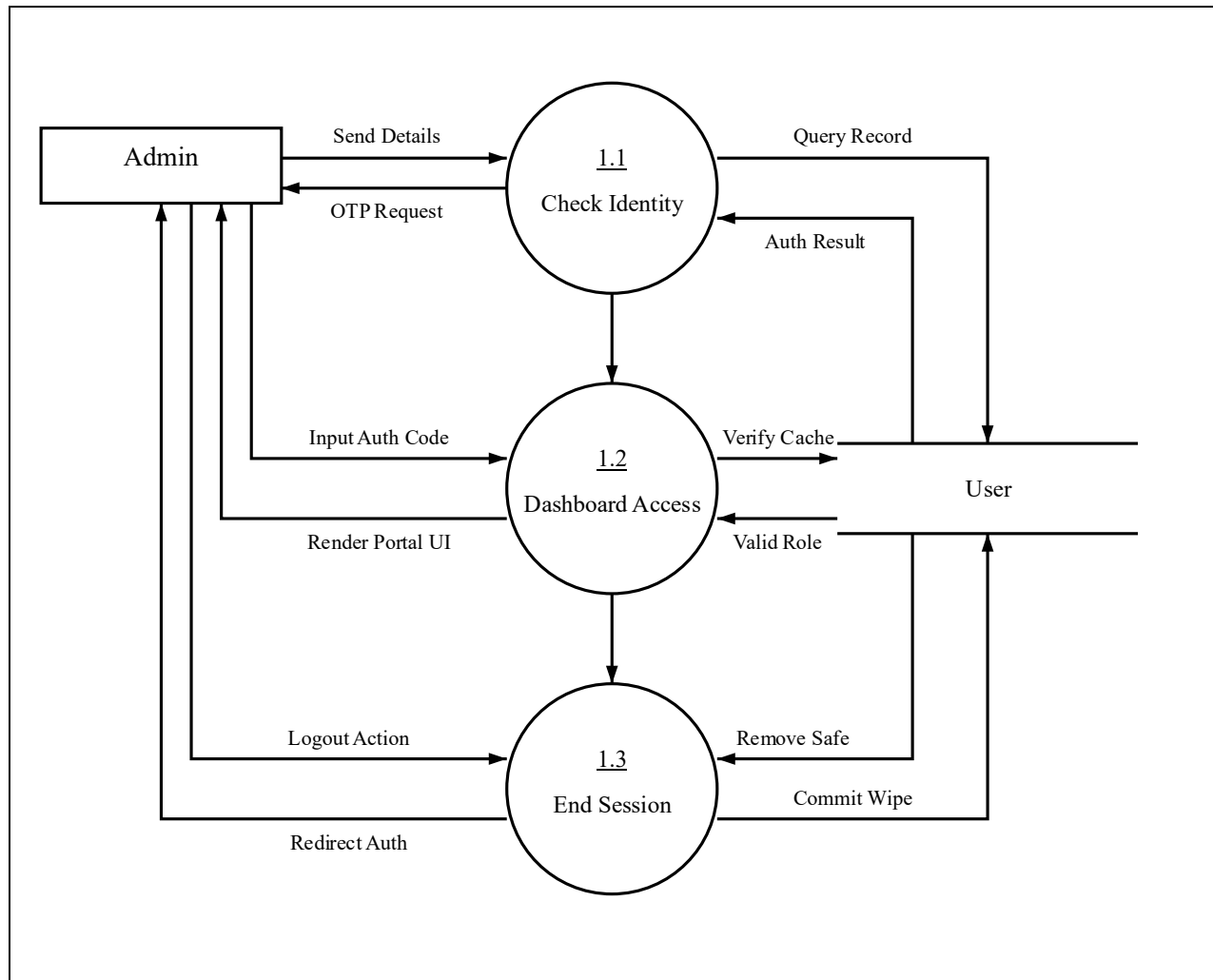
❖ **First Level DFD 4.0 Screen Applications (Recruiter):**



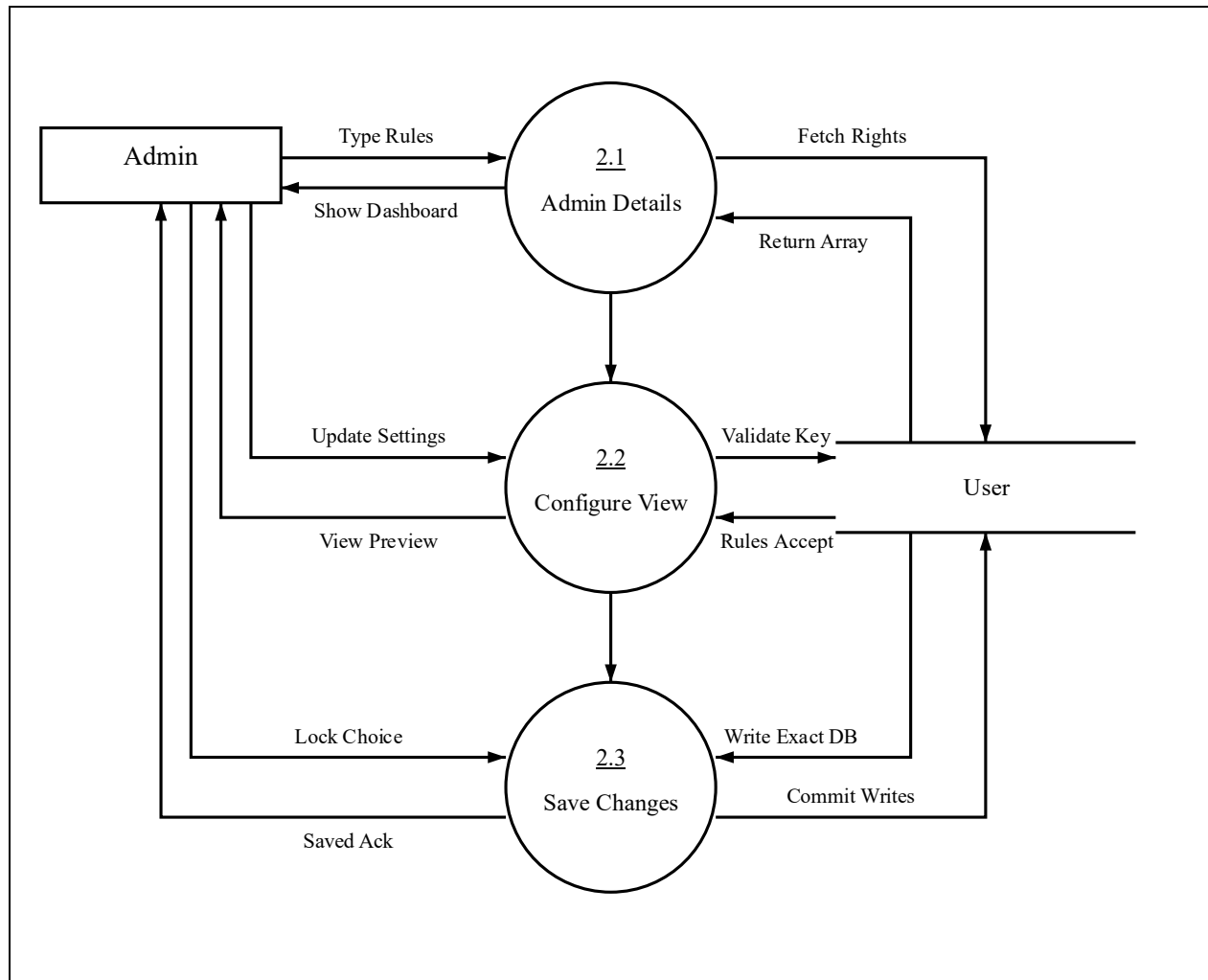
❖ **First Level DFD 5.0 Manage Interviews (Recruiter):**



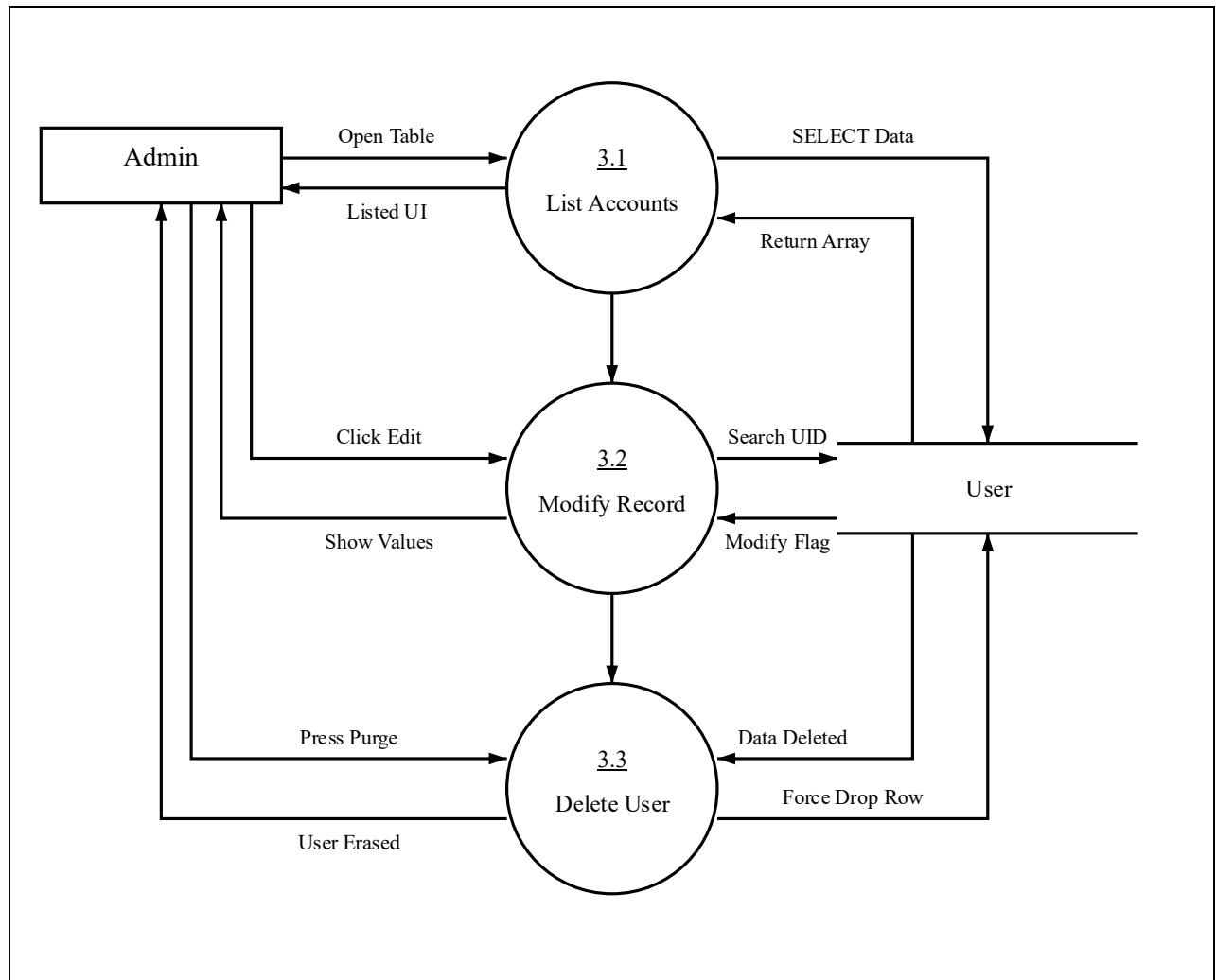
❖ **First Level DFD 1.0 Login (Admin):**



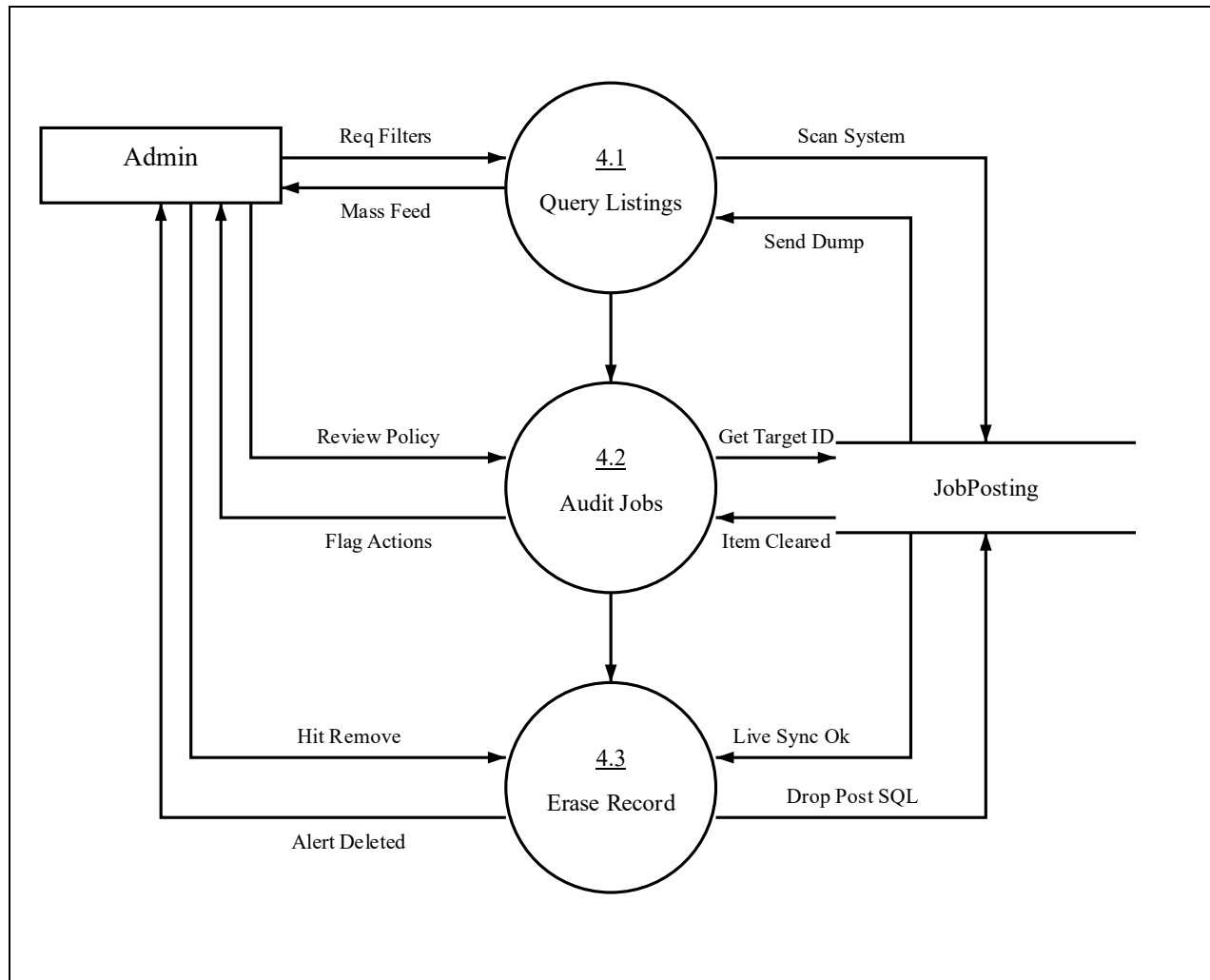
❖ **First Level DFD 2.0 Manage Profile (Admin):**



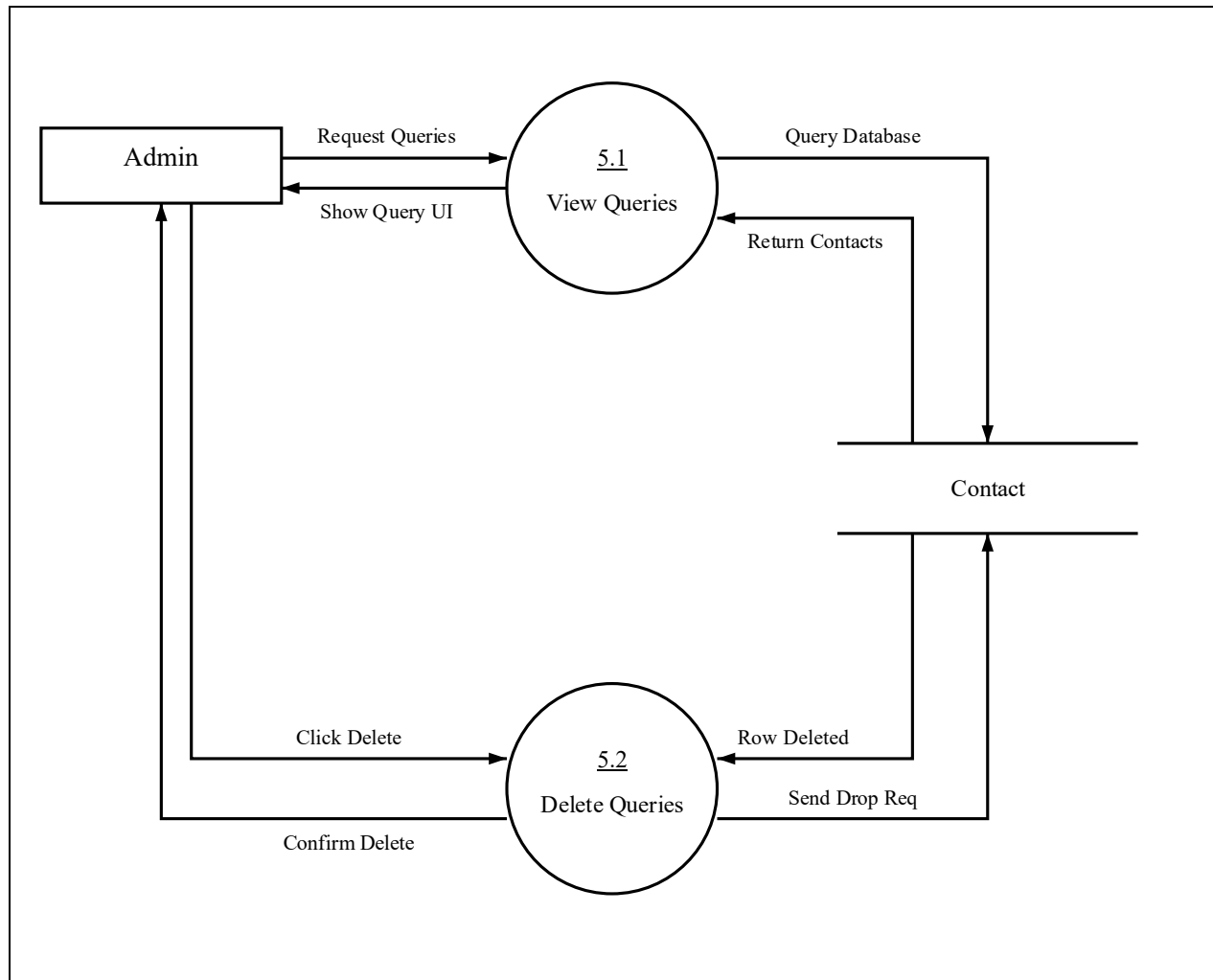
❖ **First Level DFD 3.0 Manage Users (Admin):**



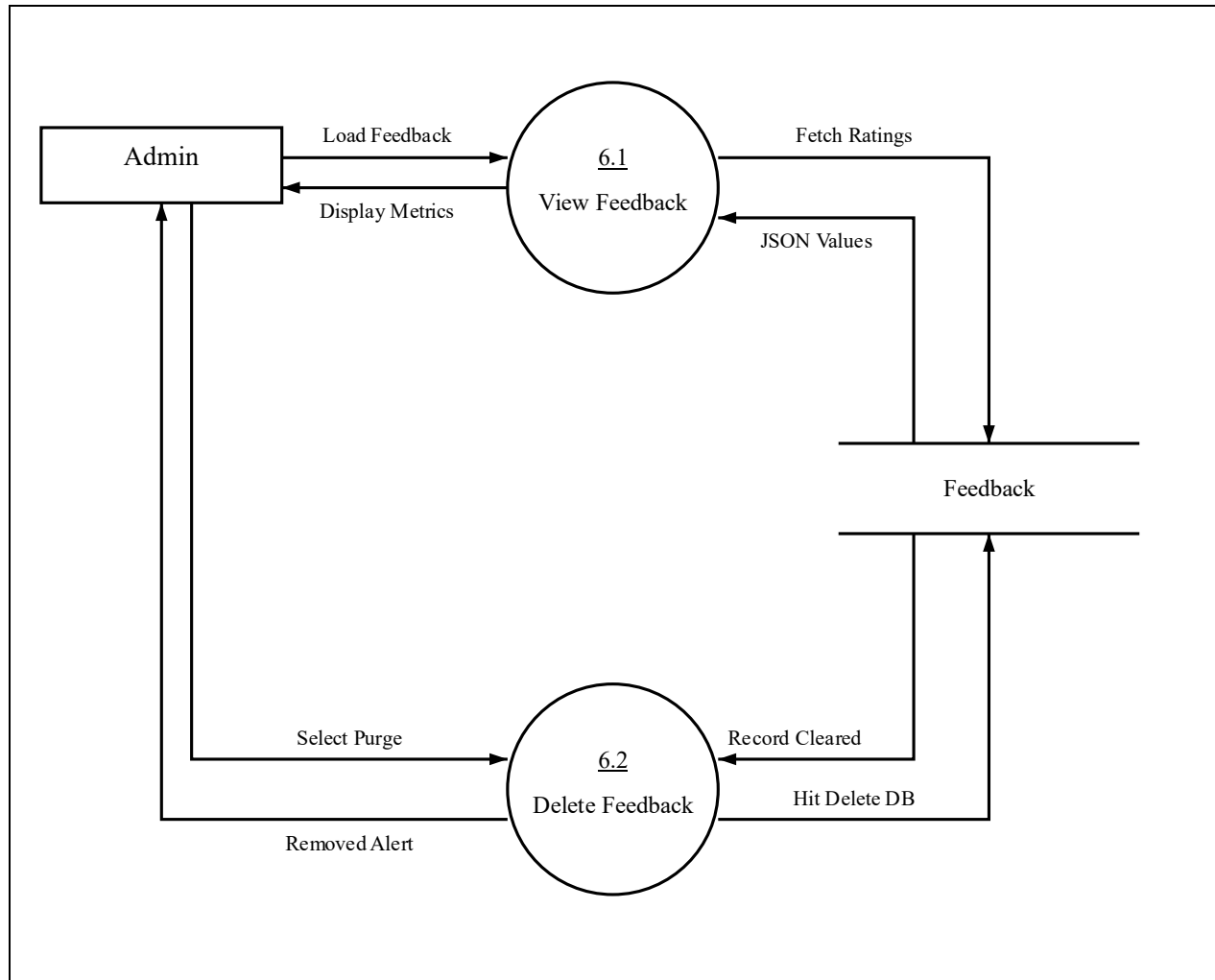
❖ **First Level DFD 4.0 Manage Jobs (Admin):**



❖ **First Level DFD 5.0 Manage Contact (Admin):**



❖ **First Level DFD 6.0 Manage Feed Back (Admin):**



DATA DICTIONARY

❖ **TABLE 1: User (Login & Authentication):**

Description: Stores core user credentials and access permissions.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique ID for each user
2	username	Varchar(150)	Unique, Not Null	Chosen username for login
3	password	Varchar(128)	Not Null (Hashed)	Encrypted password string
4	email	Varchar(254)	Not Null	Primary contact email address
5	is_candidate	Boolean	Default=False	True if the user is a Candidate
6	is_recruiter	Boolean	Default=False	True if the user is a Recruiter
7	professional_title	Varchar(150)	Nullable	Current role/designation of the user
8	profile_pic	Varchar(100)	Nullable	File path to uploaded profile image

❖ **TABLE 2: JobPosting (Job Management):**

Description: Stores details of all active and past job postings.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique ID for the job posting
2	recruiter_id	Integer	Foreign Key (User), Not Null	ID of the recruiter who posted it
3	title	Varchar(200)	Not Null	Job Designation/Title
4	job_type	Varchar(20)	Choices, Default='FULL_TIME'	Employment type (e.g., Remote, Contract)
5	location	Varchar(100)	Not Null	Geographical location of the job
6	required_skills	TextField	Not Null	Comma-separated list of technical skills
7	salary_range	Varchar(100)	Nullable	Ex: "\$80k - \$100k"
8	status	Varchar(10)	Choices, Default='OPEN'	Current status (Open/Closed/Pending)
9	created_at	DateTime	Auto Now Add	Timestamp when the job was published

❖ **TABLE 3: Candidate (Applicant Profiles):**

Description: Contains detailed profile data for job seekers.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Candidate ID
2	user_id	Integer	Foreign Key (User), Unique, Nullable	Link back to the Authentication user
3	full_name	Varchar(200)	Not Null	Real name of the candidate
4	experience_years	Float	Default=0	Years of professional experience
5	resume_file	Varchar(100)	Nullable	File path to uploaded PDF resume
6	github_url	Varchar(500)	Nullable	Link to applicant's GitHub profile
7	skills_extracted	TextField	Nullable	Skills parsed automatically by AI
8	current_location	Varchar(100)	Default='Remote'	Candidate's residing city/state

❖ **TABLE 4: Application (Hiring Pipeline):**

Description: Tracks the progression of a candidate applying for a specific job.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Application ID tracking number
2	job_id	Integer	Foreign Key (JobPosting), Not Null	ID of the target job role
3	candidate_id	Integer	Foreign Key (Candidate), Not Null	ID of the applying candidate
4	ai_score	Float	Default=0.0	AI generated score based on resume match
5	status	Varchar(20)	Choices, Default='APPLIED'	Current pipeline stage (e.g., HR Round)
6	ai_insights	JSONField	Default={}	AI matching data and keyword gaps
7	applied_at	DateTime	Auto Now Add	Timestamp of initial application submission

❖ **TABLE 5: Interview / Assessment (Evaluation Logs):**

Description: Stores technical test inputs, results, and interview schedules.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Interview/Assessment log ID
2	application_id	Integer	Foreign Key (Application), Not Null	Link to the specific candidate application
3	interview_type	Varchar(20)	Choices, Default='AI_BOT'	Type of interview (Aptitude, Coding, Technical)
4	status	Varchar(20)	Choices, Default='SCHEDULED'	Status of assessment (Completed/Cancelled)
5	ai_confidence_score	Float	Default=0.0	Marks obtained in technical assessment
6	code_final	TextField	Nullable	The actual coding answer submitted by candidate
7	scheduled_time	DateTime	Nullable	Scheduled interview date and time
8	feedback	TextField	Nullable	Recruiter's manual remarks and grading

❖ **TABLE 6: Contact (Support Queries):**

Description: Stores queries and messages sent by users via the 'Contact Us' page.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Query tracking ID
2	name	Varchar(100)	Not Null	Name of the person making the inquiry
3	email	Varchar(255)	Not Null	Contact email address
4	subject	Varchar(255)	Not Null	Subject line of the query message
5	message	TextField	Not Null	Detailed text content of the problem/query
6	user_type	Varchar(50)	Nullable	Role of the user (e.g., Candidate or Recruiter)
7	query_date	DateTime	Auto Now Add	Timestamp when the query was submitted

❖ **TABLE 7: FeedBack (User Experience Feedback):**

Description: Captures reviews and general feedback regarding the ATS platform.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Feedback tracking ID
2	name	Varchar(100)	Not Null	Name of the user providing feedback
3	contact_no	Varchar(20)	Not Null	User's phone/contact number
4	email	Varchar(255)	Not Null	Email address for feedback follow-ups
5	feedback	TextField	Not Null	Detailed constructive feedback text
6	user_type	Varchar(50)	Not Null	Role making the submission (Candidate, Admin, etc.)
7	feedback_date	DateTime	Auto Now Add	Timestamp when the feedback was generated

❖ **TABLE 8: Question (Technical Question Bank):**

Description: Stores technical and behavioral questions generated by AI or recruiters for assessments.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Question ID
2	recruiter_id	Integer	Foreign Key (User), Not Null	Recruiter who saved/generated it
3	text	TextField	Not Null	The actual interview question
4	options	JSONField	Default=[]	Multiple choice options if applicable
5	correct_answer	Varchar(255)	Not Null	Code logic or exact text answer
6	category	Varchar(20)	Choices, Default='GENERAL'	Tech Stack (Python, Java, Rust, etc.)
7	difficulty	Varchar(10)	Choices, Default='medium'	Assessment level (Easy/Medium/Hard)
8	created_at	DateTime	Auto Now Add	Timestamp when added to the bank

❖ **TABLE 9: Offer (Contract Generation):**

Description: Tracks the final offer letter generation and candidate acceptance.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Offer ID tracking number
2	application_id	Integer	Foreign Key, Unique, Not Null	Links to the specific Application
3	salary_offered	Varchar(50)	Default='Not Specified'	Finalized salary/CTC details
4	designation	Varchar(100)	Not Null	Finalized job joining designation
5	joining_date	Date	Nullable	Agreed commencement date
6	offer_file	Varchar(100)	Nullable	File path to the generated PDF Offer Letter
7	status	Varchar(20)	Choices, Default='GENERATED'	Current status (Sent/Accepted/Rejected)
8	created_at	DateTime	Auto Now Add	Timestamp tracking generation

❖ **TABLE 10: Notification (Real-time System Alerts):**

Description: Tracks all pushed notifications to users across the platform.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Notification ID
2	user_id	Integer	Foreign Key (User), Not Null	Target user receiving the alert
3	title	Varchar(255)	Not Null	Headline of the notification
4	message	TextField	Not Null	Detailed text content of the alert
5	link	Varchar(255)	Nullable	Clickable redirect URL (e.g., job link)
6	type	Varchar(20)	Choices, Default='INFO'	Type of alert (Success, Error, Warning)
7	is_read	Boolean	Default=False	Tracks if user has seen the notification
8	created_at	DateTime	Auto Now Add	Timestamp when notification triggered

❖ **TABLE 11: EmailTemplate (Dynamic Email Management):**

Description: Automatically stores templates for sending out dynamic SMTP emails.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Template ID
2	name	Varchar(100)	Unique, Not Null	Internal code identifier for rendering
3	subject	Varchar(255)	Not Null	Subject line configuration for email
4	body_content	TextField	Not Null	Raw HTML formatting for the email body
5	is_active	Boolean	Default=True	Status of email template in production
6	created_at	DateTime	Auto Now Add	Timestamp tracking creation

❖ **TABLE 12: ActivityLog (Recruiter Admin Logs):**

Description: Maintains an audit trail of all crucial administrative actions within the ATS.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Log Tracking ID
2	user_id	Integer	Foreign Key (User), Not Null	Which Recruiter performed the action
3	action	Varchar(255)	Not Null	Exact technical action taken
4	details	TextField	Nullable	Any extended log variables or reasons
5	timestamp	DateTime	Auto Now Add	Exact date and time of the occurrence

❖ **TABLE 13: ProctoringLog (Anti-Cheat Evidence Security):**

Description: Keeps records and images of any suspicious activity during assessments.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Log ID
2	application_id	Integer	Foreign Key, Not Null	Associated candidate application test
3	log_type	Varchar(20)	Choices	Classification (Violation, Screenshot)
4	image	Varchar(100)	Nullable	Physical file path to the snapshot taken
5	details	TextField	Nullable	Explanation of the cheating heuristic trigger
6	timestamp	DateTime	Auto Now Add	Timestamp matching assessment test time

❖ **TABLE 14: EvaluationScorecard (Collaborative Interview Form)**

Description: Digitally stores multi-recruiter grading notes and aggregated feedback details.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique grading sheet tracking ID
2	application_id	Integer	Foreign Key, Not Null	Links to the candidate's active application
3	recruiter_id	Integer	Foreign Key (User), Not Null	Specific interviewer giving feedback
4	overall_comments	TextField	Nullable	Qualitative assessment wrap-up remarks
5	is_final	Boolean	Default=False	Is this the definitive hiring decision rating?
6	created_at	DateTime	Auto Now Add	Processing timestamp of evaluation form

❖ **TABLE 15: AIUsageLog (Artificial Intelligence Token Tracker):**

Description: Audits exact costs, queries, and response delays from Gemini/Hugging Face.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique Model API Transaction ID
2	user_id	Integer	Foreign Key, Nullable	Recruiter who executed the AI feature
3	feature	Varchar(50)	Choices, Not Null	Which module originated call (e.g. Resume Parser)
4	model_used	Varchar(50)	Default='gemini-2.0'	Backend engine currently configured
5	total_tokens	Integer	Default=0	Sum of prompt tokens + completion tokens
6	cost_estimate	Float	Default=0.0	Calculated monetary tracking of model load
7	status	Varchar(20)	Default='SUCCESS'	Result check from Google/HF servers

❖ **TABLE 16: SourcingMatch (AI Candidate Matchmaker):**

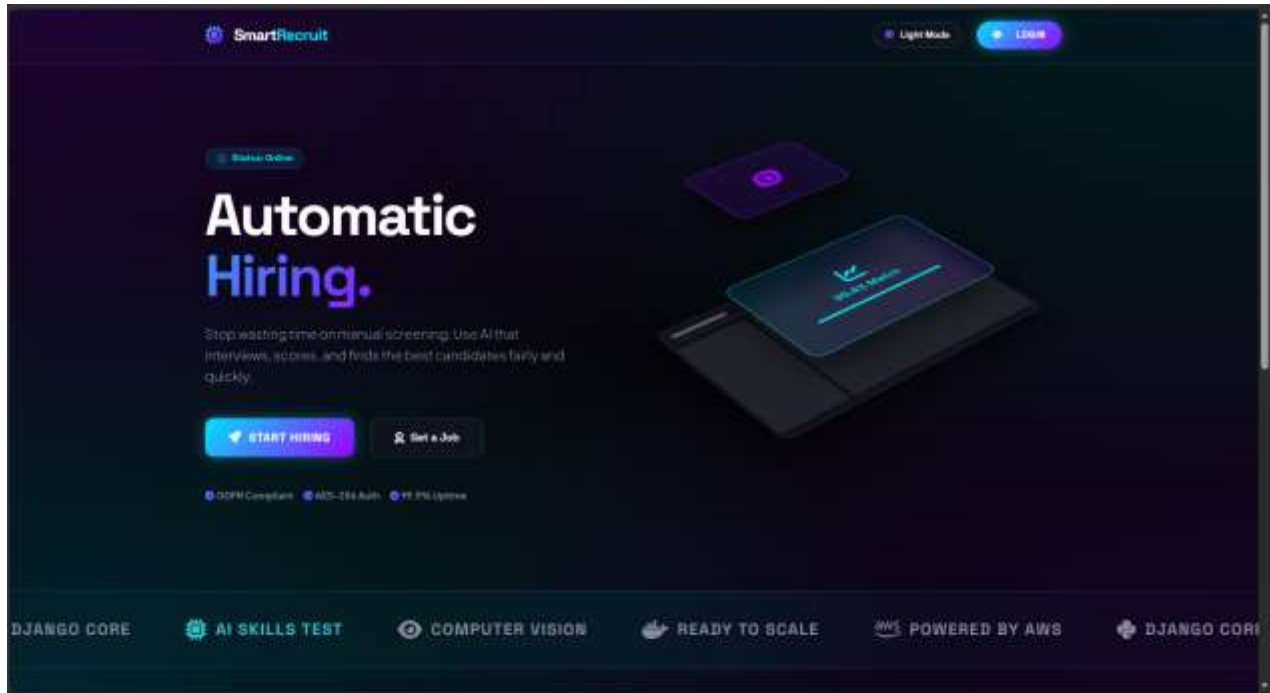
Description: Holds pre-computed percentage matching analytics between Candidates and Job requirements.

Sr. No	Field Name	Data Type	Constraints	Description
1	id	Integer	Primary Key, Auto Increment	Unique algorithmic match ID
2	job_id	Integer	Foreign Key, Not Null	Active system Job Listing designation
3	candidate_id	Integer	Foreign Key, Not Null	System matched Candidate profile
4	match_score	Float	Not Null	Computed match success percentage
5	fit_rationale	TextField	Not Null	AI-driven reason why they are a strong match
6	is_shortlisted	Boolean	Default=False	Has recruiter marked them actively?
7	created_at	DateTime	Auto Now Add	Algorithmic sweep occurrence timestamp

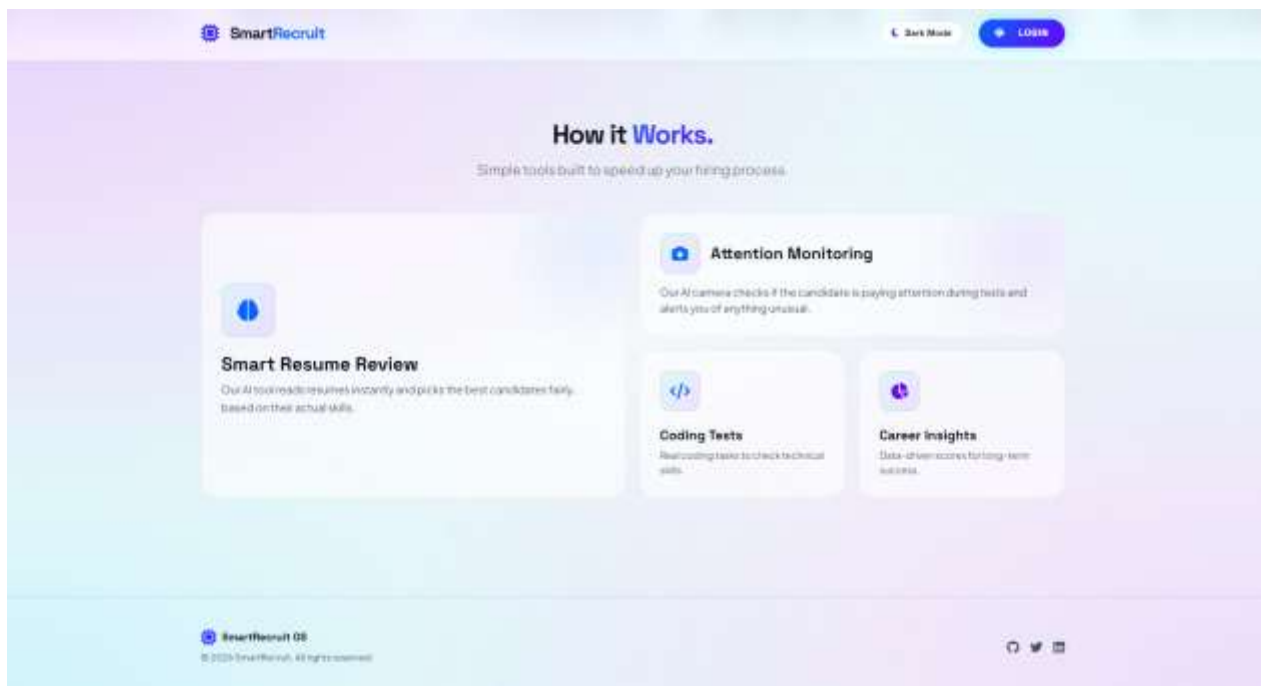
SYSTEM OUTPUT

❖ Input-Output Screenshots [Candidate]:

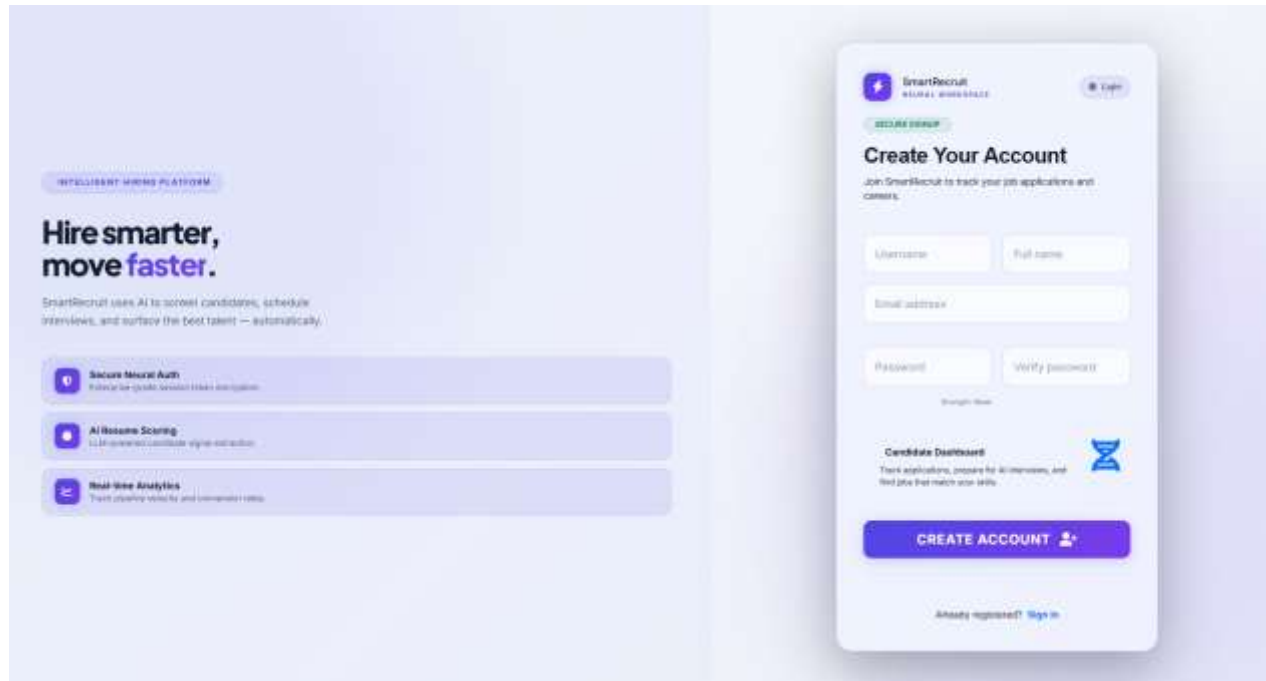
- **Landing Page (Dark Theme):** User Gets to see this Page Whenever the Site is Opened.



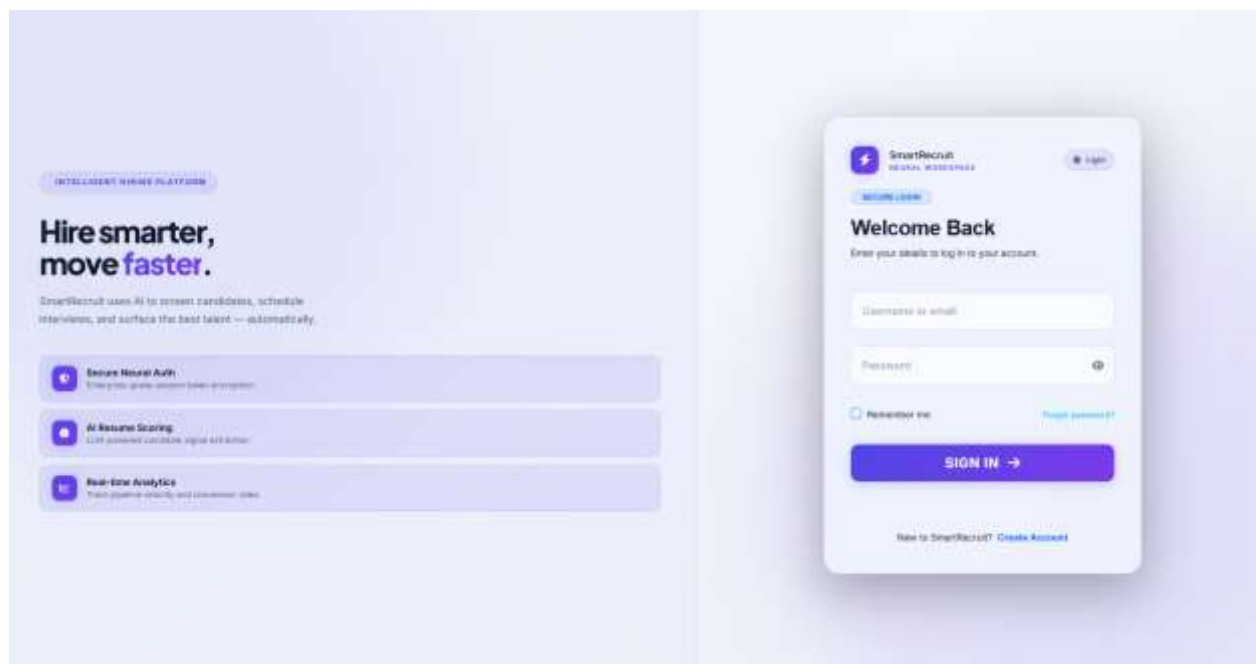
- **Landing Page Scrolled (Light Theme):** User Gets to see this Page Whenever the Site is Opened.



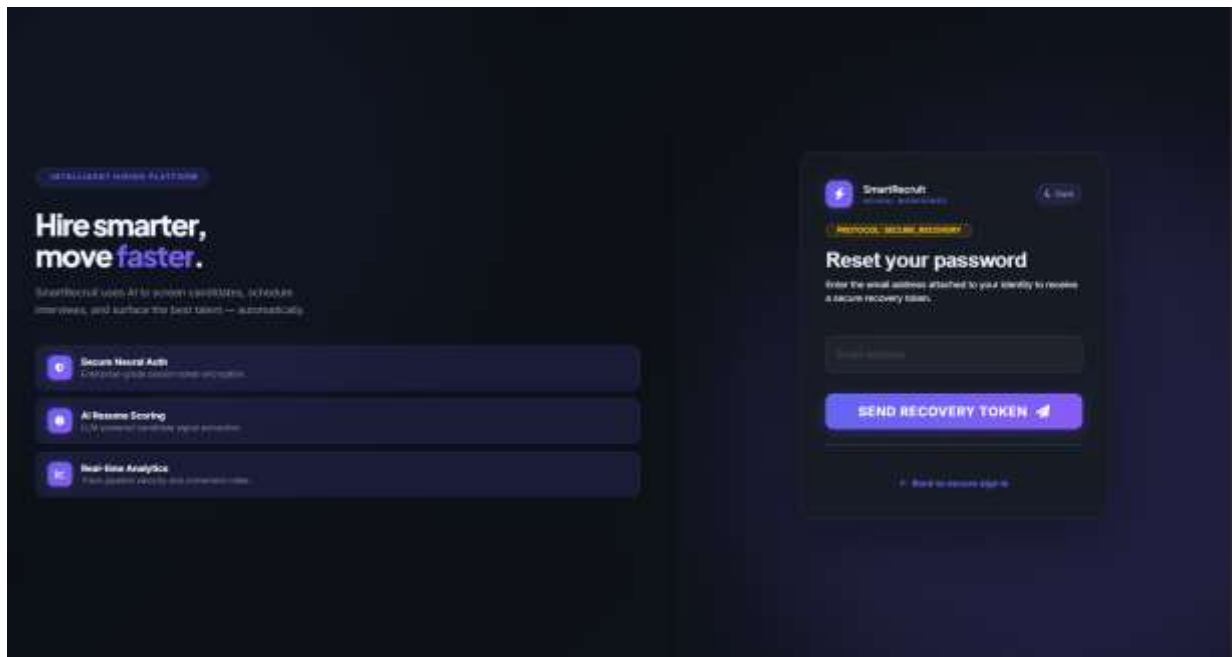
- **Register Page:** Here User Can Fill the Details and Register as a Candidate to Access the System.



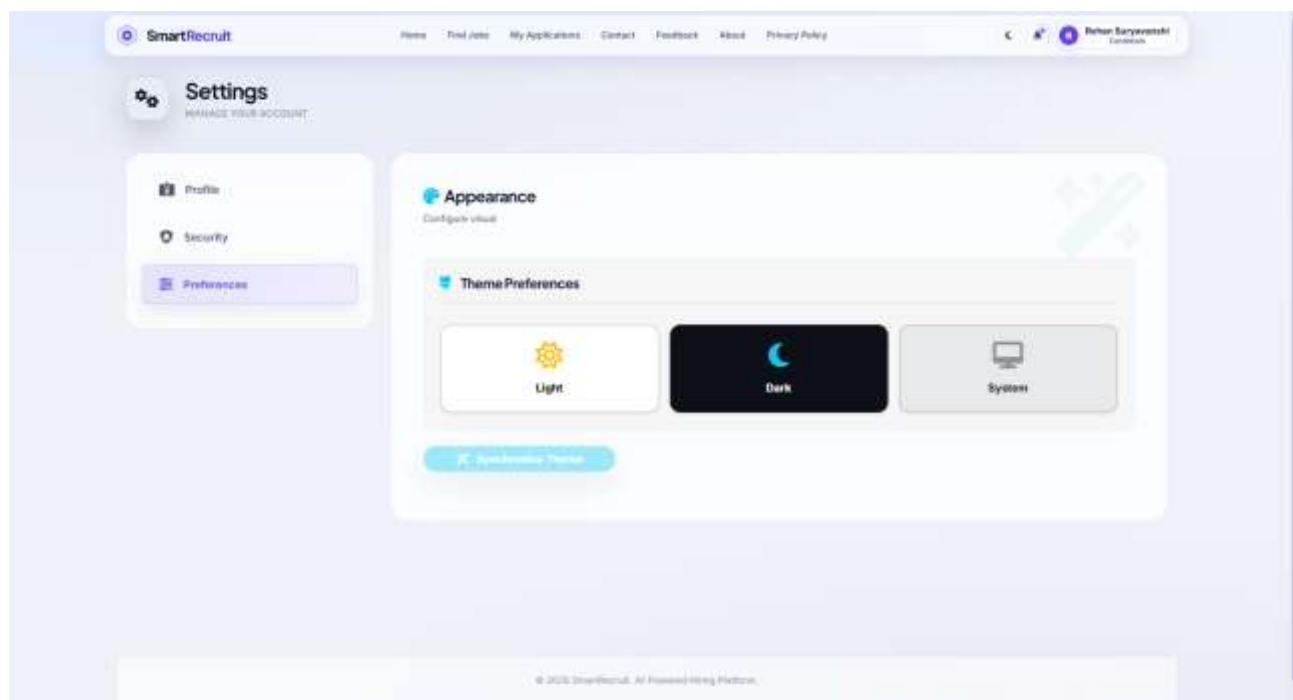
- **Login Page:** Here User Can Fill Account Credentials and Login to the System.



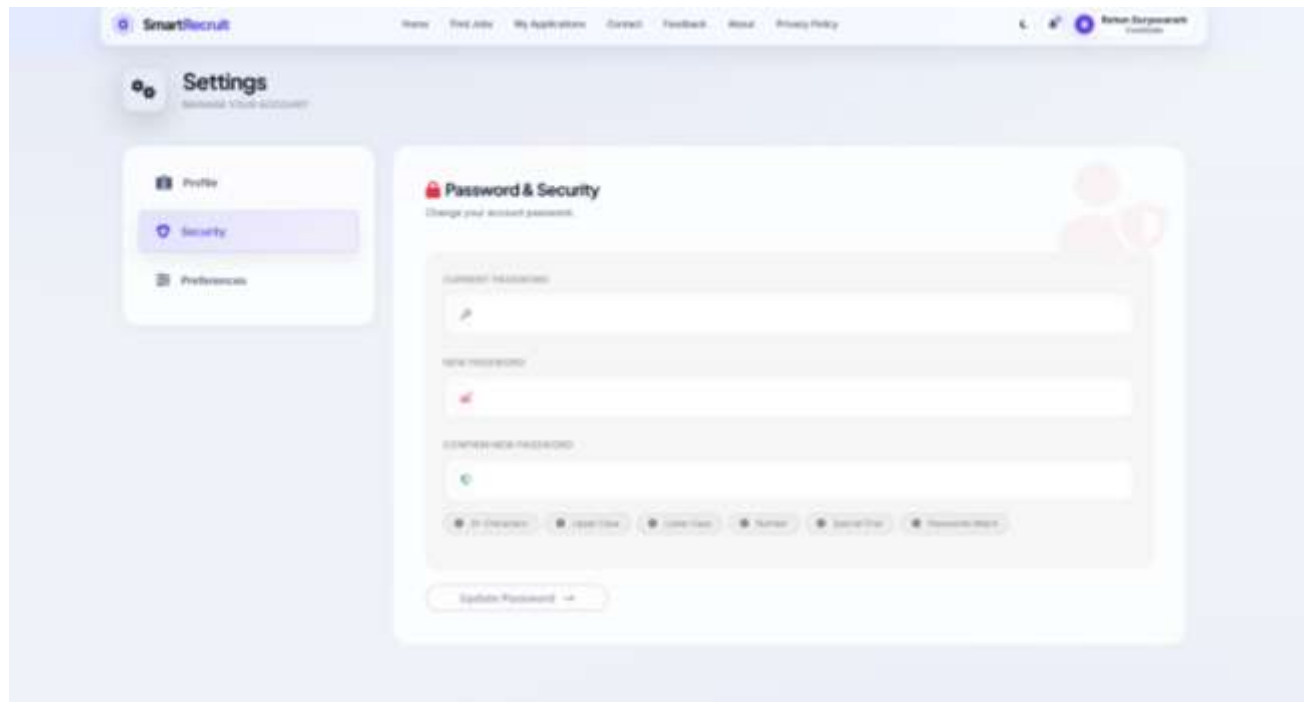
- **Forgot Password Page:** Here User Can send the Password Recovery Link to the Registered Email to Change the Forgotten Password.



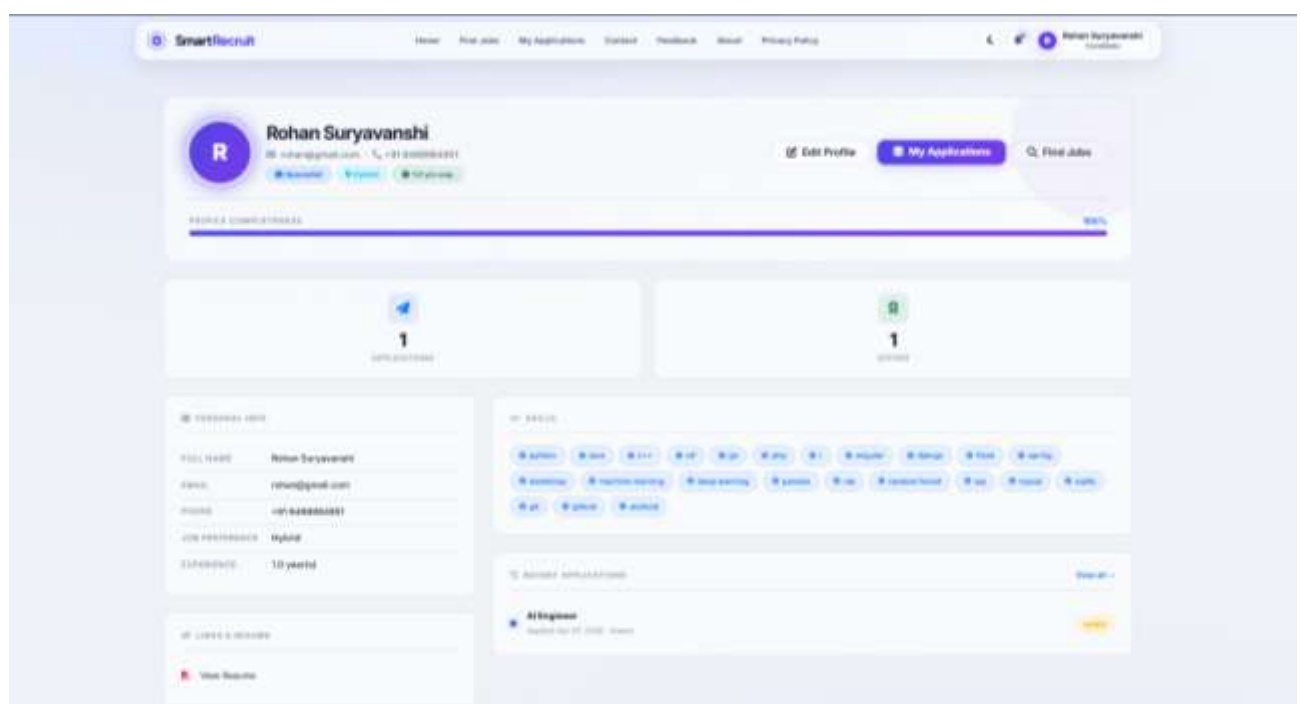
- **Preferences Page:** Here User Can Change the UI to System Default, Light Theme or Dark Theme.



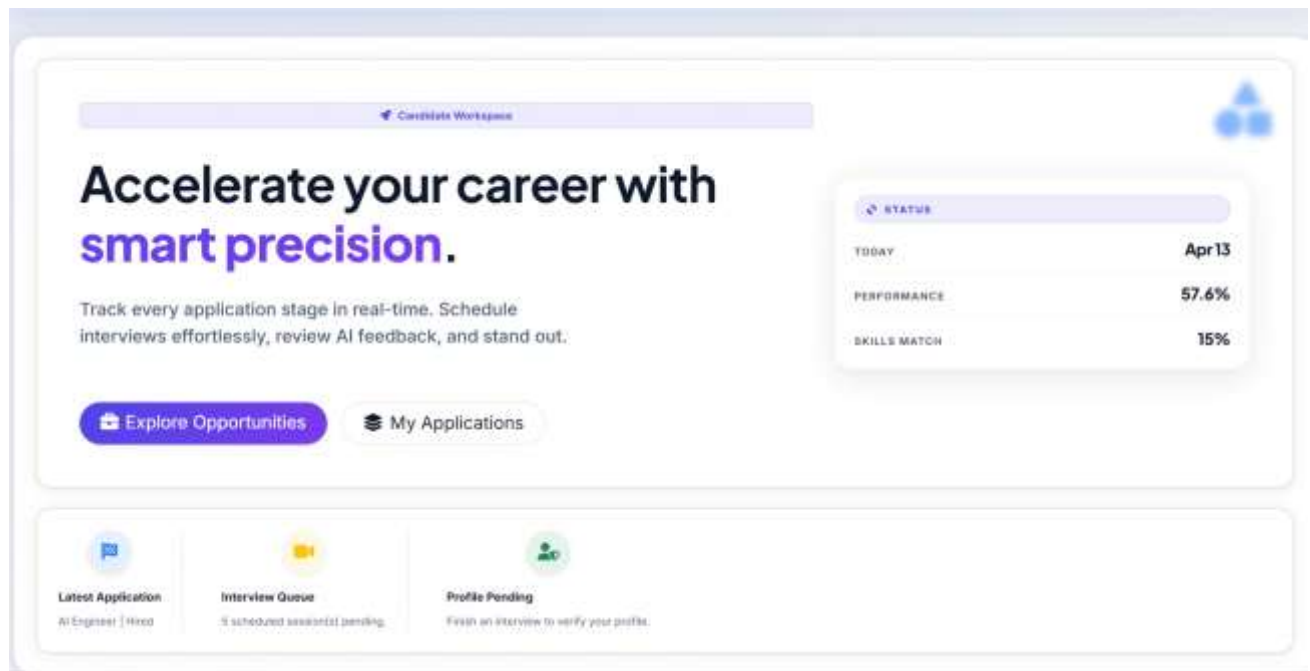
- **Change Password Page:** Here User Can Change His / Her Account Password.



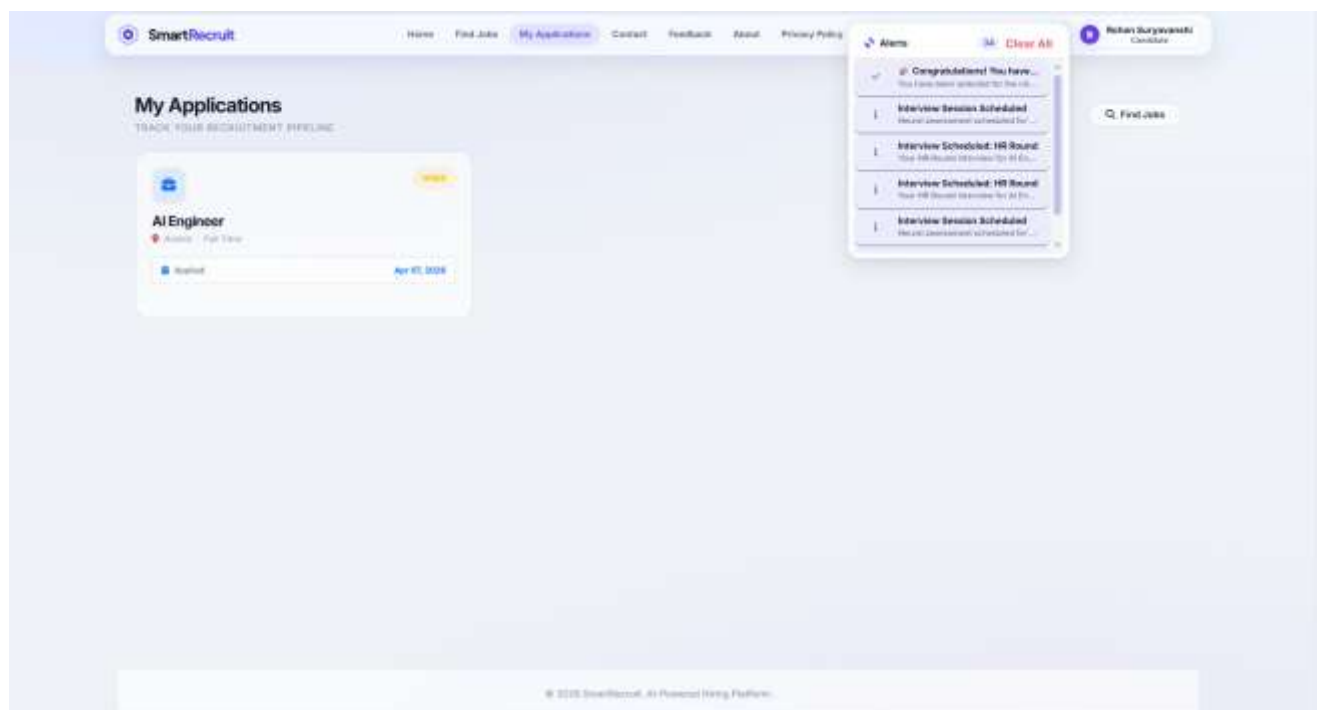
- **My Profile Page:** Here User Can View and edit His / Her Profile Details.



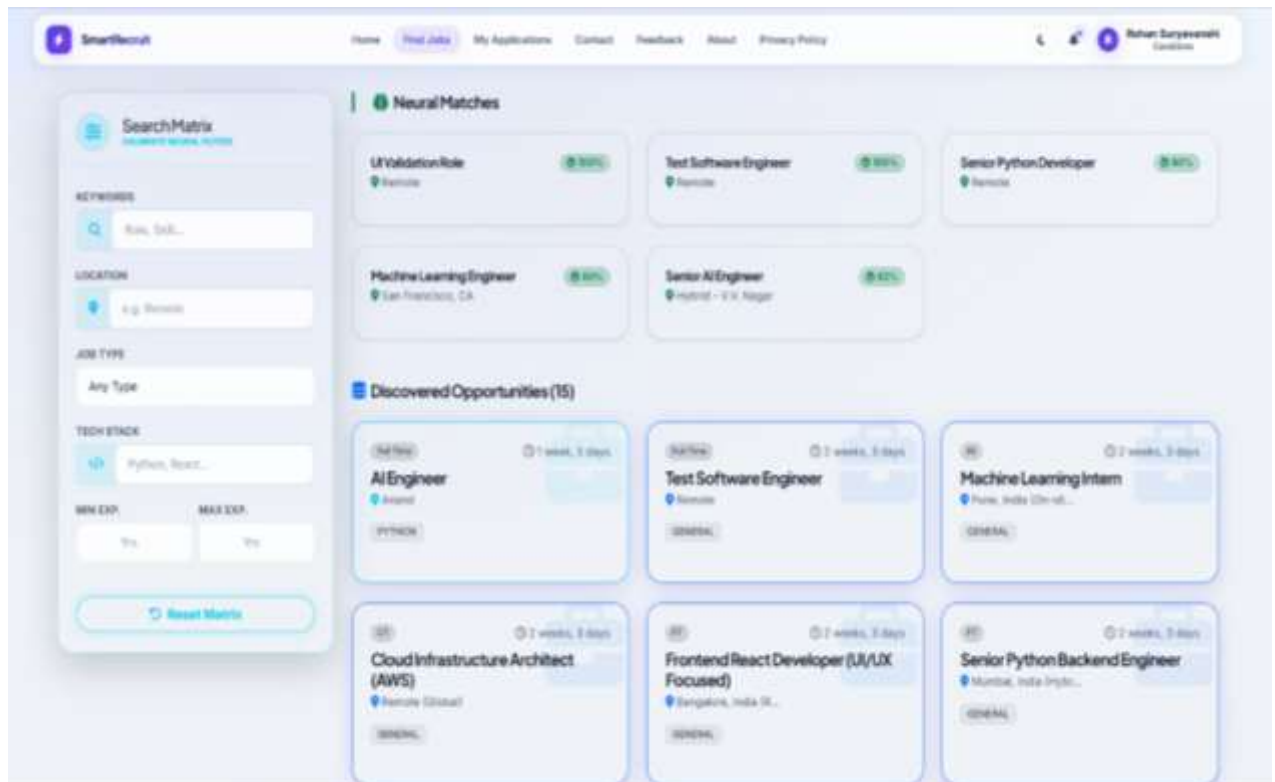
- **Dashboard:** Candidate will View this page immediately after logging into the website.



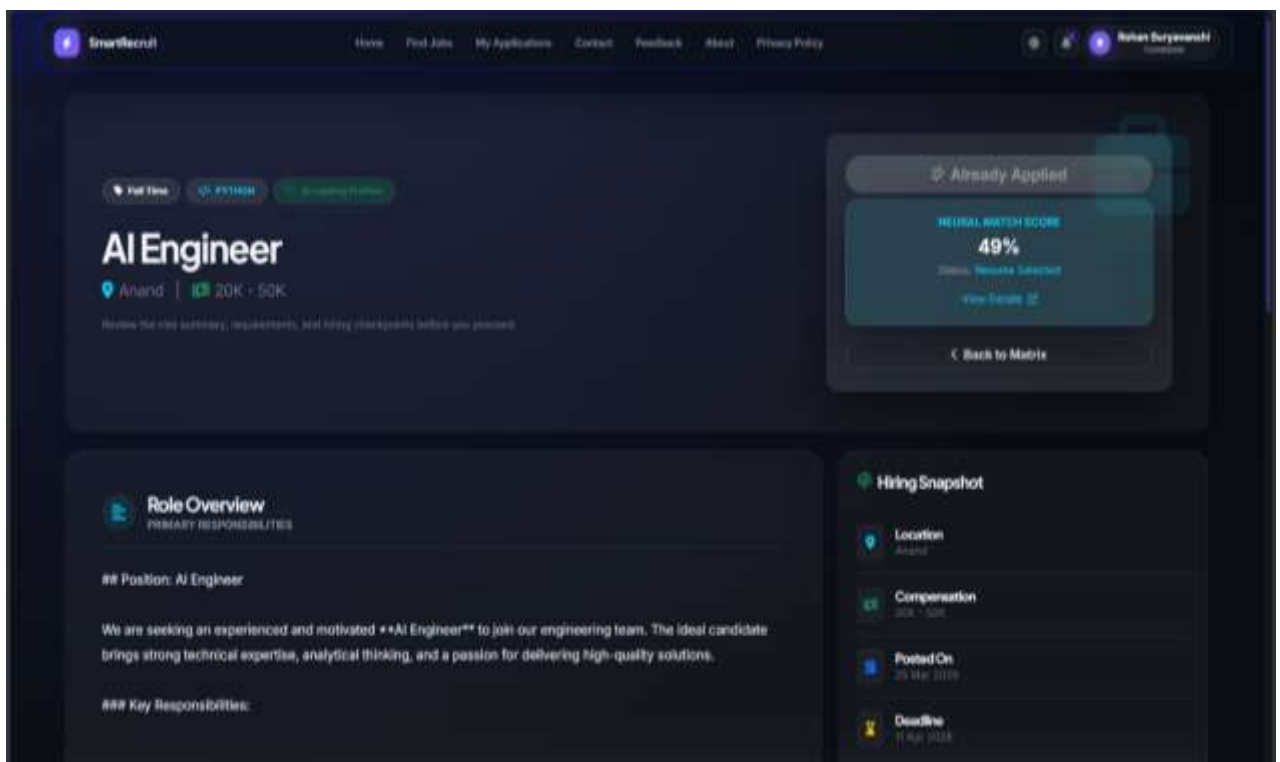
- **My Applications (Notification Panel):** Candidate can see the Broadcast Messages Sent by the System Admin Here.



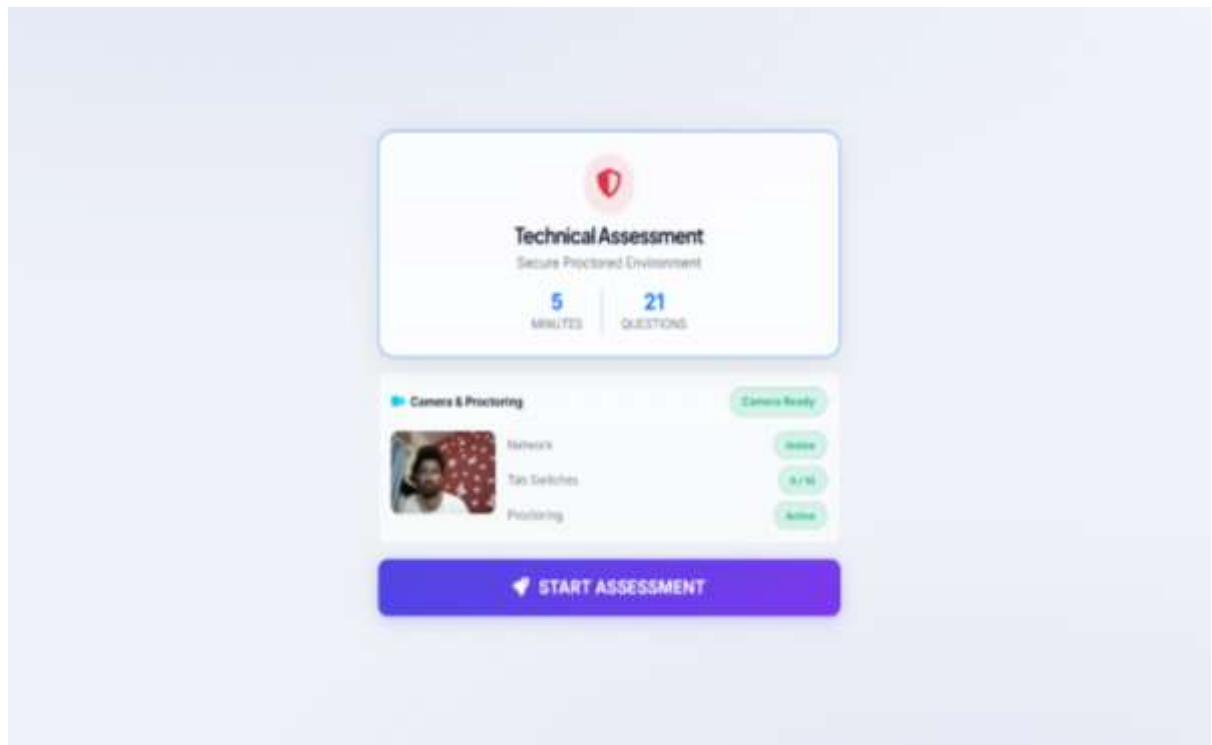
- **Job List Page:** Candidate can View Job Lists on this Page.



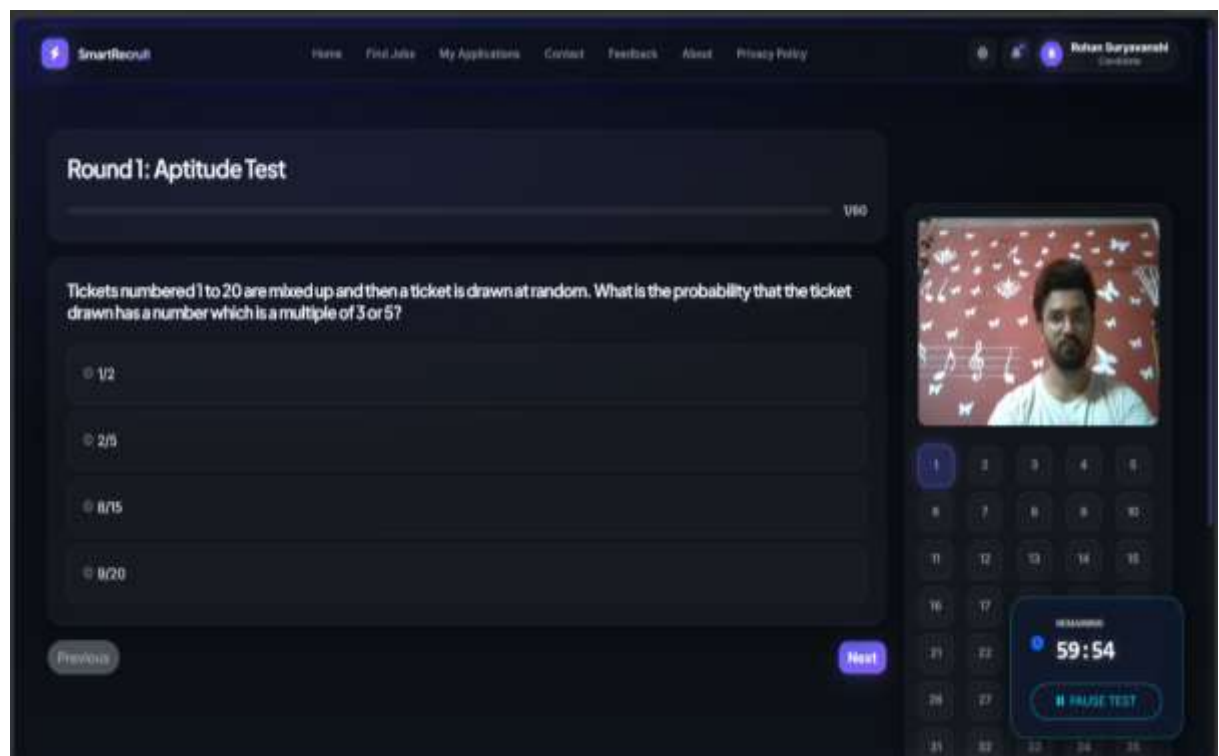
- **Job Detail Page:** Candidate can View Job Details on this Page.



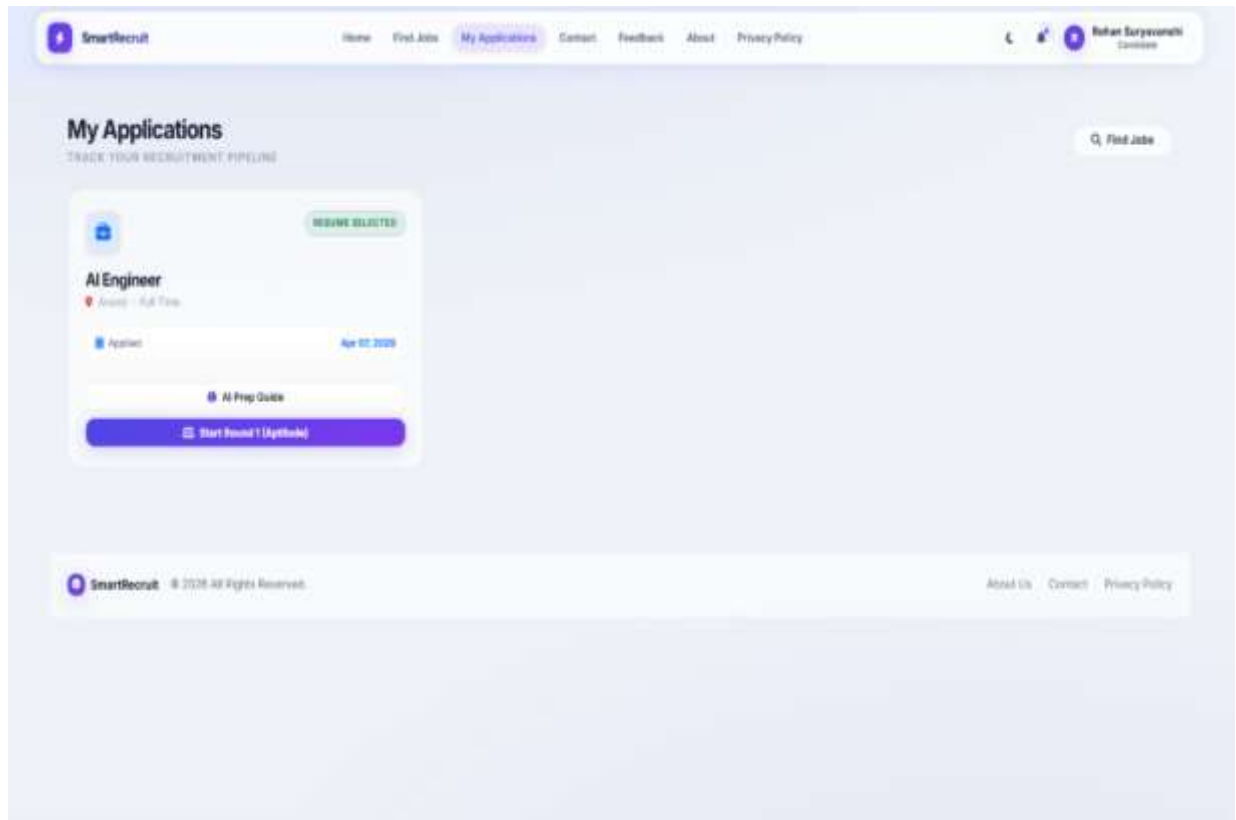
- **Take Assessment Page:** Candidate can View the Environment and set everything up before starting the Actual Interview on Take Assessment Page.



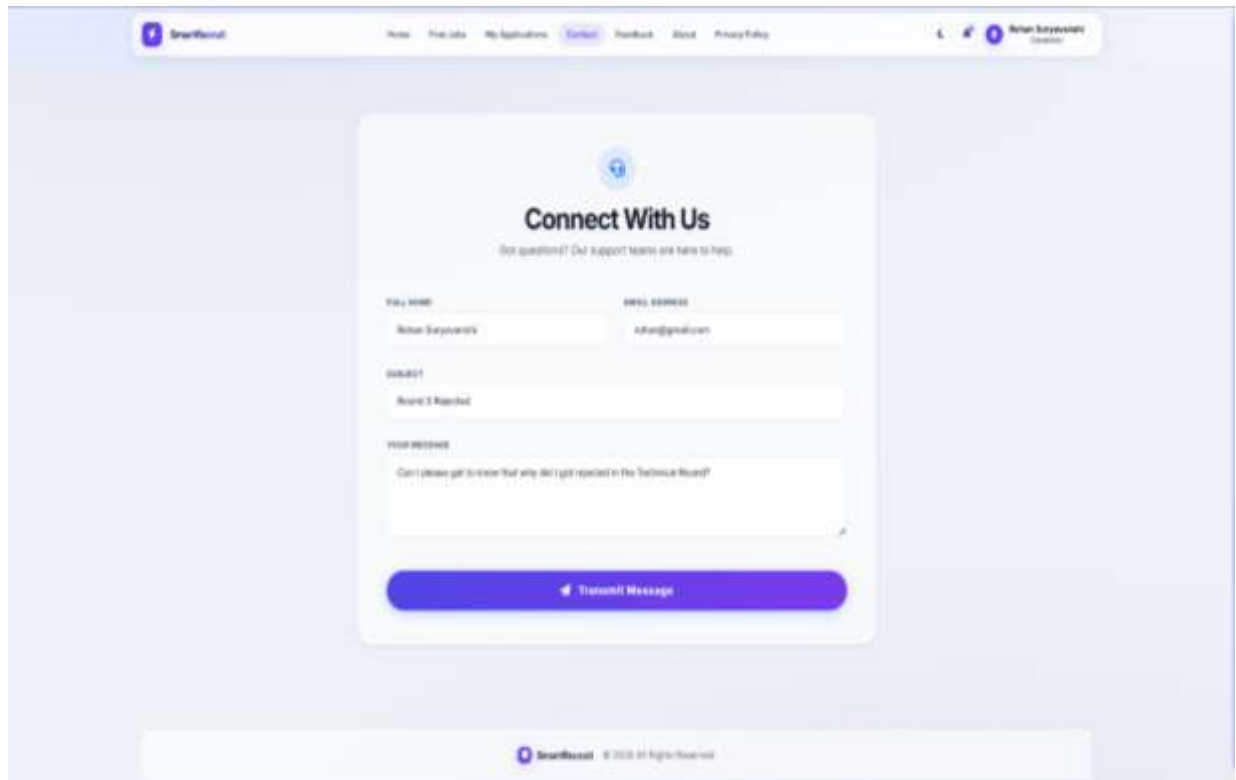
- **Assessment Page:** Candidate can Appear for Interviews on Assessment Page.



- **My Applications Page:** Candidate can View the Status of Job Applications here.



- **Contact Page:** Candidate can Submit Contact Queries via this Page.

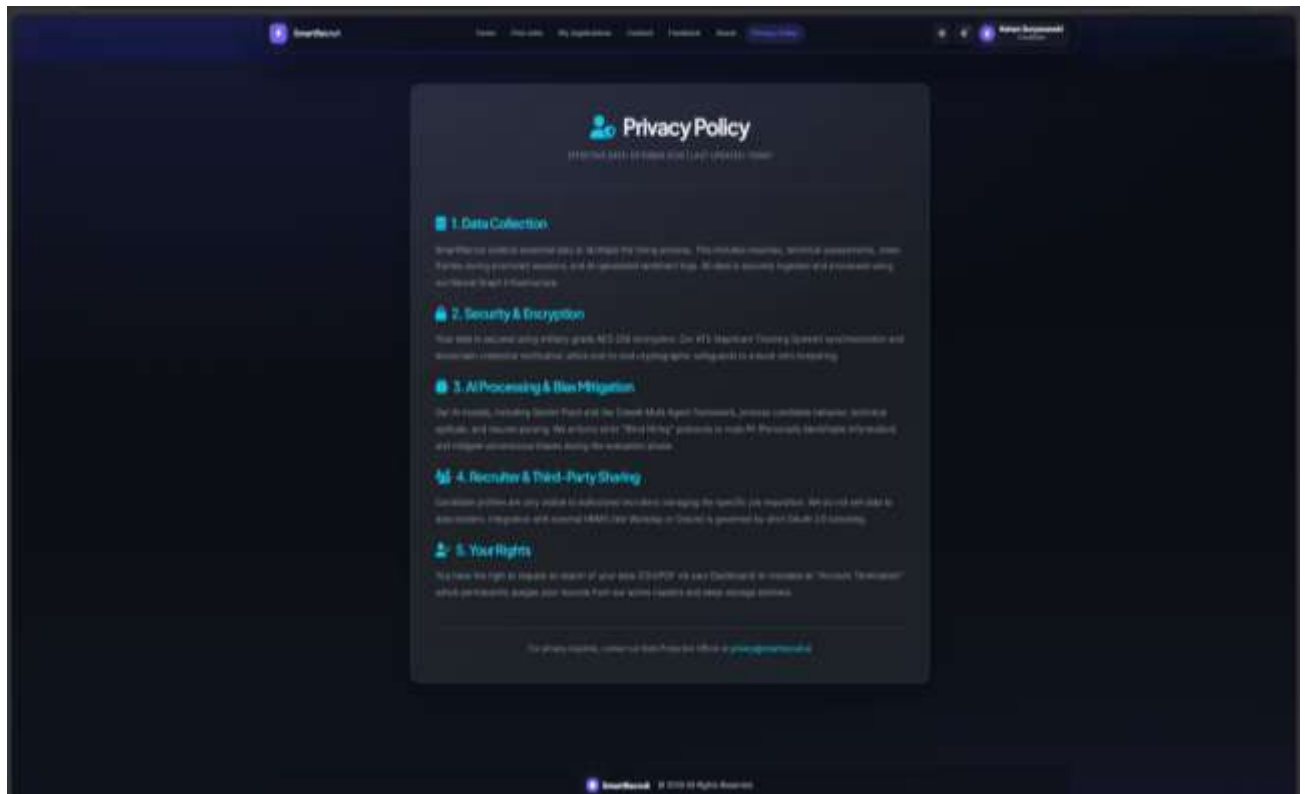


- **Feed Back Page:** Candidate can Submit Feed Backs via this Page.

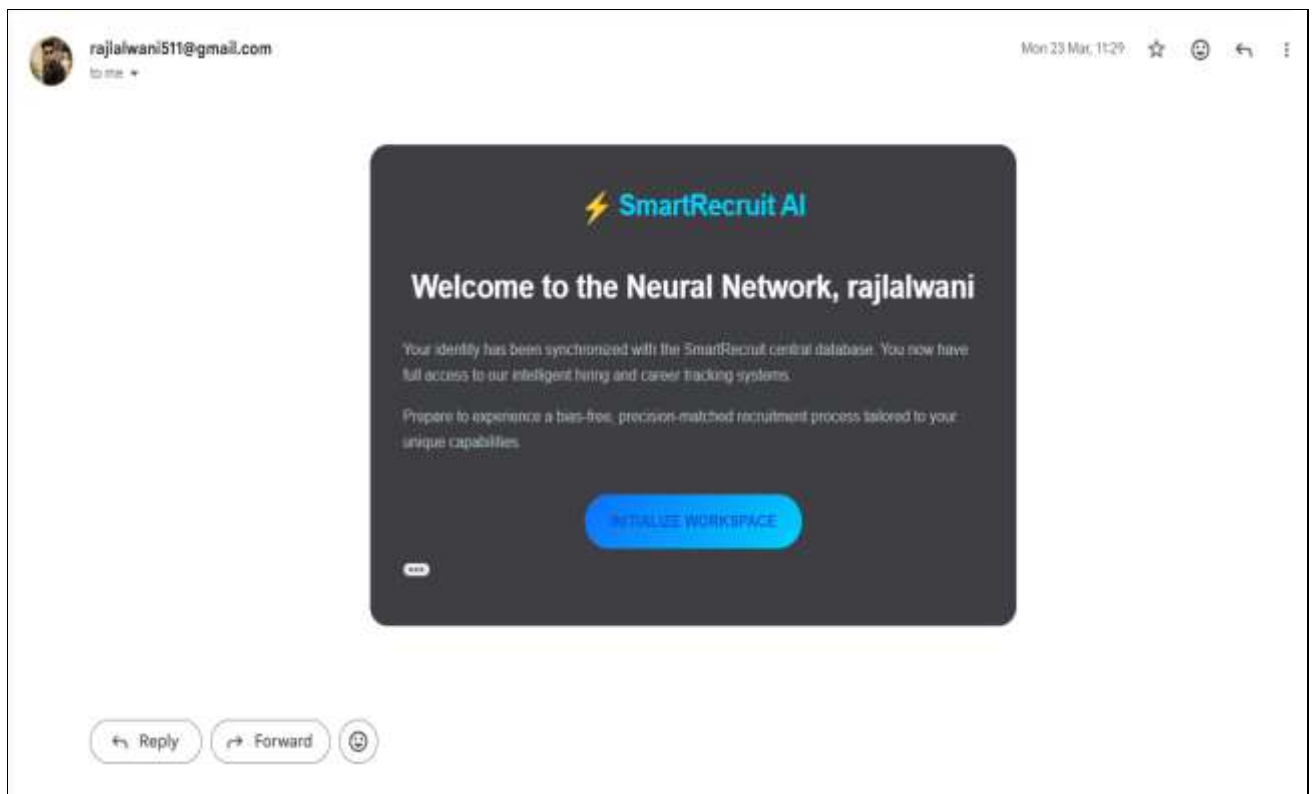
- **About Page:** Candidate can View the About US Page.



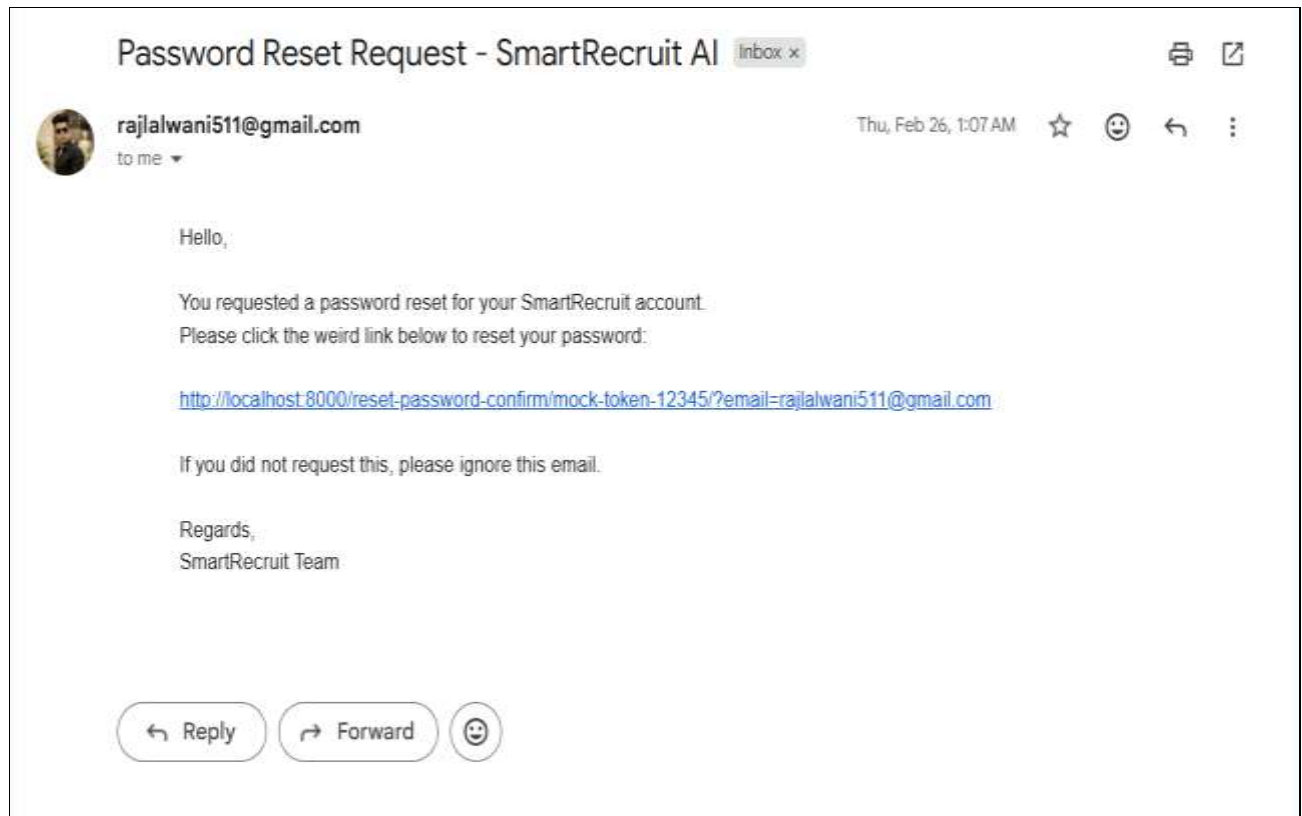
- **Privacy Policy Page:** Candidate can View the Privacy Polic on this Page.



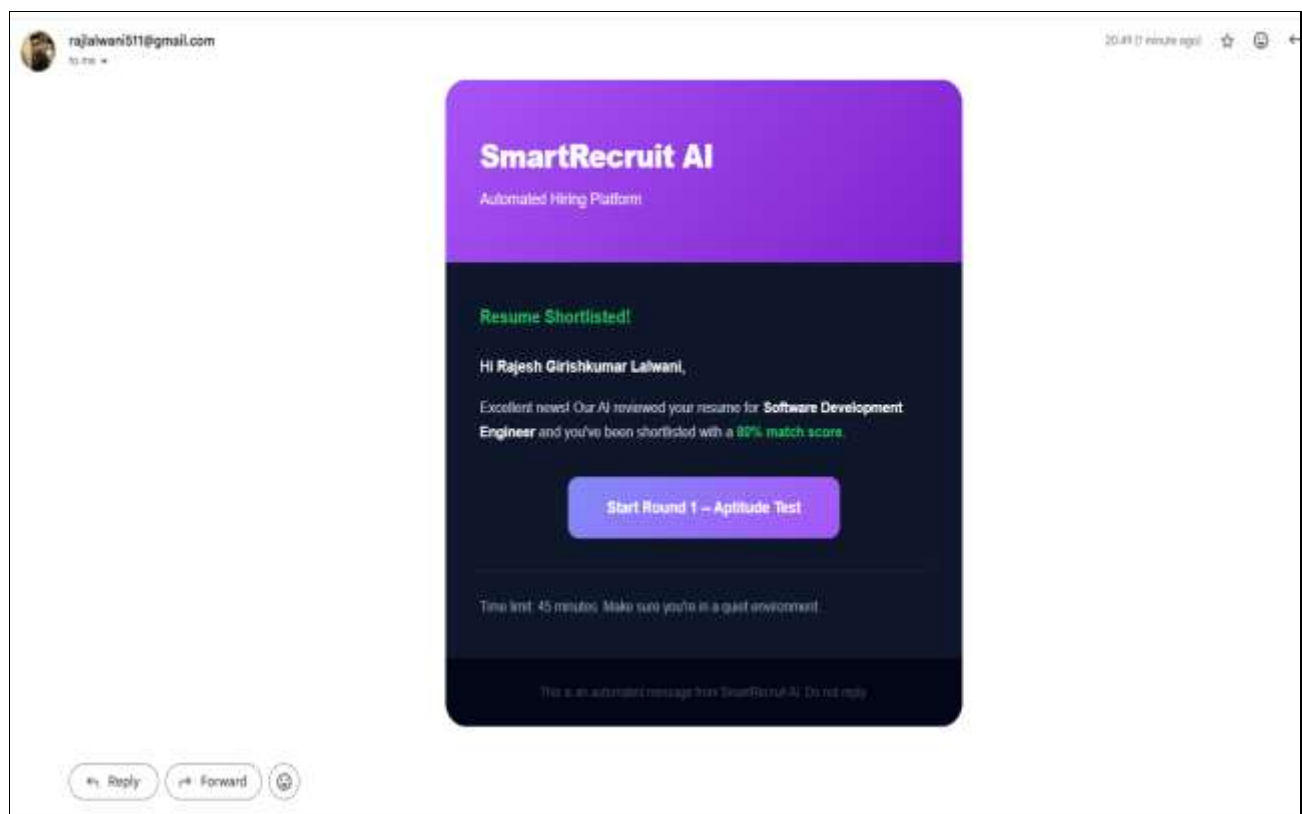
- **Welcome Email:**



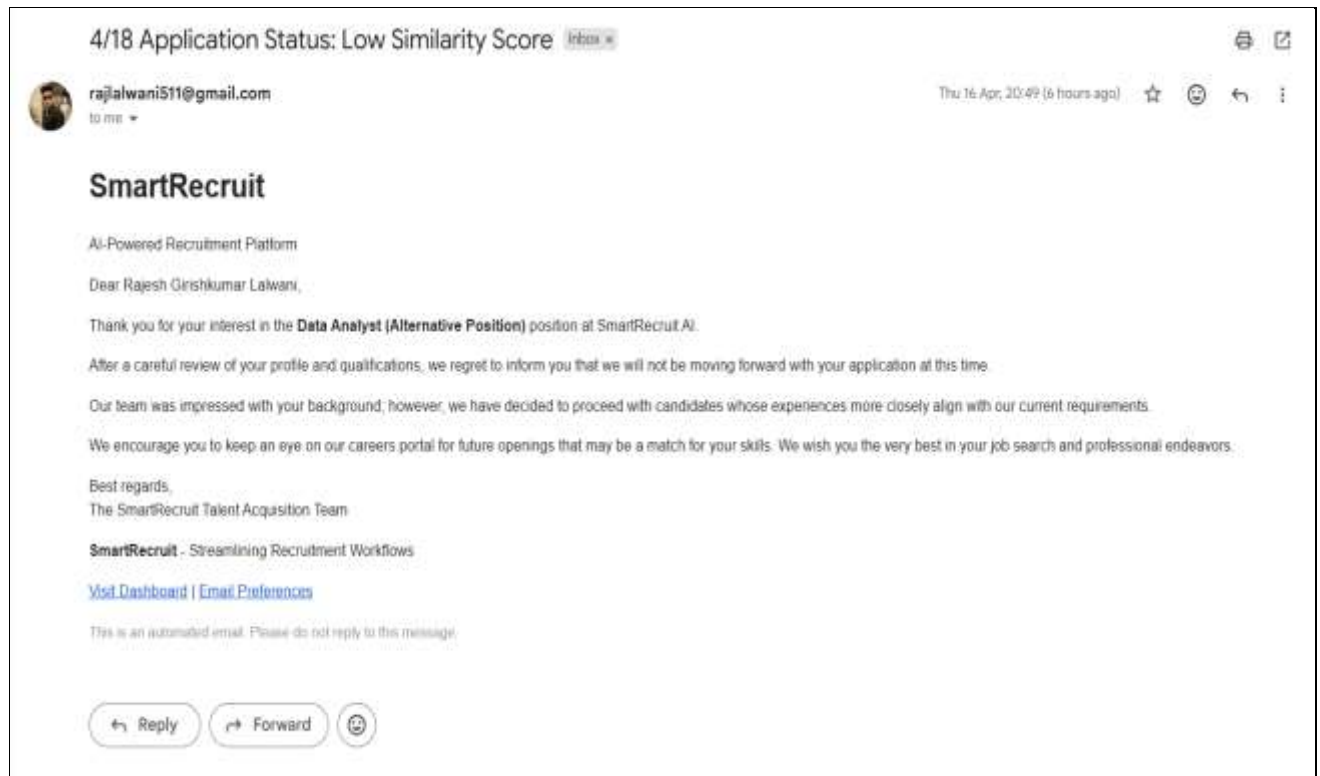
- **Password Reset Request Email:**



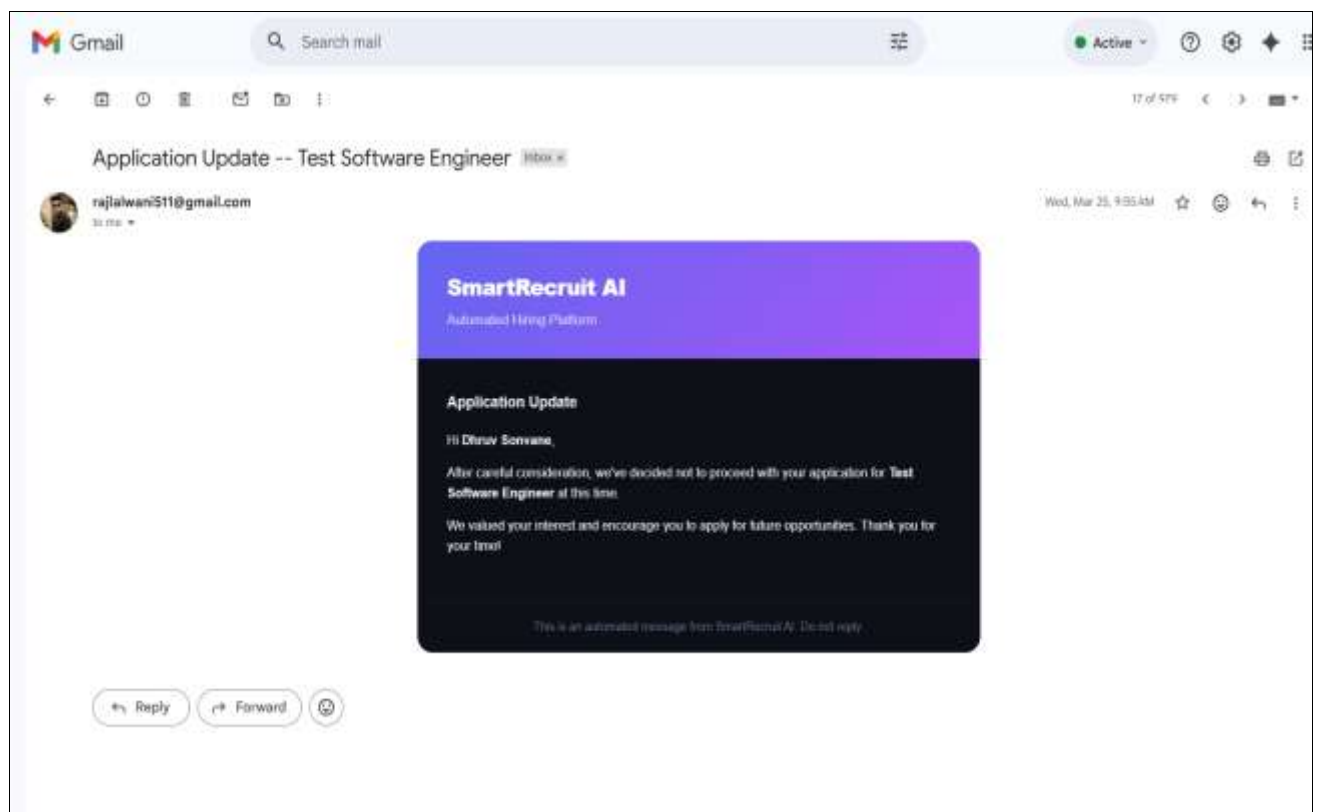
- **Resume Shortlisting Email:**



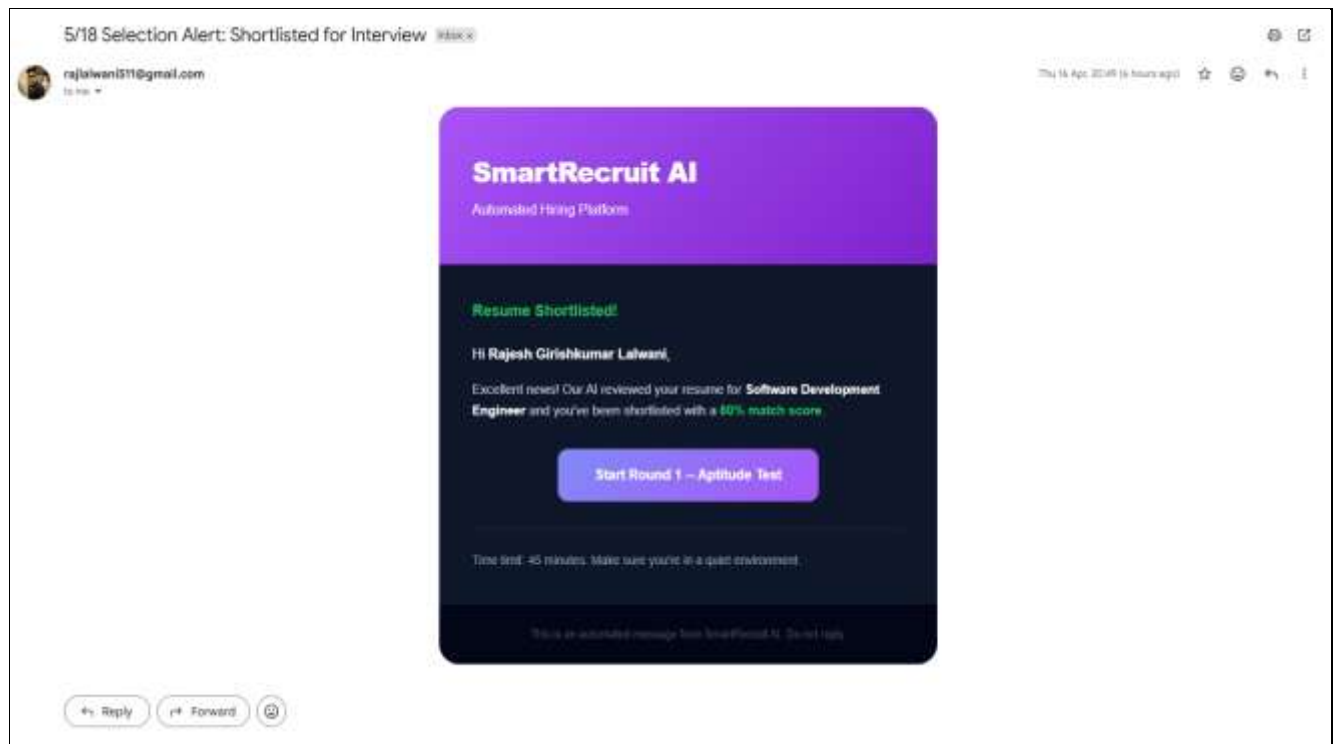
- **Resume Low Similarity Score Email:**



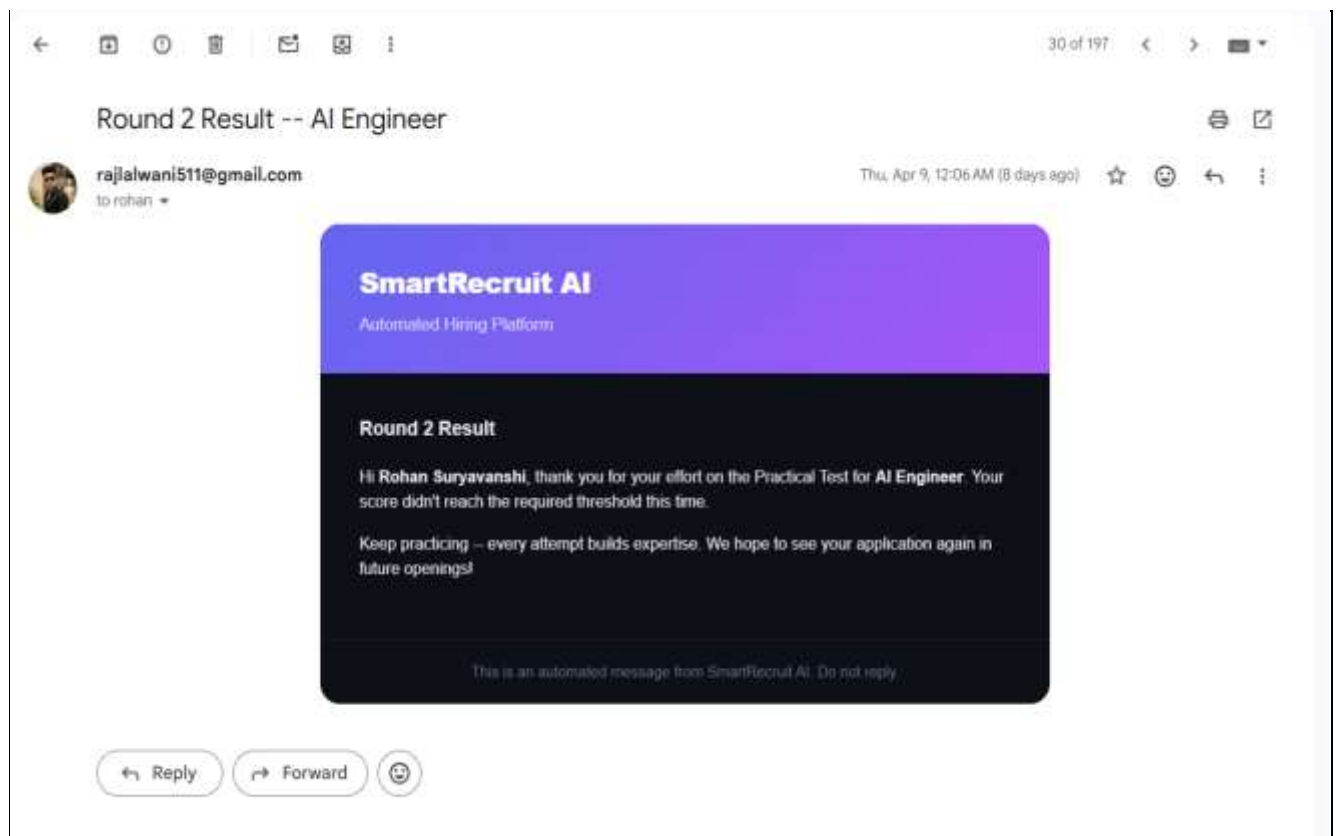
- **Resume Rejection Email:**



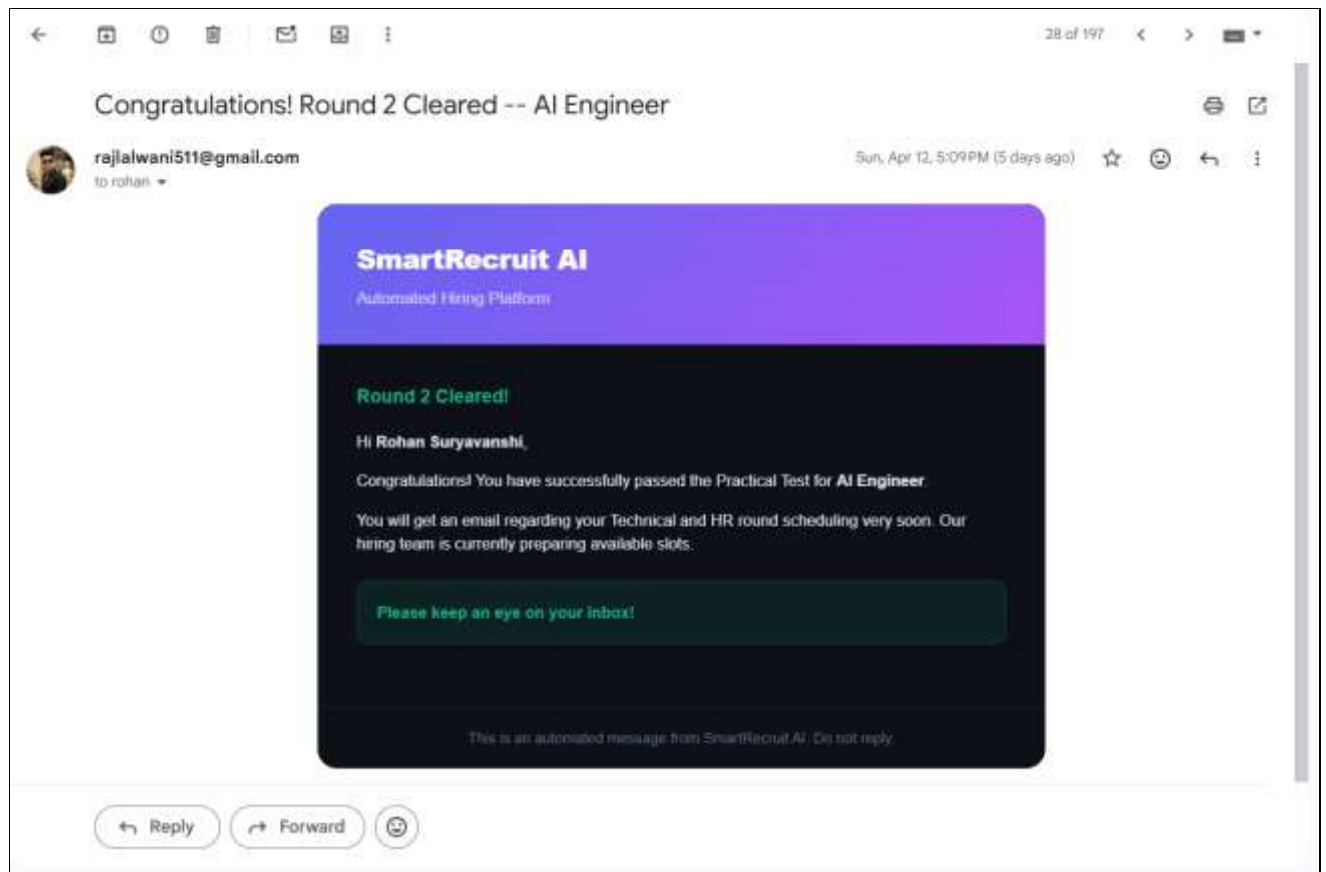
- **Resume Shortlisting For Round 1 Email:**



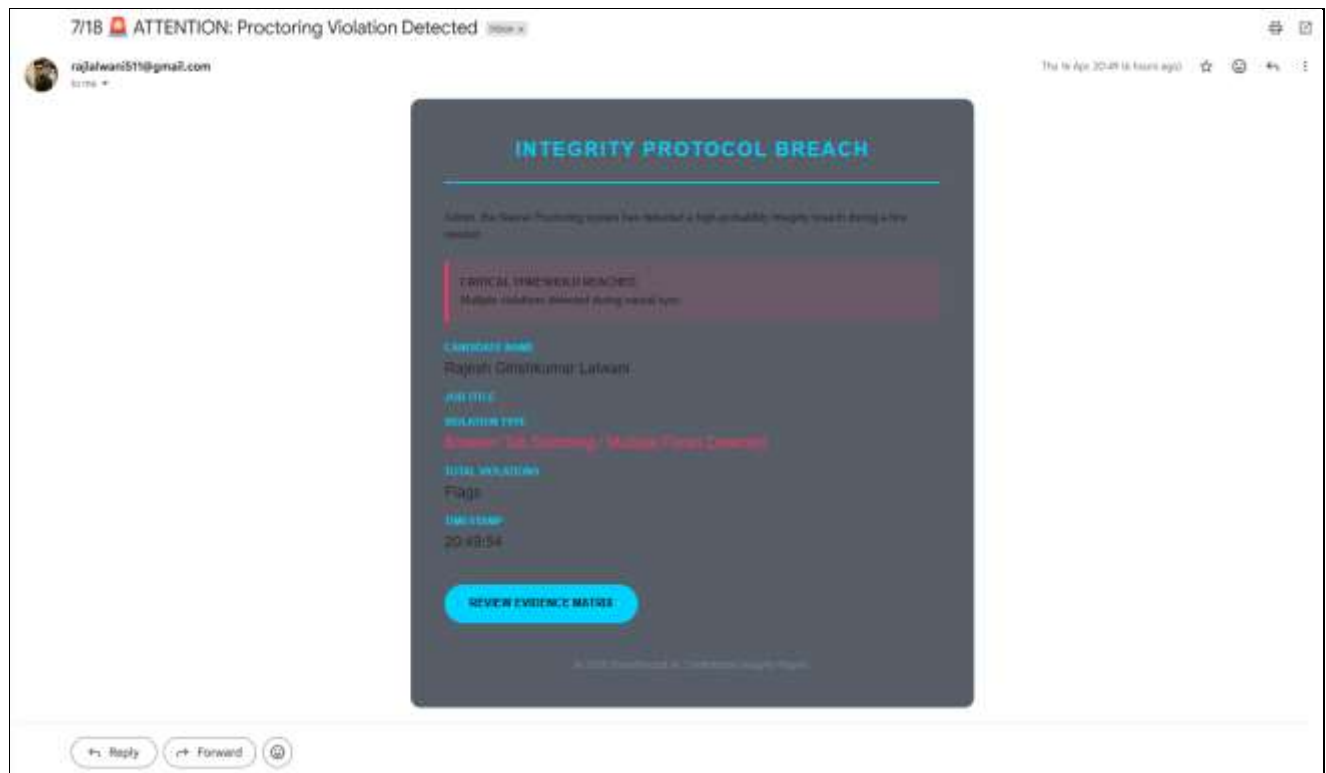
- **Round 2 Rejection Email:**



- **Round 2 Cleared Email:**



- **Proctoring Violation Detection Email:**



- **Round 3 (Technical Round) Scheduling Email:**



Interview Scheduled

Dear Rohan Suryavanshi,

Congratulations! Your application for AI Engineer has progressed to the next stage. We are pleased to inform you that your Technical Interview has been scheduled. Please find the details below.

POSITION	AI Engineer
ROUND	Technical Interview
DATE & TIME	02:00 PM
MEETING LINK	https://meet.google.com/reschedule-link
COMPANY	IR Info Tech (SmartRecruit)

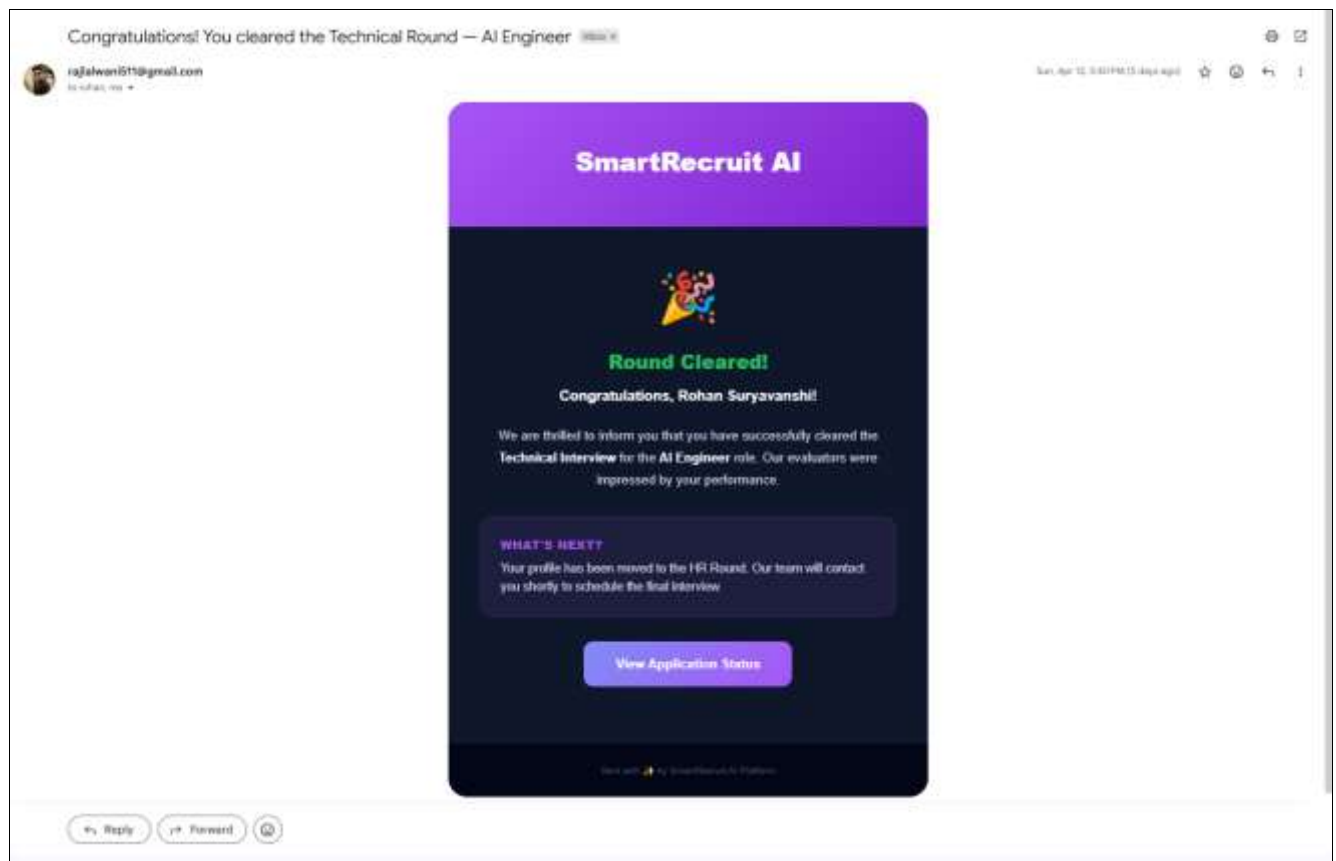
 **Instructions:** Please join the meeting 5 minutes before the scheduled time. Ensure your camera and microphone are working properly. Be in a quiet environment for the best experience. A calendar invite (.ics) is attached to this email for your reference.

Join Interview

If you have any questions or need to reschedule, please reach out to your recruiter promptly. We look forward to speaking with you.

© 2026 IR Info Tech Pvt Ltd — SmartRecruit AI Platform
This is an automated notification. Please do not reply to this email.

- **Round 3 (Technical Round) Cleared Email:**



- **Round 4 (HR Round) Scheduling Email:**

 SMARTRECRUIT

Interview Scheduled

Dear Rohan Suryavanshi,

Congratulations! Your application for AI Engineer has progressed to the next stage. We are pleased to inform you that your HR Interview has been scheduled. Please find the details below.

POSITION	AI Engineer
ROUND	HR Interview
DATE & TIME	11:00 AM
MEETING LINK	https://meet.google.com/hr-link
COMPANY	IR Info Tech (SmartRecruit)

 **Instructions:** Please join the meeting 5 minutes before the scheduled time. Ensure your camera and microphone are working properly. Be in a quiet environment for the best experience. A calendar invite (.ics) is attached to this email for your reference.

[Join Interview](#)

If you have any questions or need to reschedule, please reach out to your recruiter promptly. We look forward to speaking with you.

© 2026 IR Info Tech Pvt Ltd — SmartRecruit AI Platform
This is an automated notification. Please do not reply to this email.

- **Round 4 Cleared & Official Offer of Employment Email:**

IR Info Tech
Private Limited

IR Info Tech Private Limited
Level 5, Tech Park, Cyber City, Bengaluru, Karnataka – 560100
CIN: U72900KA2020PTC000000 | hr@irinfotech.com | www.irinfotech.com

OFFER OF EMPLOYMENT
IR Info Tech Private Limited · Human Resources Division

Date of Issue: **April 12, 2026**

Ref: **IIT/HR/OFR/2026/0067**

To,

Rohan Suryavanshi

Subject: **Formal Offer of Employment – AI Engineer**

Dear Rohan Suryavanshi,

It is with great pleasure that **IR Info Tech Private Limited** extends this formal **Offer of Employment** to you. Following a comprehensive and competitive selection process, the Company is confident that your qualifications, professional experience, and personal attributes are well-suited to the role being offered.

We are pleased to welcome you to the IR Info Tech family and look forward to your contributions to our shared mission of building world-class technology solutions.

Position: AI Engineer

Salary: **8 LPA**

Joining Date: May 05, 2026

Please find your official, detailed **Offer Letter** attached to this email as a PDF document. It contains the complete breakdown of your compensation structure, employment terms, and benefits.

3. OFFER ACCEPTANCE

To accept this offer, please sign and return a copy of this letter or click the button below to confirm your acceptance through the candidate portal.

Please confirm your acceptance on or before:

Three (3) business days from the date of issue of this letter

Failure to confirm acceptance within the above period may result in this offer being treated as declined and the position being offered to an alternate candidate.

We are genuinely excited about the prospect of you joining the IR Info Tech team. We believe this will be the beginning of a long, mutually rewarding, and impactful association.

Accept Offer & View Full Details →

Warm regards,

Human Resources Team
IR Info Tech Private Limited
hr@irinfotech.com

IR Info Tech Private Limited
Level 5, Tech Park, Cyber City, Bengaluru, Karnataka – 560100, India
CIN: U72900KA2020PTC000000 · www.irinfotech.com

This email and any attachments are strictly confidential and intended solely for the named recipient.

• Official Offer of Employment PDF II:

IR Info Tech

Private Limited

IR Info Tech Private Limited

Level 5, Tech Park, Cyber City, Bengaluru, Karnataka – 560100

CIN: U72900KA2020PTC000000 | hr@irinfotech.com | www.irinfotech.com

Annual Gross Cost to Company (CTC)

• 8 LPA

Payable as per the Company's payroll cycle, subject to applicable statutory deductions (TDS, PF, ESI, Professional Tax, etc.) as per prevailing laws.

Compensation Component	Basis	Indicative Breakup
Basic Salary	Monthly Fixed	~40% of Annual CTC
House Rent Allowance (HRA)	Monthly Fixed	~20% of Annual CTC
Special / Flexible Allowance	Monthly Fixed	~30% of Annual CTC
Employer Statutory Contributions (PF, ESIC as applicable)	Monthly	~10% of Annual CTC
Total Annual Cost to Company (CTC)		• 8 LPA

The detailed salary structure, payslip format, reimbursement policies, and benefit enrollment documents will be shared during the induction and onboarding process. The above breakup is indicative only.

In addition to the above compensation, you will be entitled to the Company's standard benefits including **Group Health Insurance** (self + family), **Earned Leave** (18 days per annum), **Sick Leave** (12 days per annum), Provident Fund contributions, and access to the Company's **Performance Bonus Program** as applicable to your grade and role.

3. Terms & Conditions

1. This offer is contingent upon the successful completion of all applicable pre-employment background verification checks, including but not limited to identity verification, academic credentials, prior employment history, and criminal record verification, as required by company policy.
2. You are required to submit original documents for verification on or before your date of joining, including proof of identity (Aadhaar / Passport), educational qualification certificates, experience letters from previous employers, and your last drawn salary documents.
3. During the probationary period, either party may terminate the employment with a notice period of seven (7) calendar days. Upon confirmation, the standard notice period of sixty (60) days shall apply.
4. You will be required to execute a Non-Disclosure Agreement (NDA) and an Intellectual Property Assignment Agreement before or at the time of joining. All work product created during the tenure of employment shall be the exclusive property of the Company.
5. You are expected to comply fully with all Company policies, the Code of Business Conduct, information security practices, and applicable regulatory requirements in force from time to time.

- **Official Offer of Employment PDF III:**

IR Info Tech Private Limited	IR Info Tech Private Limited Level 5, Tech Park, Cyber City, Bengaluru, Karnataka – 560100 CIN: U72900KA2020PTC000000 hr@irinfotech.com www.irinfotech.com
---	---

6. This offer is personal and non-transferable. Any misrepresentation of qualifications or experience discovered at any stage — including after joining — may result in the immediate withdrawal of this offer or termination of employment.

7. This letter constitutes a conditional offer and does not create a binding employment contract until the formal Employment Agreement has been executed by both parties. The employment agreement and associated policy documents will be provided during the onboarding process.

4. Acceptance of Offer

We request you to review the terms of this offer carefully. To formally accept this offer, please sign and return a copy of this letter along with a scan of your Government-issued photo ID within the acceptance deadline specified below.

Please confirm your acceptance on or before:

Three (3) business days from the date of issue of this letter

Failure to confirm acceptance within the above period may result in this offer being treated as declined and the position being offered to an alternate candidate.

We are genuinely excited about the prospect of you joining the IR Info Tech team. We believe this will be the beginning of a long, mutually rewarding, and impactful association. Welcome aboard!

For and on behalf of
IR Info Tech Private Limited

Authorized Signatory

Human Resources Department

IR Info Tech Private Limited

Acceptance by Candidate

Rohan Suryavanshi

Signature & Date of Acceptance

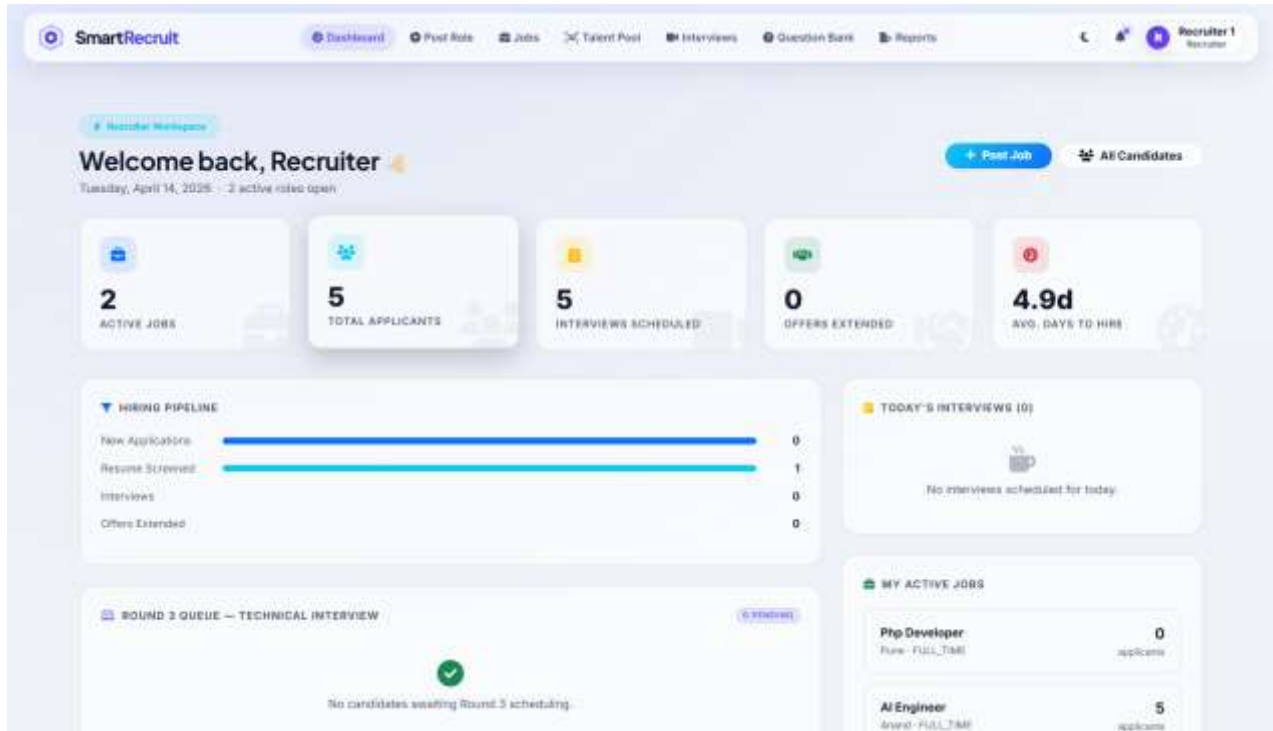
Date: _____

IT/HR/OFR/2026/0067 Issued: April 12, 2026 This document is system-generated and does not require a physical seal.

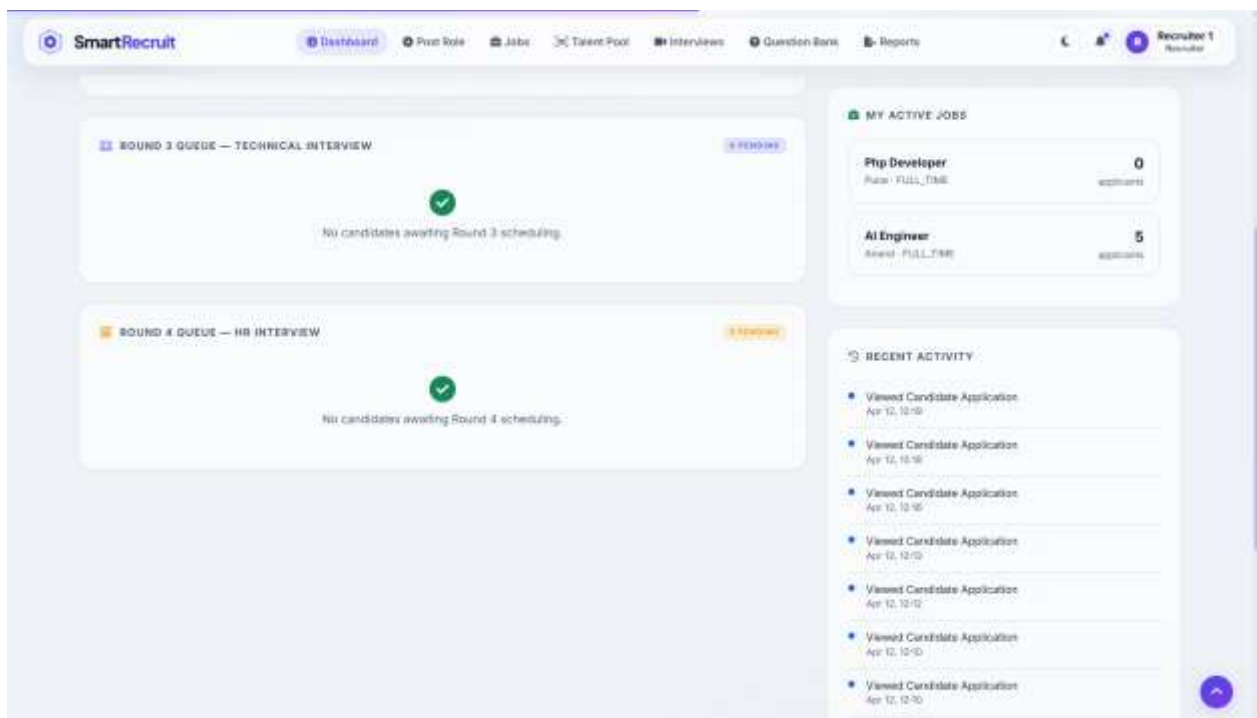
Confidential – This document is intended solely for the named recipient. | Page 3 of 3 | IR Info Tech Private Limited © 2026

❖ Input-Output Screenshots [Recruiter]:

- **Dashboard Page I:** Recruiter will View this page immediately after logging into the website.



- **Dashboard Page II [Scrolled Down]:** Recruiter will View this page immediately after logging into the website.



- **Post Job I:** Recruiter can Post New Job Roles using this Page.

SmartRecruit | Dashboard | **Post Role** | Jobs | Talent Pool | Interviews | Question Bank | Reports | Recruiter 1

POST A NEW JOB

Create a new job on the platform.

Fill in the job details, passing scores, and required skills. Our AI assistant is ready to help you write the perfect job description.

ESTIMATED TIME
3min
Average time to post a new job.

Job Details

JOB TITLE
e.g. Senior Python Developer

JOB CATEGORY
Full Time

TECH STACK FOCUS
General IT

LOCATION
e.g. Remote, New York

COMPENSATION RANGE
e.g. \$80k - \$120k

APPLICATION DEADLINE

MIN. EXPERIENCE (YEARS)

Scoring Rules

APTITUDE PASSING SCORE (%)
70.0

PRACTICAL PASSING SCORE (%)
70.0

APTITUDE DIFFICULTY
Medium

- **Post Job II [Scrolled Down]:** Recruiter can Post New Job Roles using this Page.

SmartRecruit | Dashboard | **Post Role** | Jobs | Talent Pool | Interviews | Question Bank | Reports | Recruiter 1

LOCATION
e.g. Remote, New York

COMPENSATION RANGE
e.g. \$80k - \$120k

APPLICATION DEADLINE
dd-mm-yyyy

MIN. EXPERIENCE (YEARS)
0

JOB DESCRIPTION
Detailed job description...

AI ASSISTANT | Professional | **Generate**

REQUIRED SKILLS (COMMA SEPARATED)
Comma-separated skills (e.g. Python, Django, AWS)

The system uses these exact skills to match candidates. e.g. "Python, React, Django, AWS"

APTITUDE DIFFICULTY
Medium

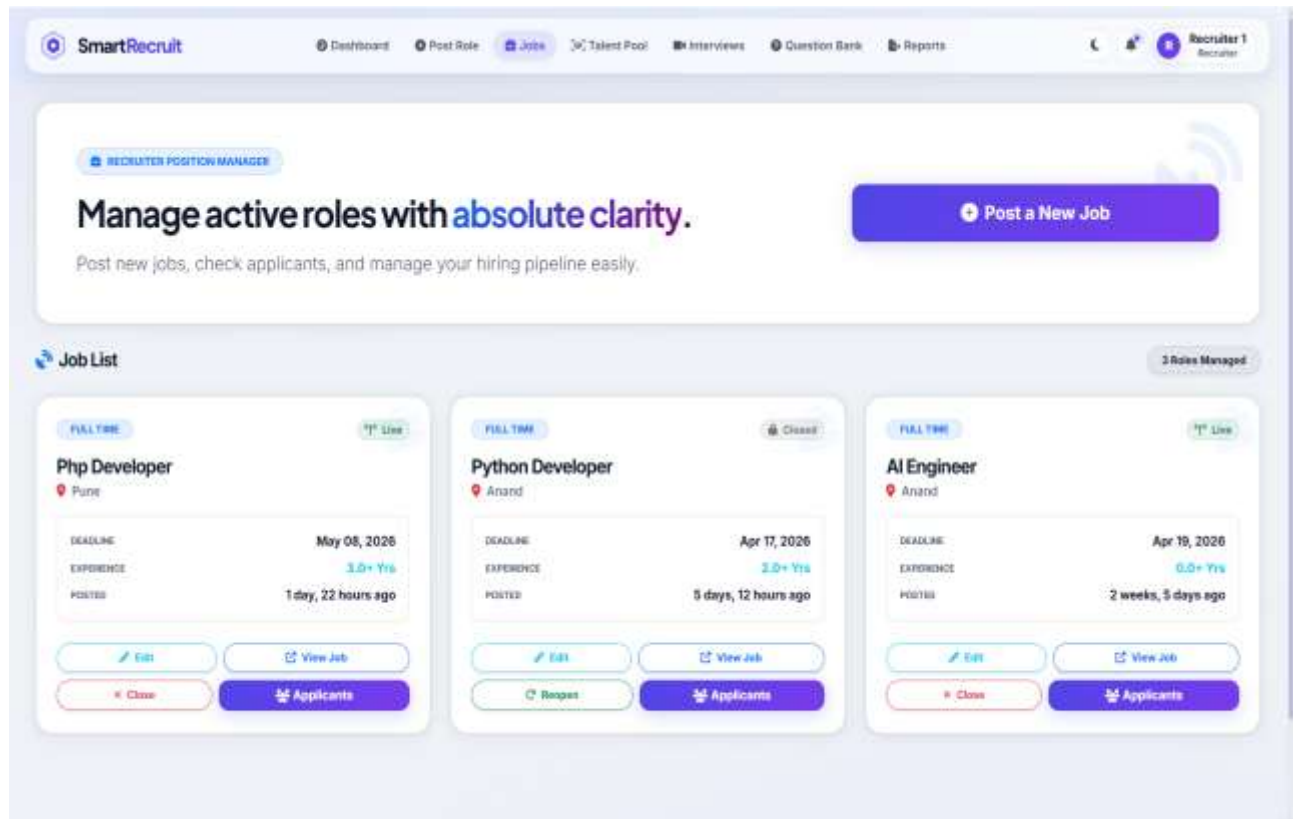
PRACTICAL DIFFICULTY
Medium

FINISH POSTING
Posting this job will make it visible to candidates and start the matching process.

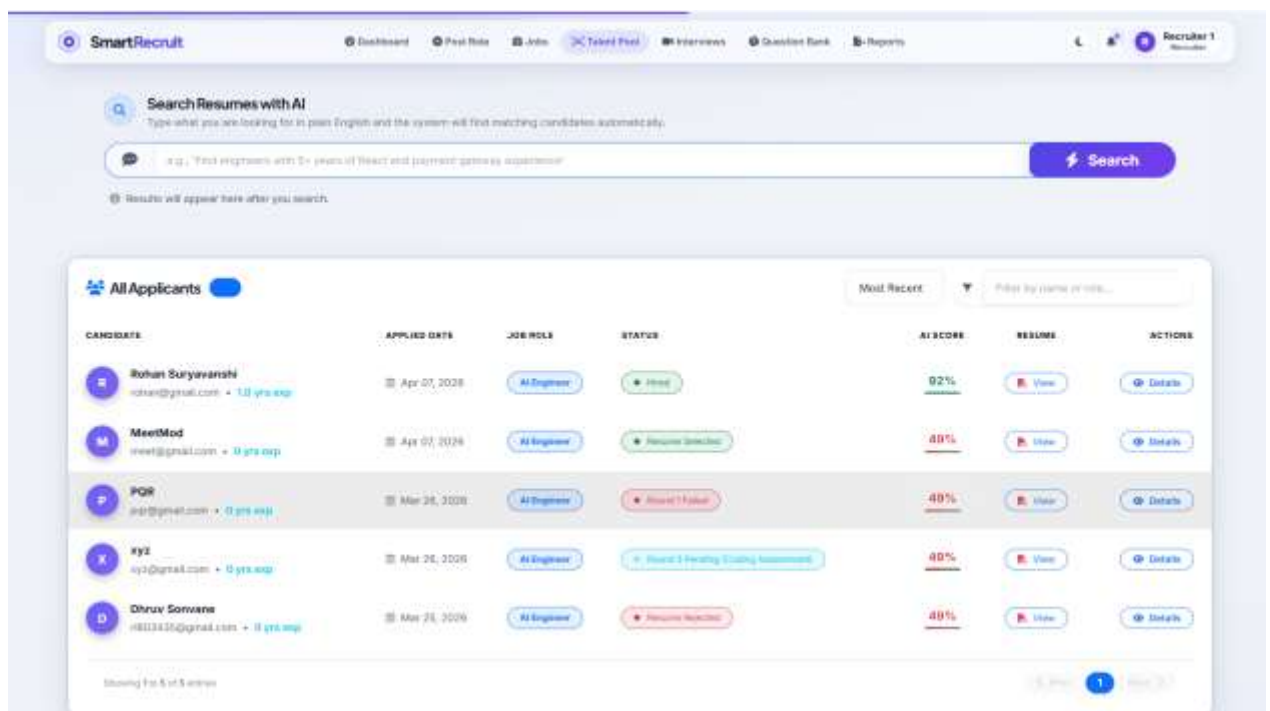
Post Job | **Cancel**

© 2024 SmartRecruit, AI-Powered Hiring Platform.

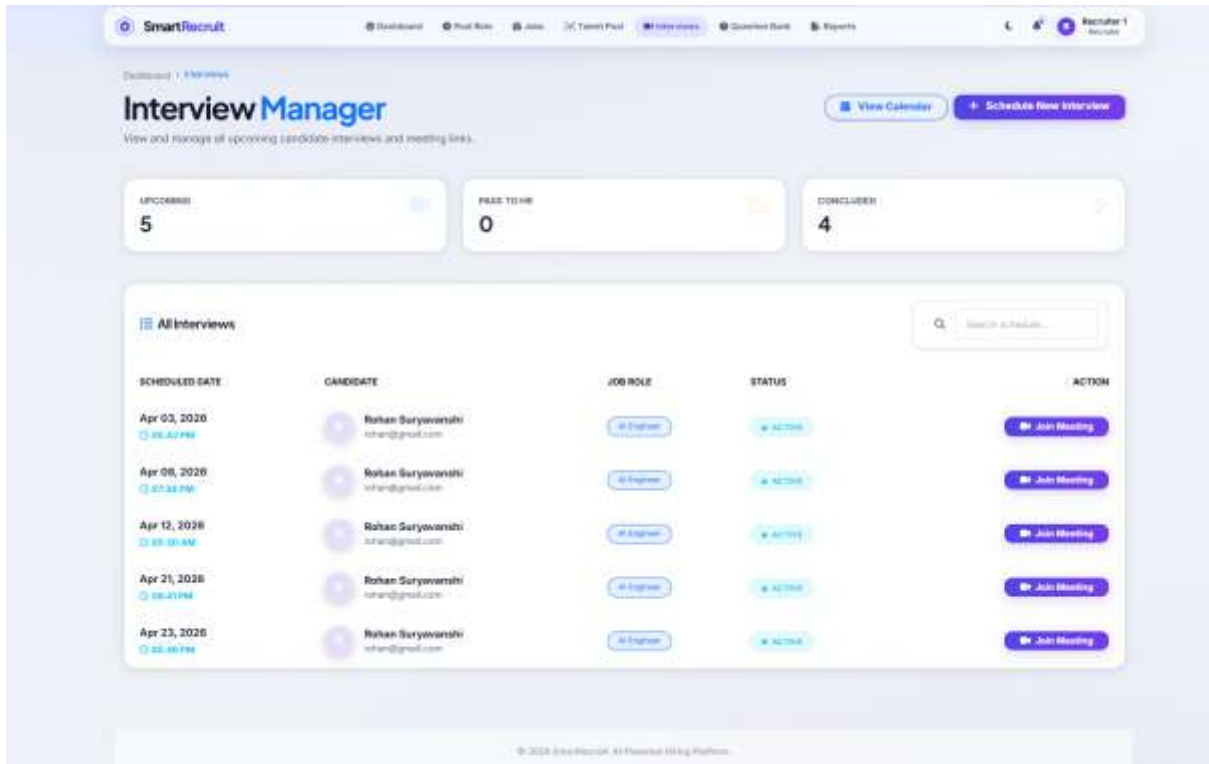
- **Manage Jobs Page:** Recruiter can Manage Posted Jobs and View it's Applicants using this Page.



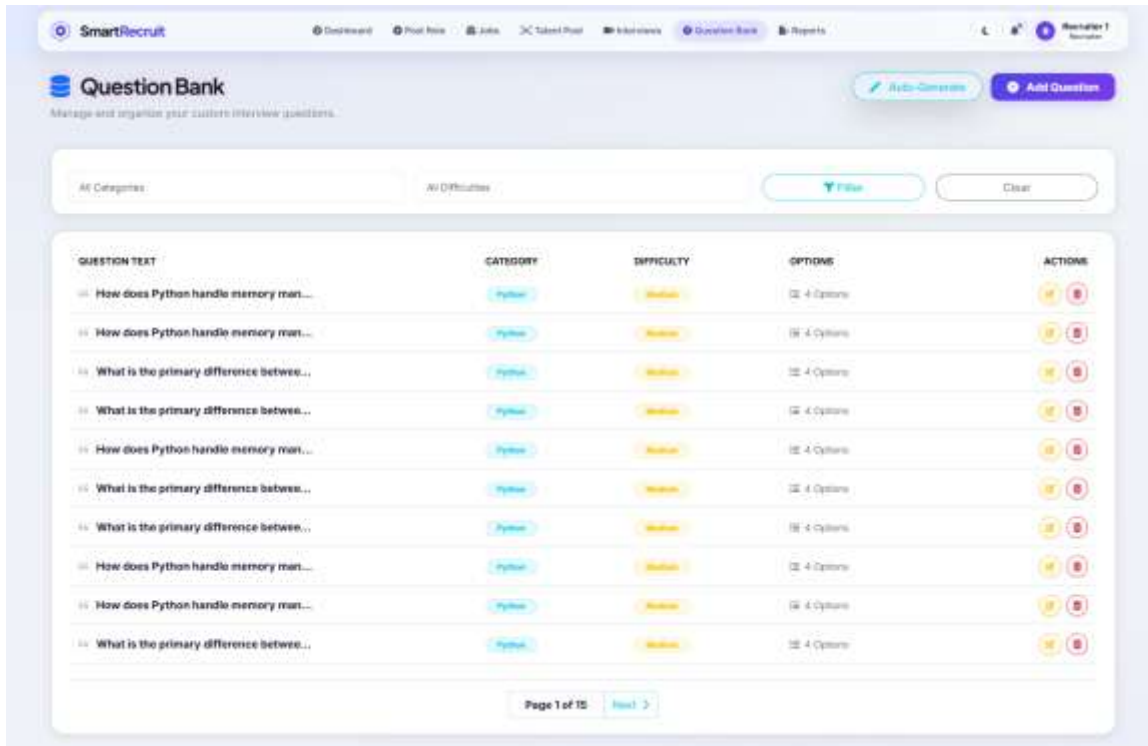
- **Talent Pool Page:** Recruiter can Manage Applicants' Job Applications, View their AI Score and Resume using this Page.



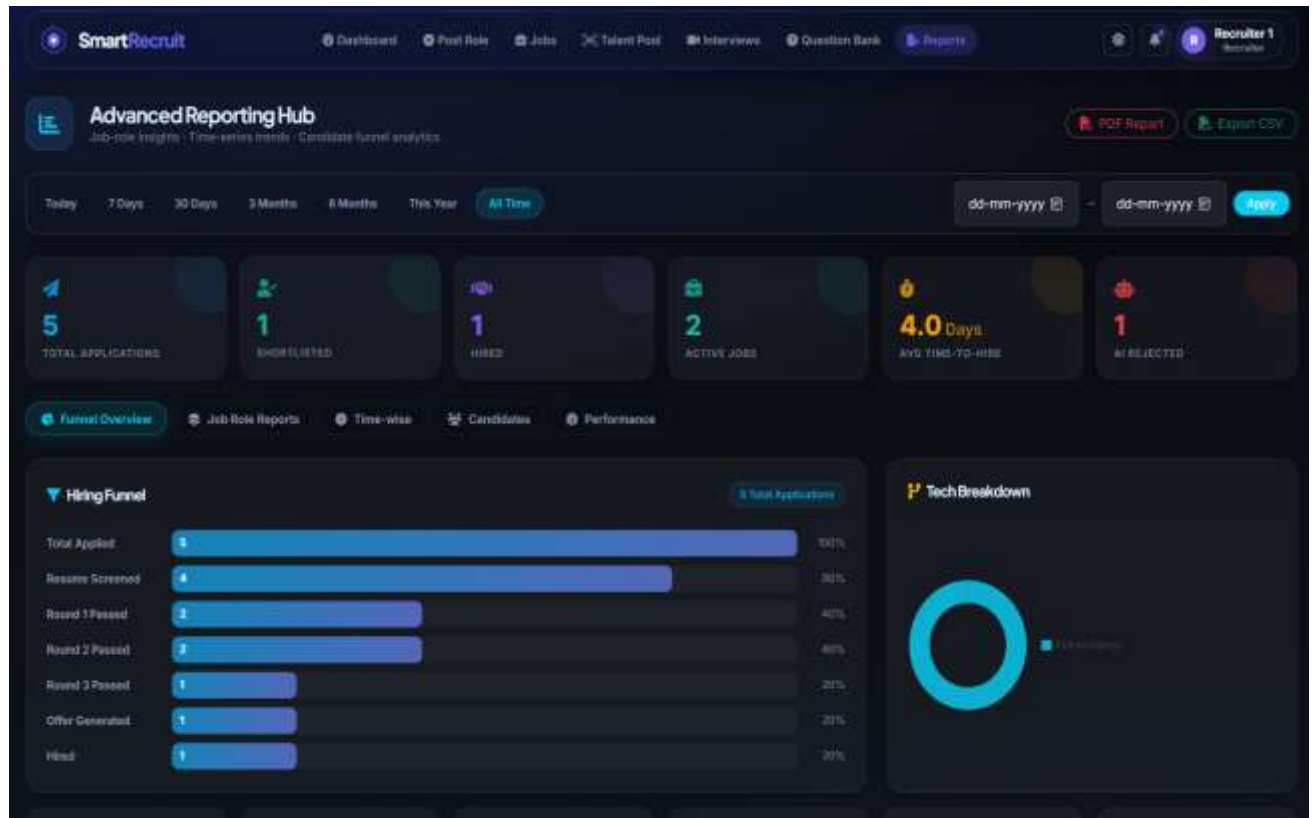
- **Manage Interviews Page:** Recruiter can Manage Applicants' Interviews using this Page.



- **Manage Question Bank Page:** Recruiter can Manage Question Bank for Round 1 & Round 2 using this Page.

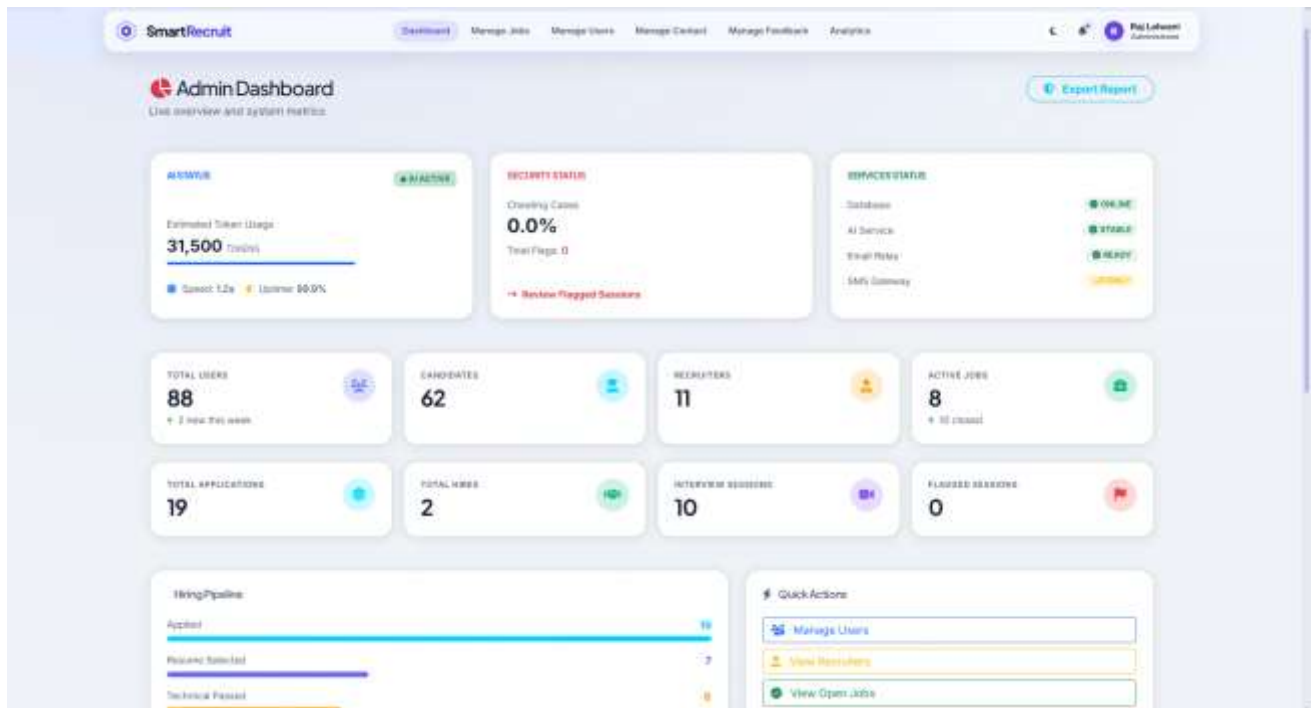


- **Analytics Page (Funnel Overview):** Recruiter can View and Download Funnel Overview Analytics Reports via this Page.

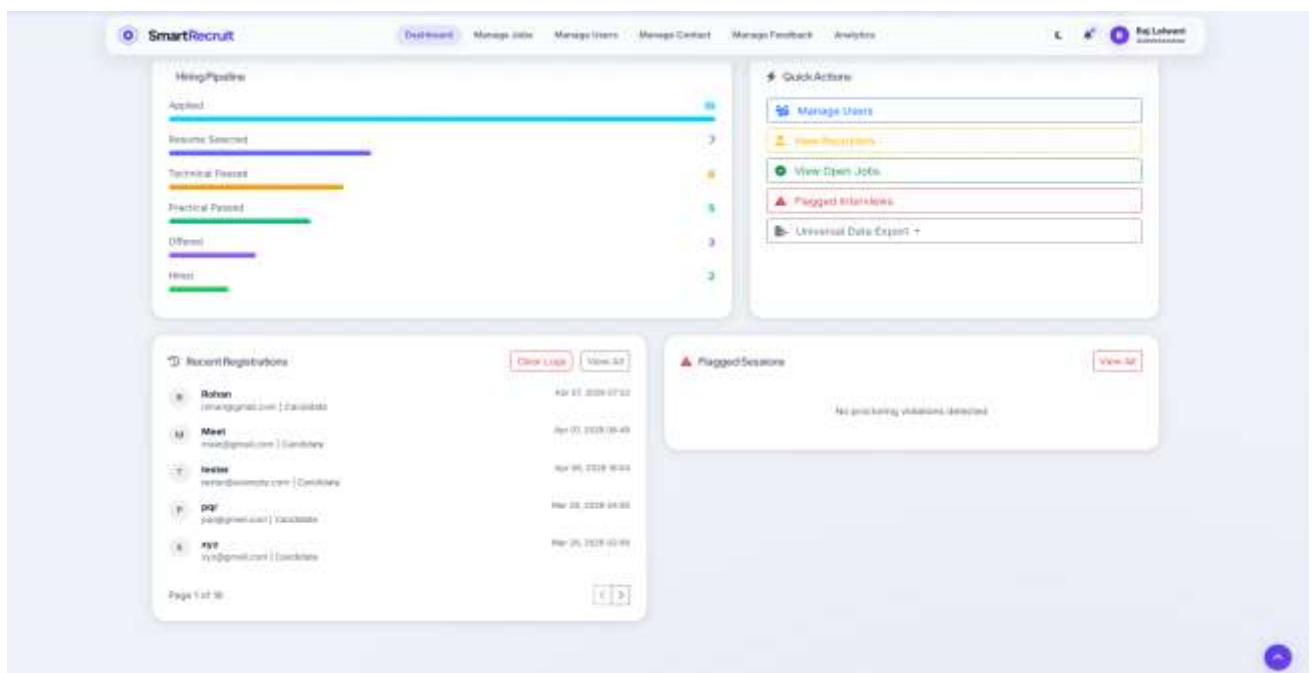


❖ Input-Output Screenshots [Admin]:

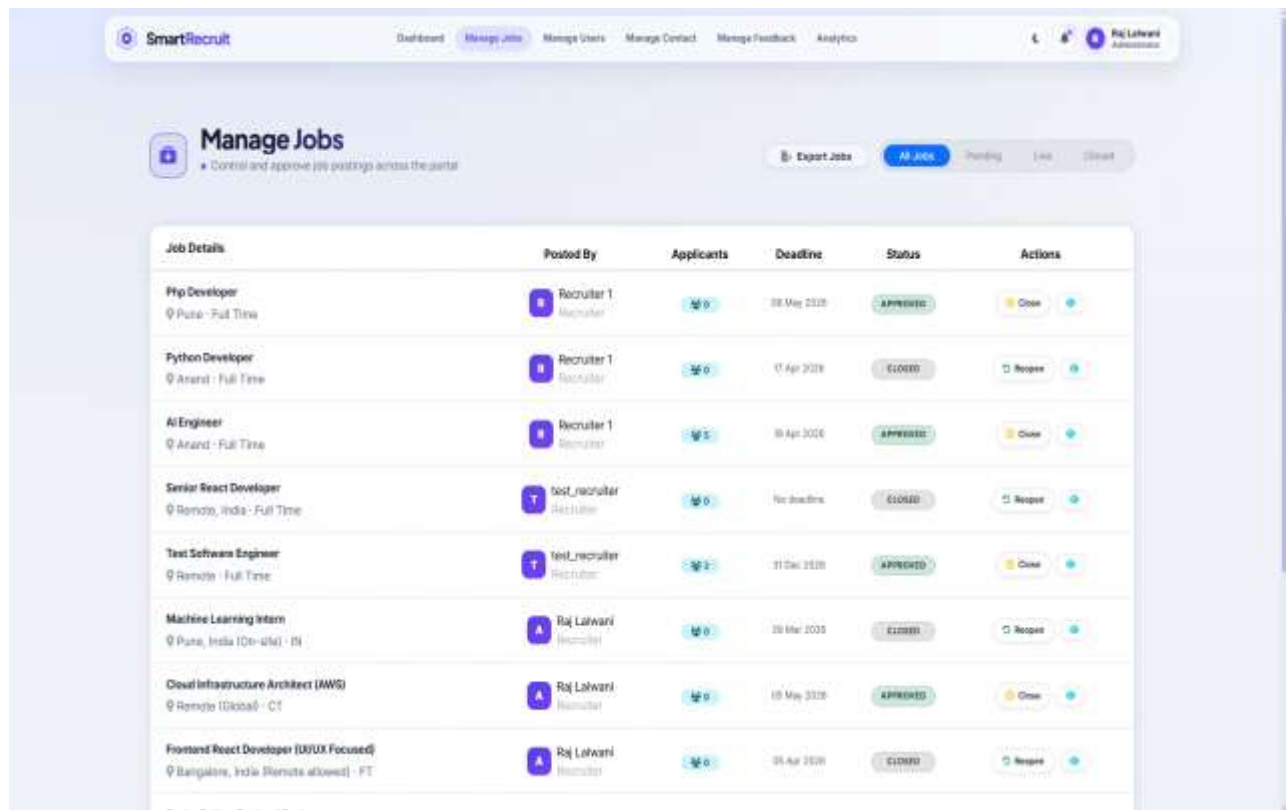
- **Dashboard 1.1:** Admin will View this page immediately after logging into the website.



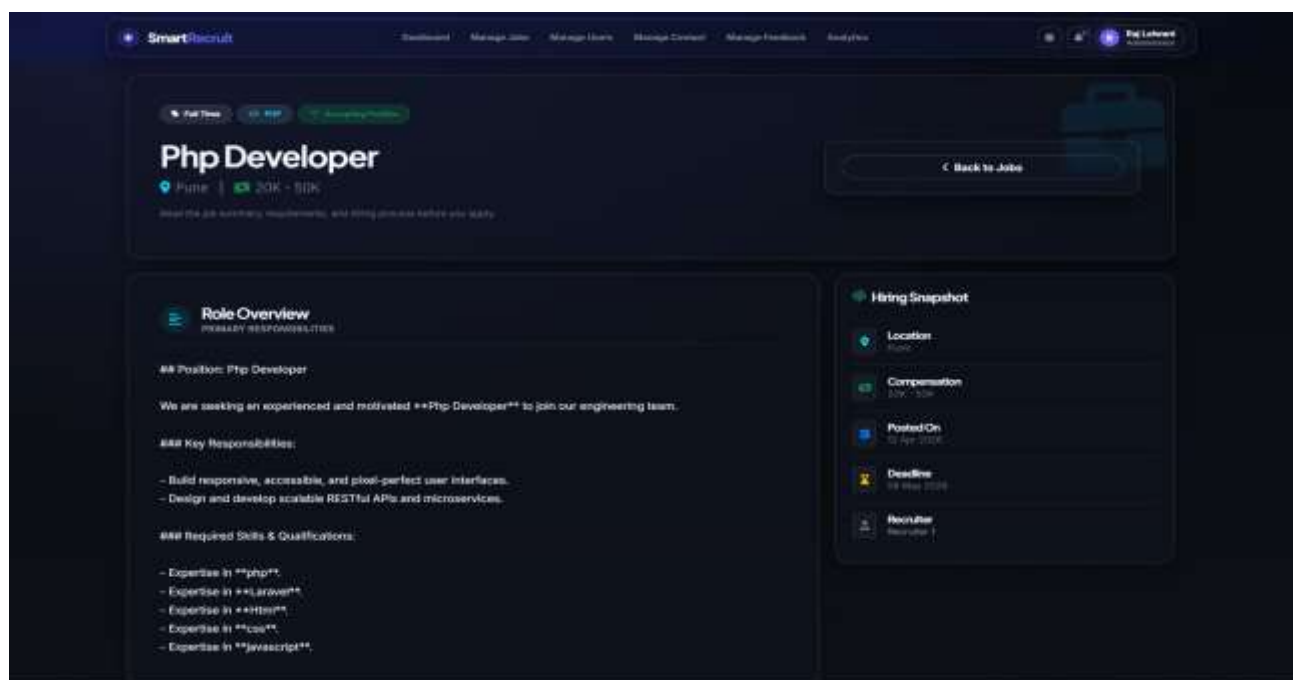
- **Dashboard 1.2:** Admin will View this page immediately after logging into the website.



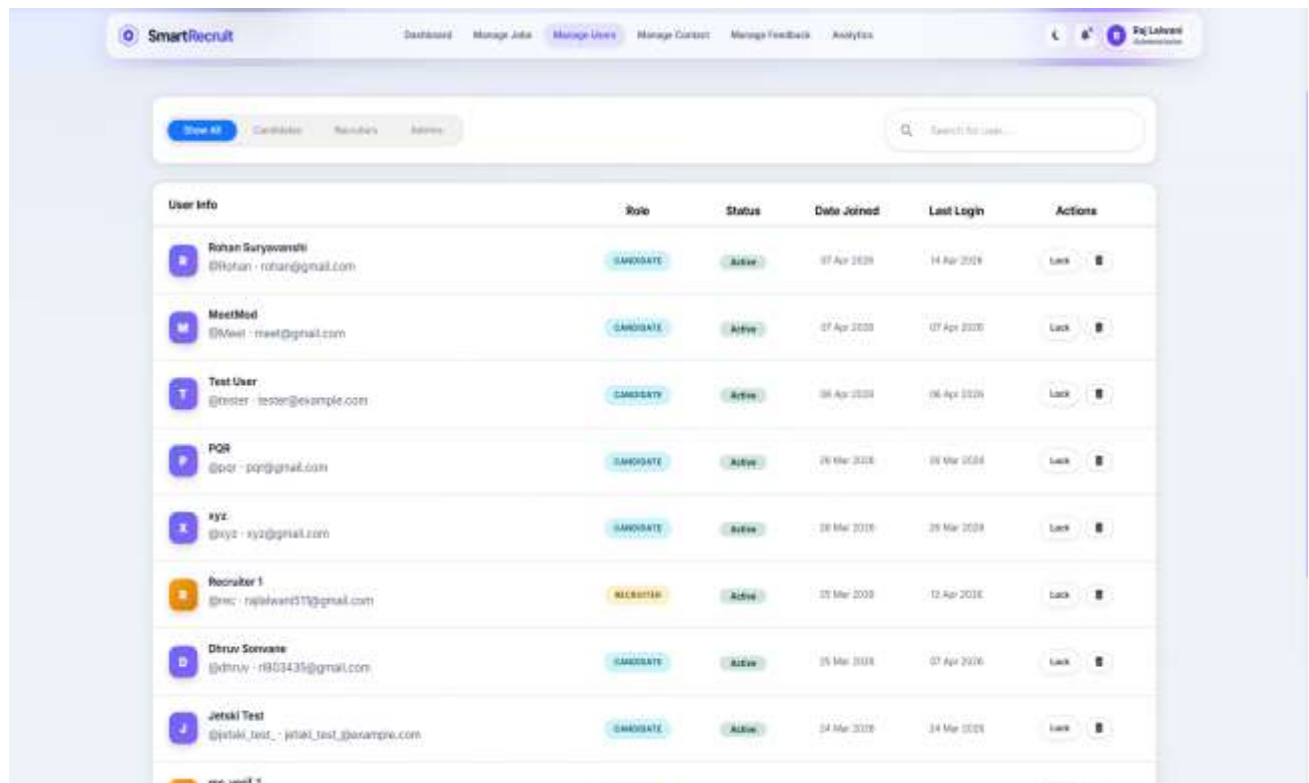
- **Manage Jobs Page (Light Theme):** Admin can Approve, Open and Close Job status along with facility of viewing jobs Page.



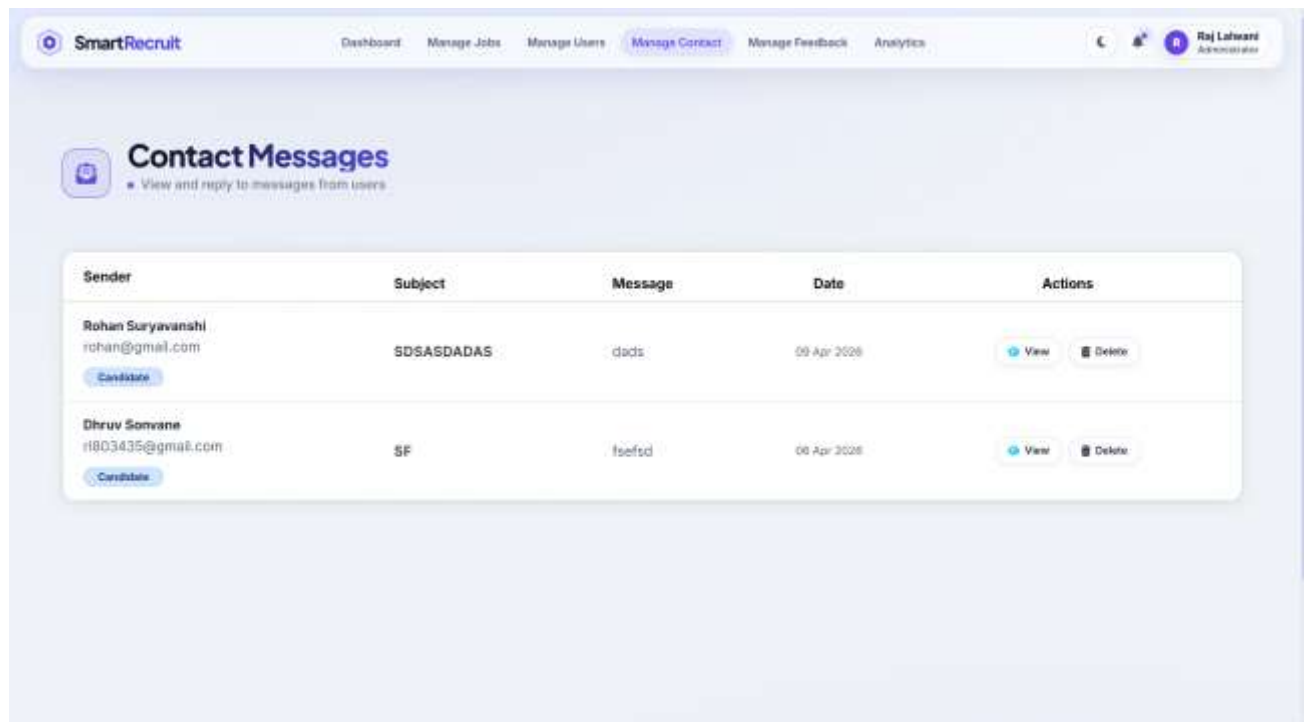
- **Manage Jobs Page (Dark Theme):** Admin can Approve, Open and Close Job status along with facility of viewing jobs Page.



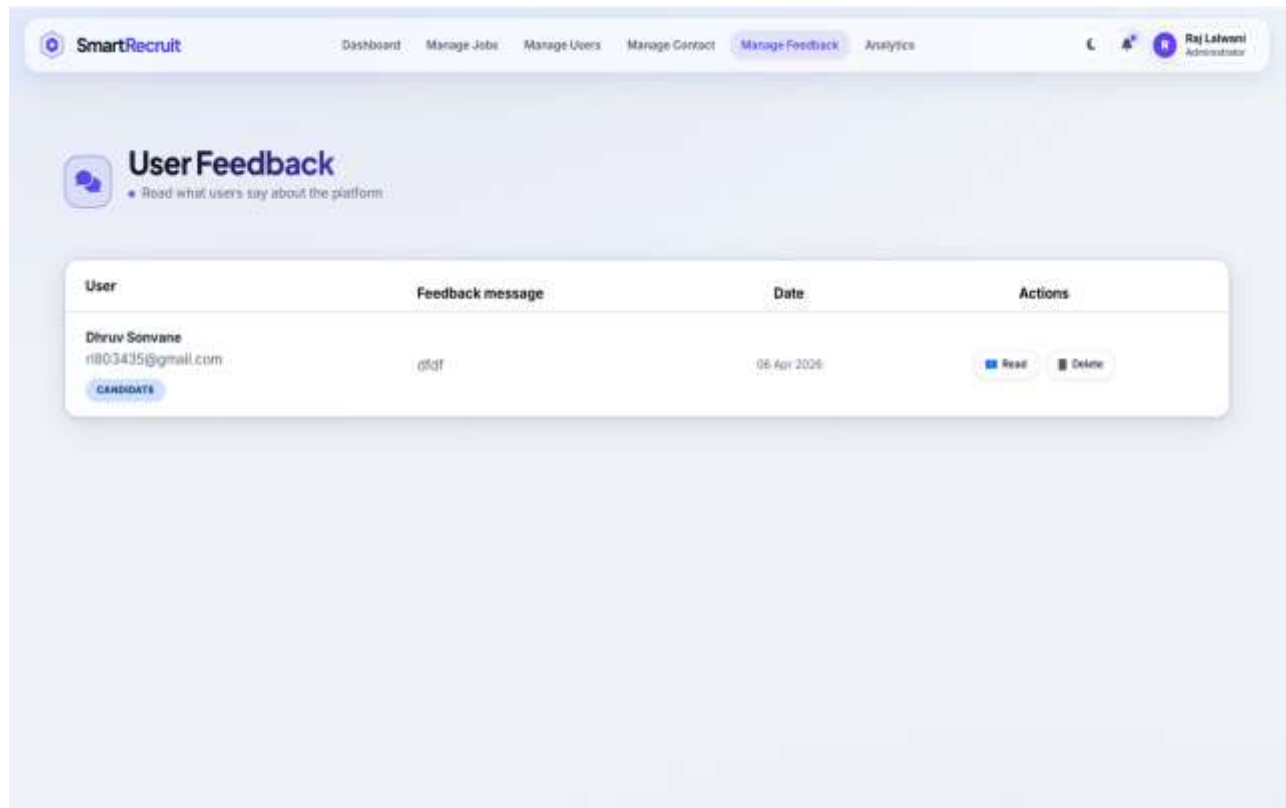
- **Manage Users Page (Light Theme):** Admin can Lock, Unlock and Delete All the Users' Account via this Page.



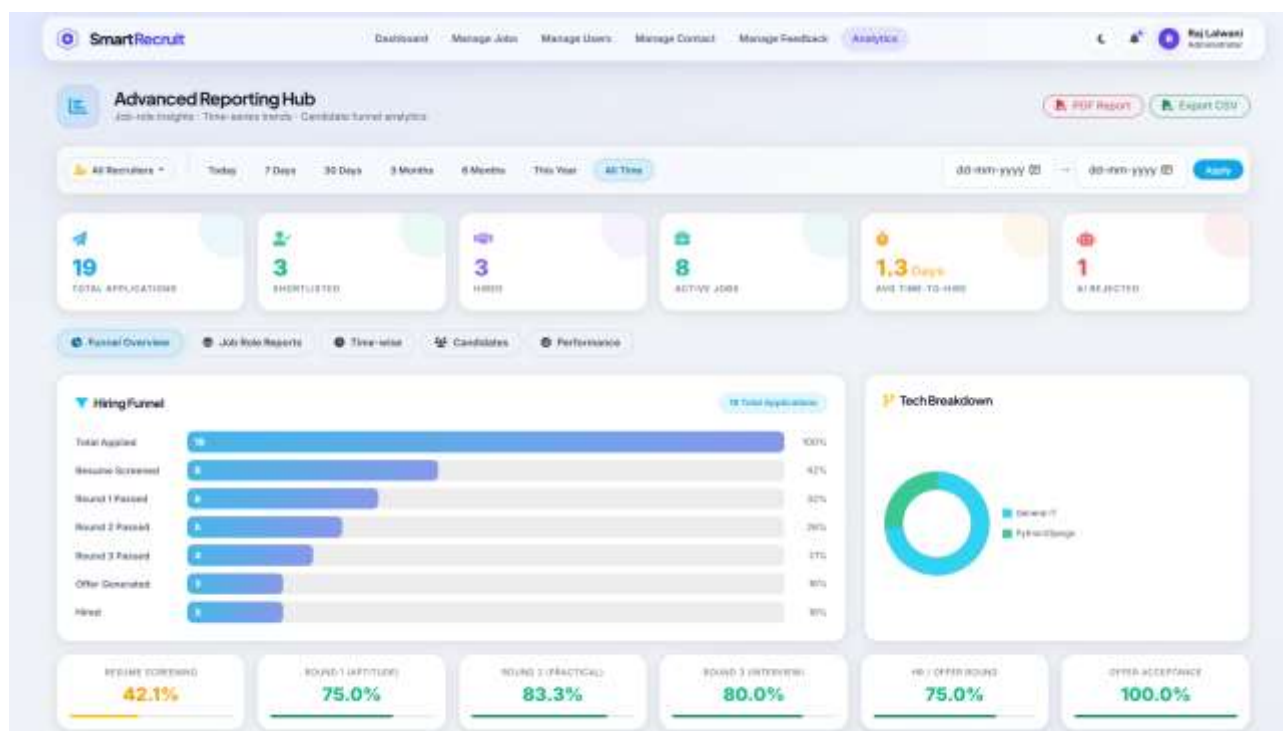
- **Manage Contact Page (Light Theme):** Admin can View and Delete All the Users' Contact Queries via this Page.



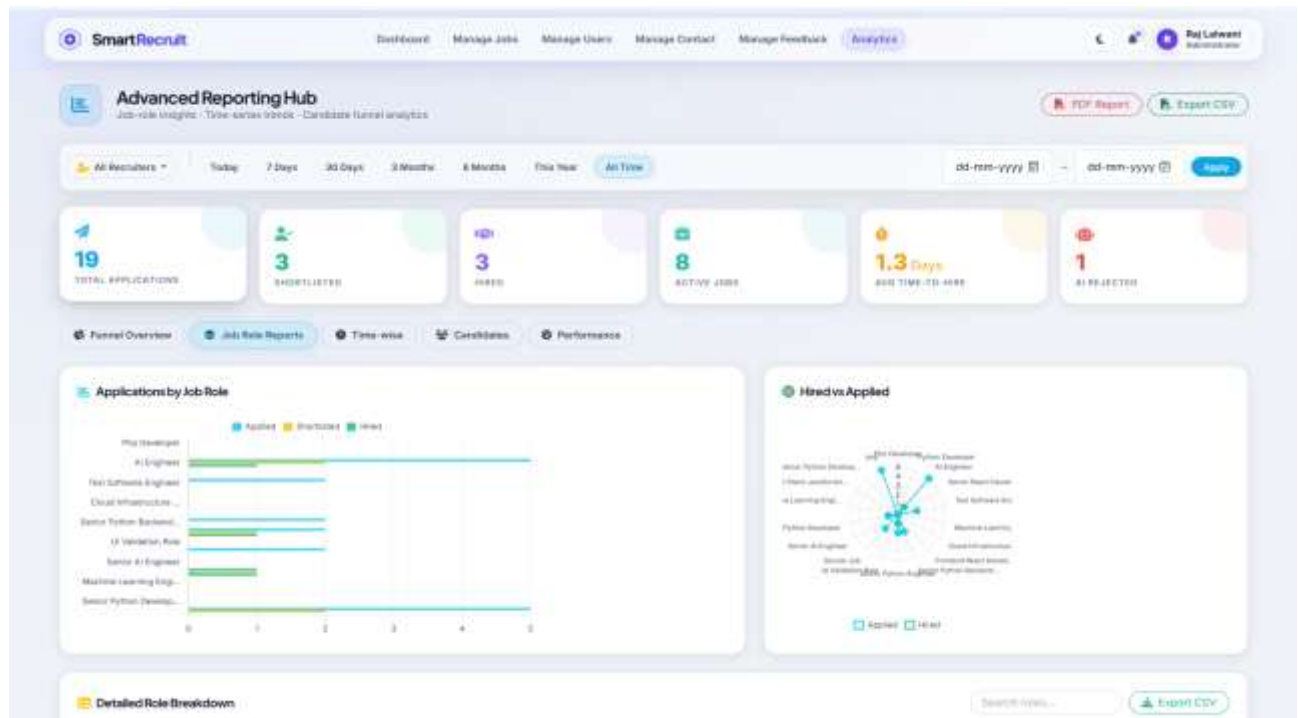
- **Manage FeedBack Page (Light Theme):** Admin can View and Delete All the Users' FeedBacks via this Page.



- **Analytics Page (Funnel Overview):** Admin can View and Download Funnel Overview Analytics Reports via this Page.



- **Analytics Page (Job Role Reports):** Admin can View and Download Job Role Reports via this Page.



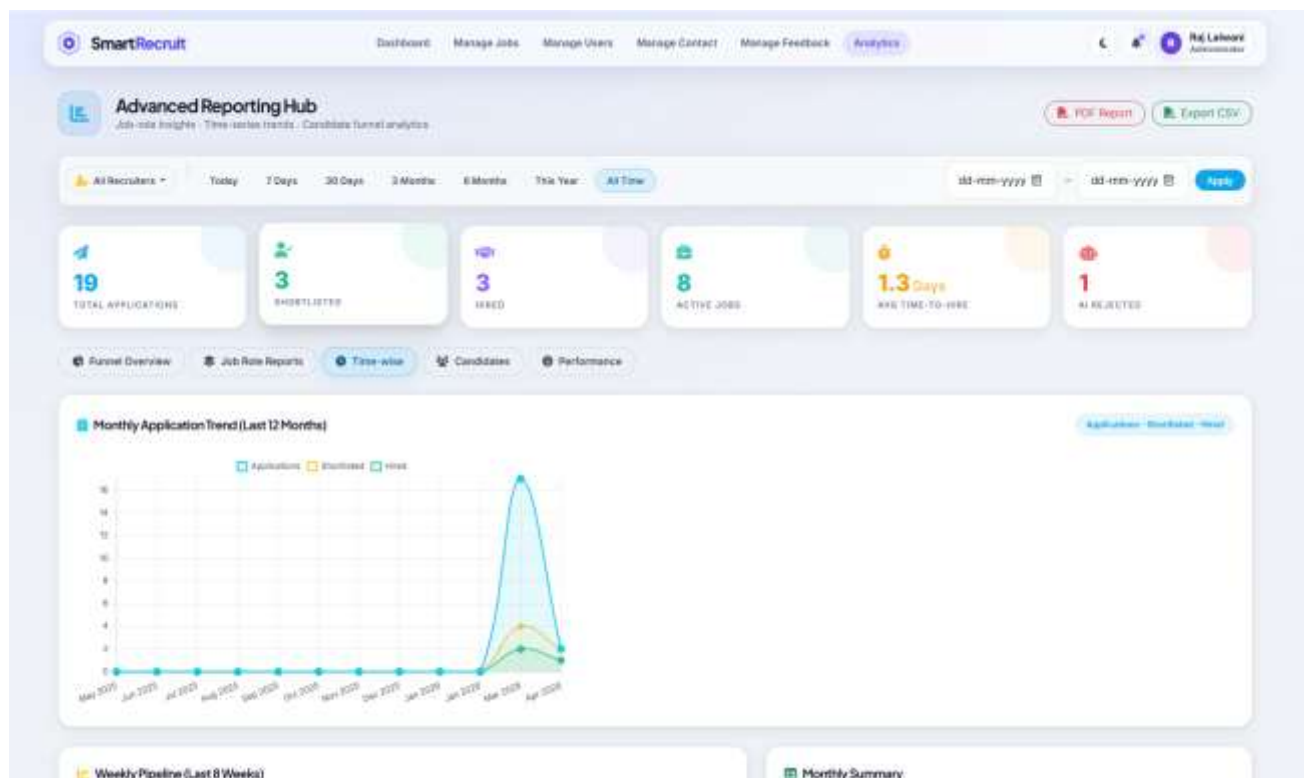
- **Analytics Page (Job Role Reports Scrolled):** Admin can View and Download Job Role Reports via this Page.

The screenshot shows the 'Detailed Role Breakdown' section of the Analytics page, displaying a table of job roles with their respective statistics. The table includes columns for Job Title, Tech Stack, Type, Applied, Shortlisted, Hired, Pass Rate, and Status. The data is as follows:

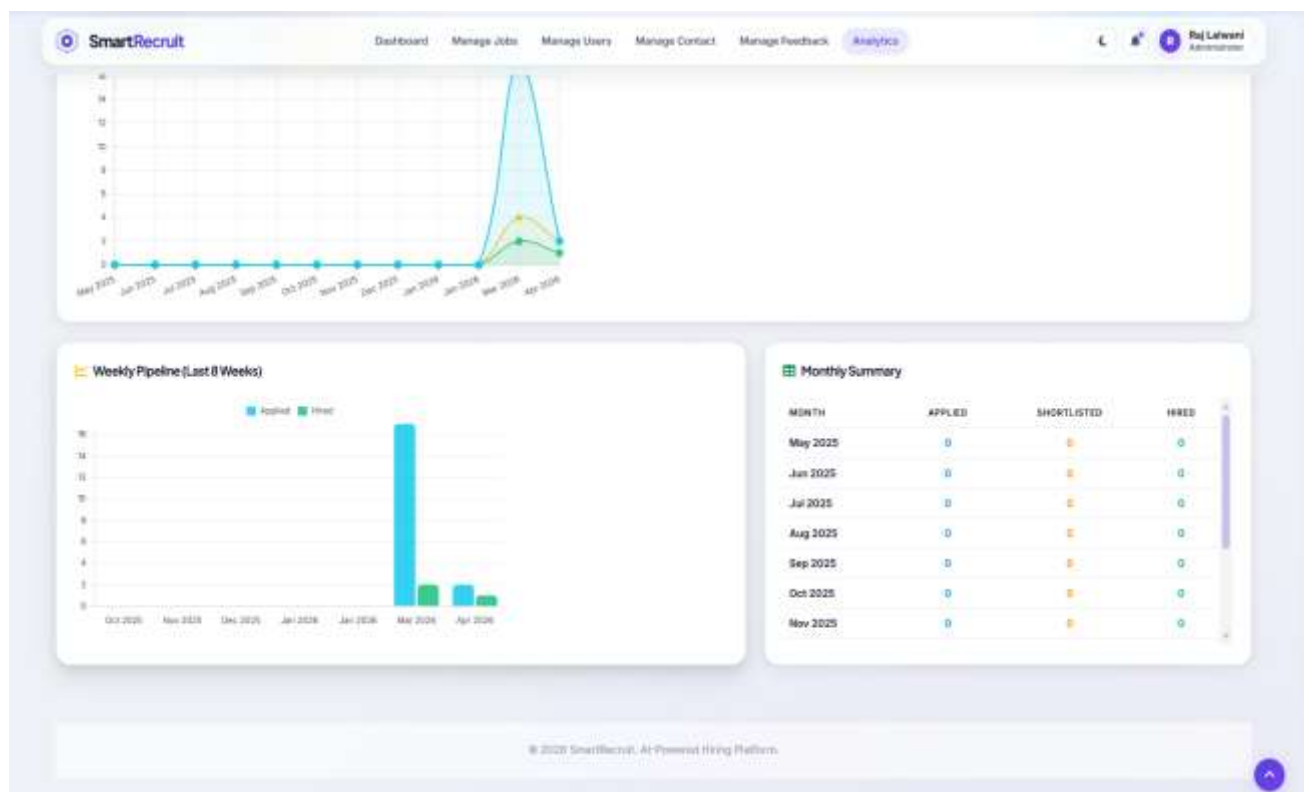
#	JOB TITLE	TECH STACK	TYPE	APPLIED	SHORTLISTED	HIRRED	PASS RATE	AVG AI SCORE	STATUS
1	Php Developer Apr 12, 2020	PHP/Laravel	Full Time	0	0	0	0%	0	Open
2	Python Developer Apr 08, 2020	Python/Django	Full Time	0	0	0	0%	0	Closed
3	AI Engineer Mar 25, 2020	Python/Django	Full Time	5	2	1	40.0%	57.8	Open
4	Senior React Developer Mar 21, 2020	General IT	Full Time	0	0	0	0%	0	Closed
5	Test Software Engineer Mar 21, 2020	General IT	Full Time	2	0	0	0.0%	0	Open

The table is paginated, showing 1-5 of 10 rows. The footer indicates '© 2024 SmartRecruit. AI-Powered Hiring Platform.'

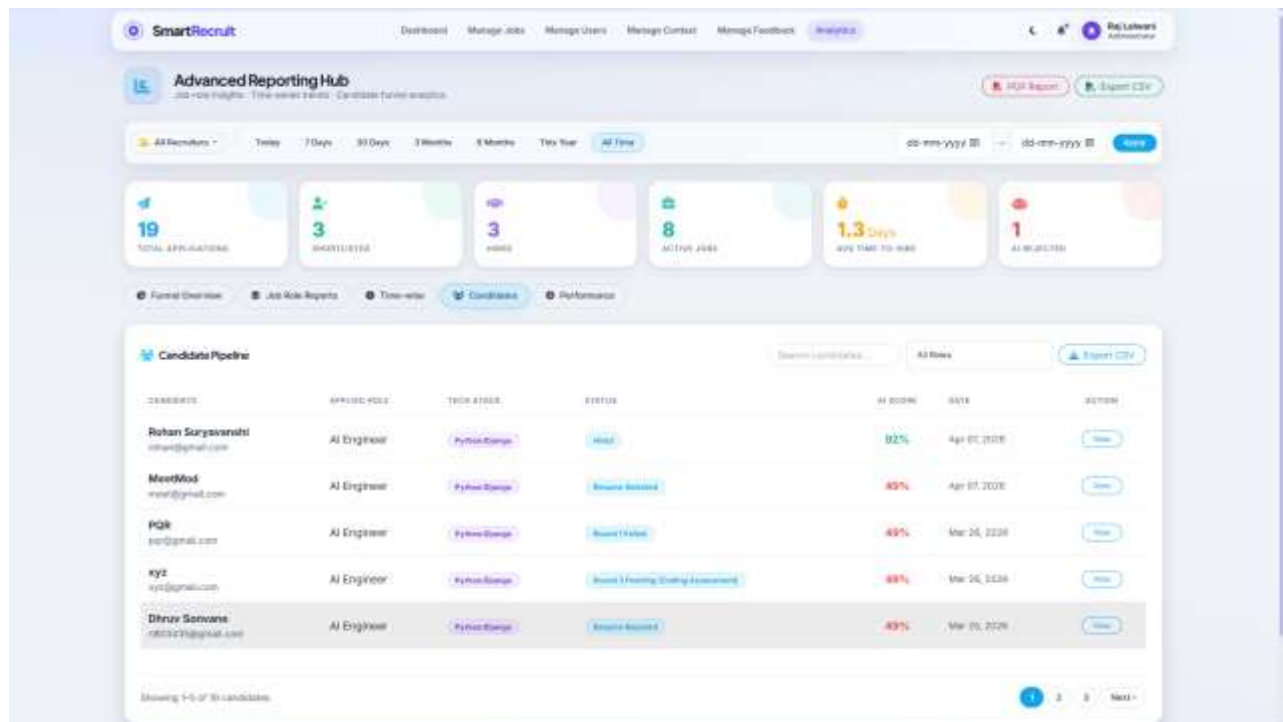
- **Analytics Page (Time wise Reports):** Admin can View and Download Time wise Reports via this Page.



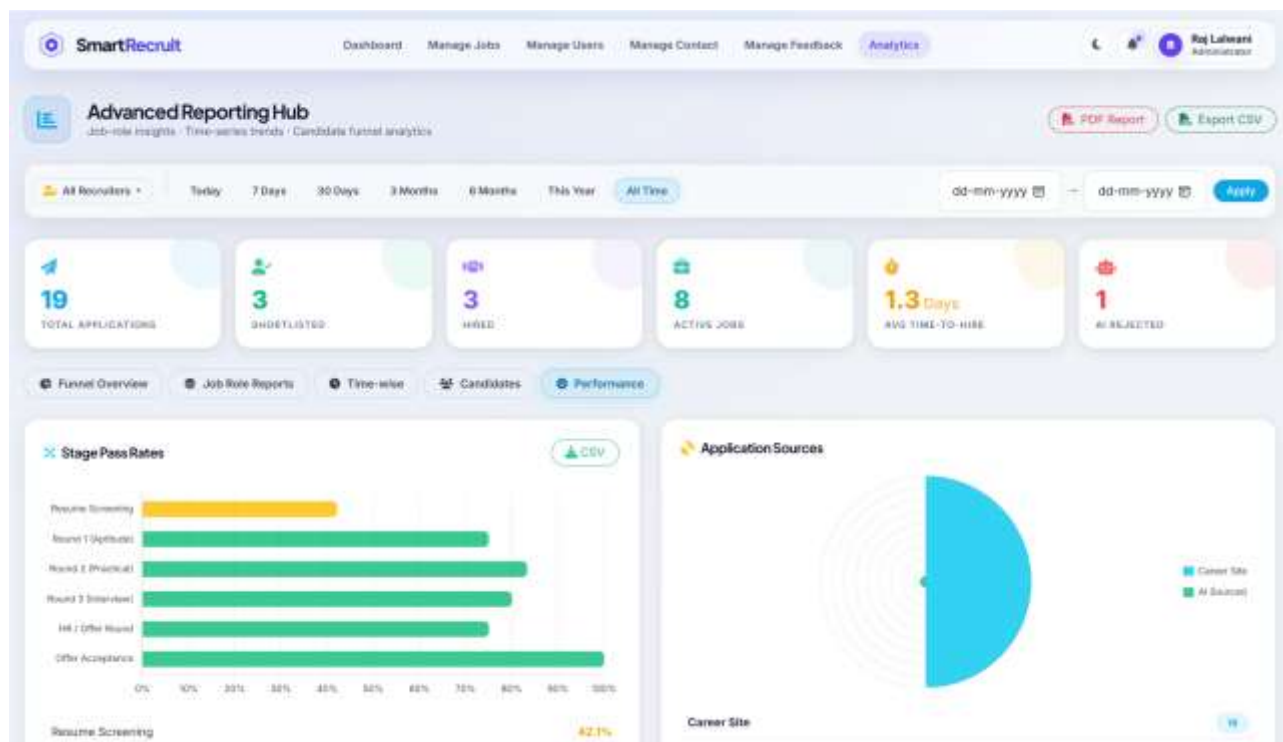
- **Analytics Page (Time wise Reports Scrolled):** Admin can View and Download Time wise Reports via this Page.



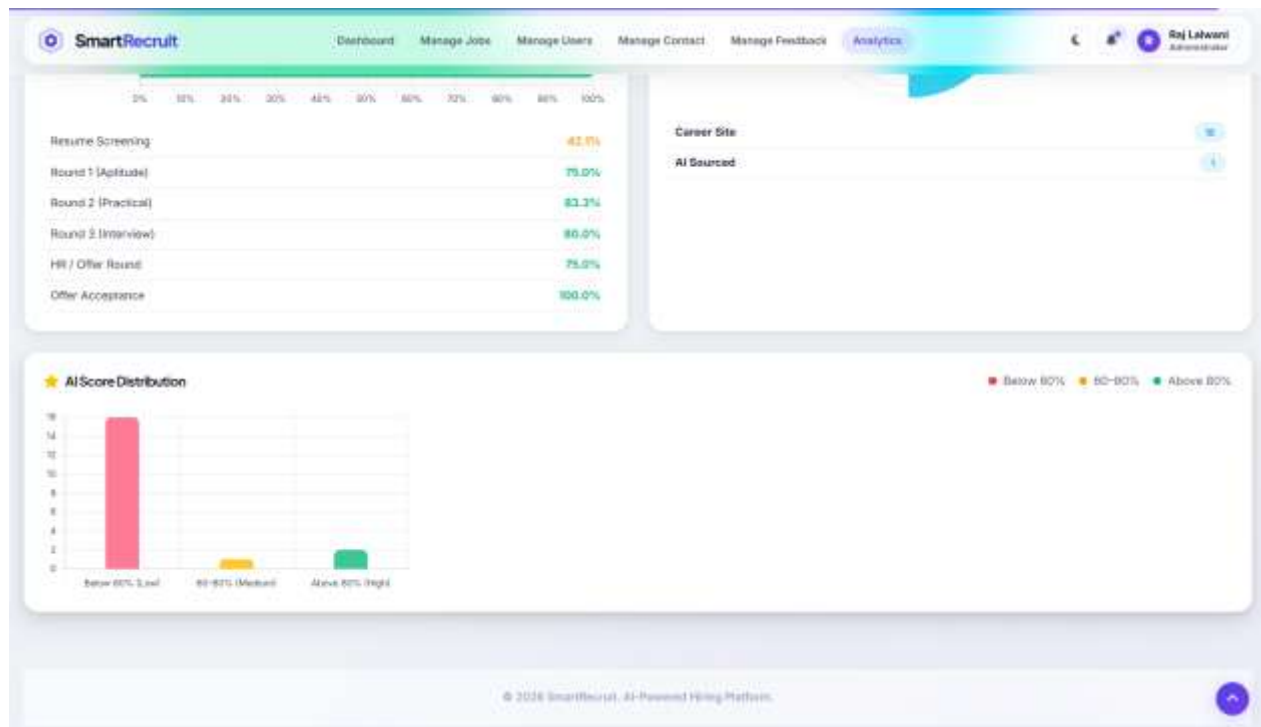
- **Analytics Page (Candidate Reports):** Admin can View and Download Candidate Reports via this Page.



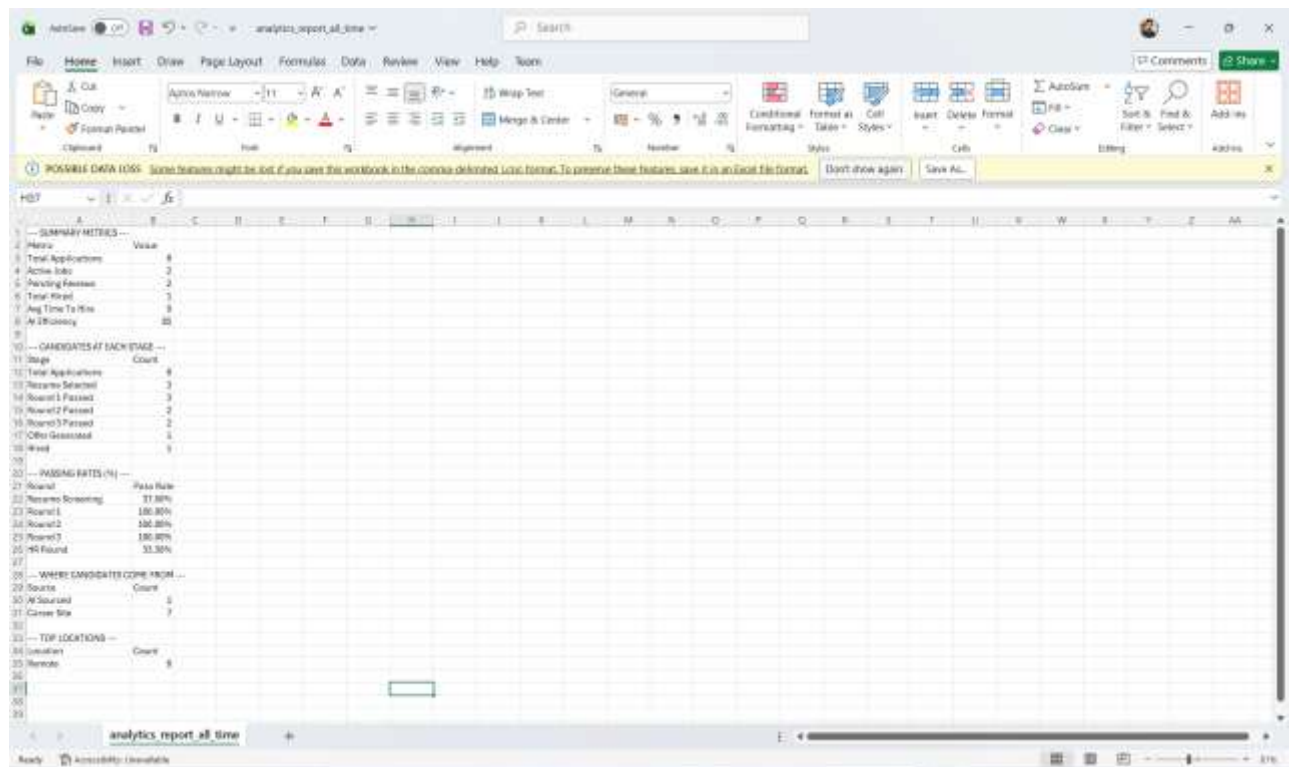
- **Analytics Page (Performance Reports):** Admin can View and Download Performance Reports via this Page.



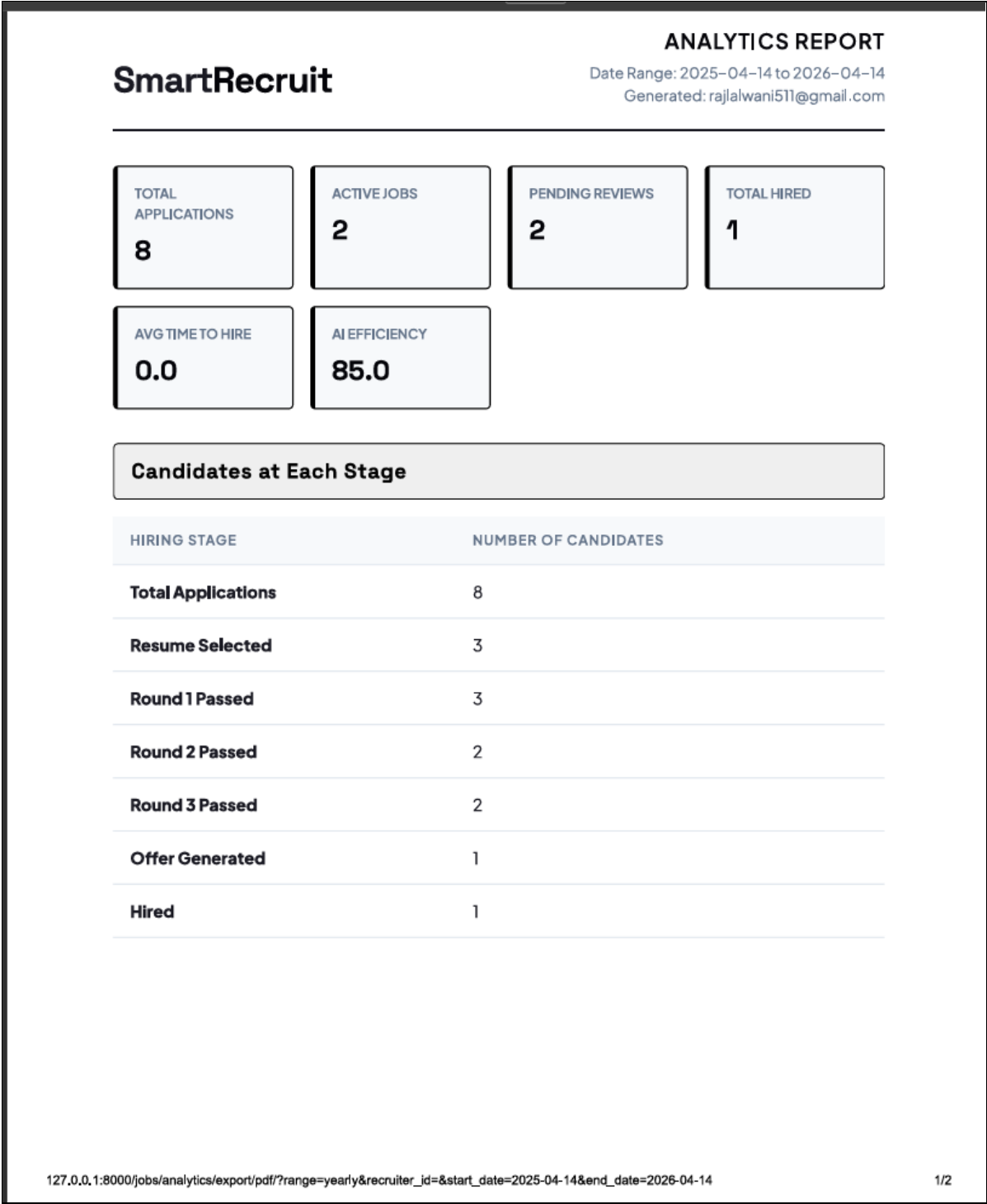
- **Analytics Page (Performance Reports Scrolled Down):** Admin can View and Download Performance Reports via this Page.



- **Analytics Report (CSV Format):**



- Analytics Report (PDF Format) Page I:



- **Analytics Report (PDF Format) Page I:**

4/14/26, 12:11 PM

Recruitment Analytics Report | SmartRecruit OS

Passing Rates

ROUND	PASS PERCENTAGE
Resume Screening	37.5%
Round 1	100.0%
Round 2	100.0%
Round 3	100.0%
HR Round	33.3%

Where Candidates Come From

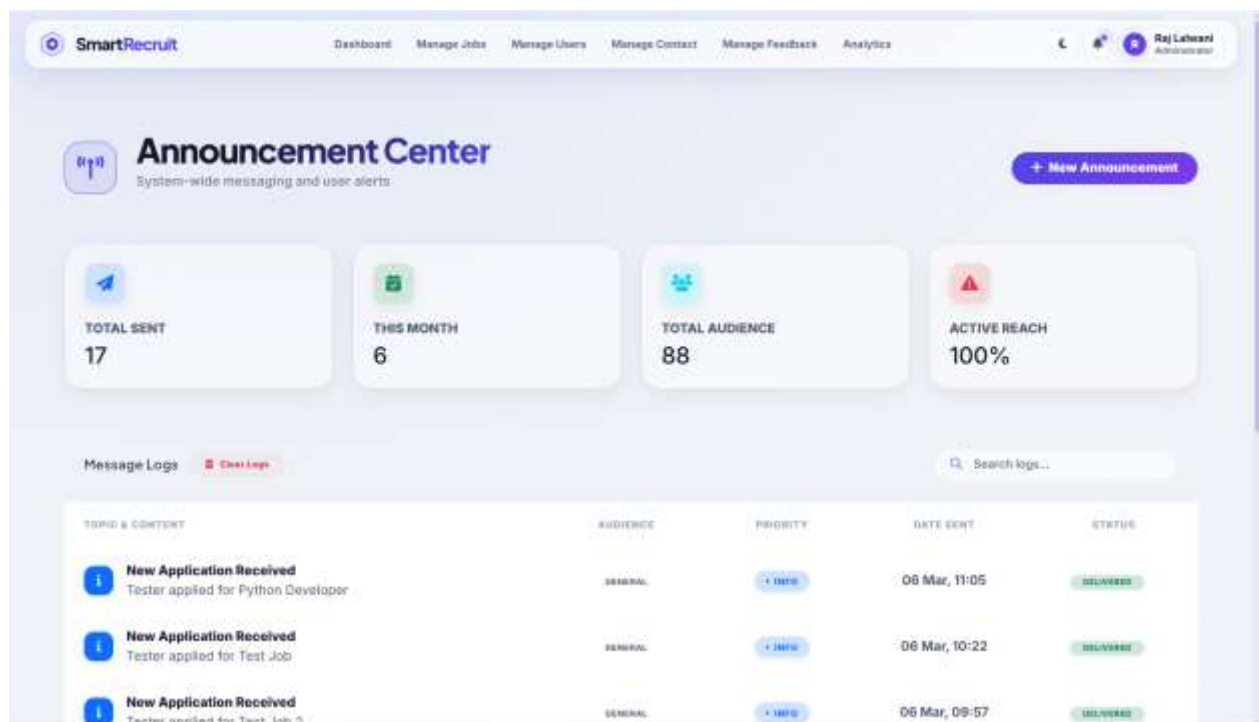
SOURCE	TOTAL CANDIDATES
AI Sourced	1
Career Site	7

SmartRecruit OS Analytics Engine | Strictly Confidential
Auto-generated Report

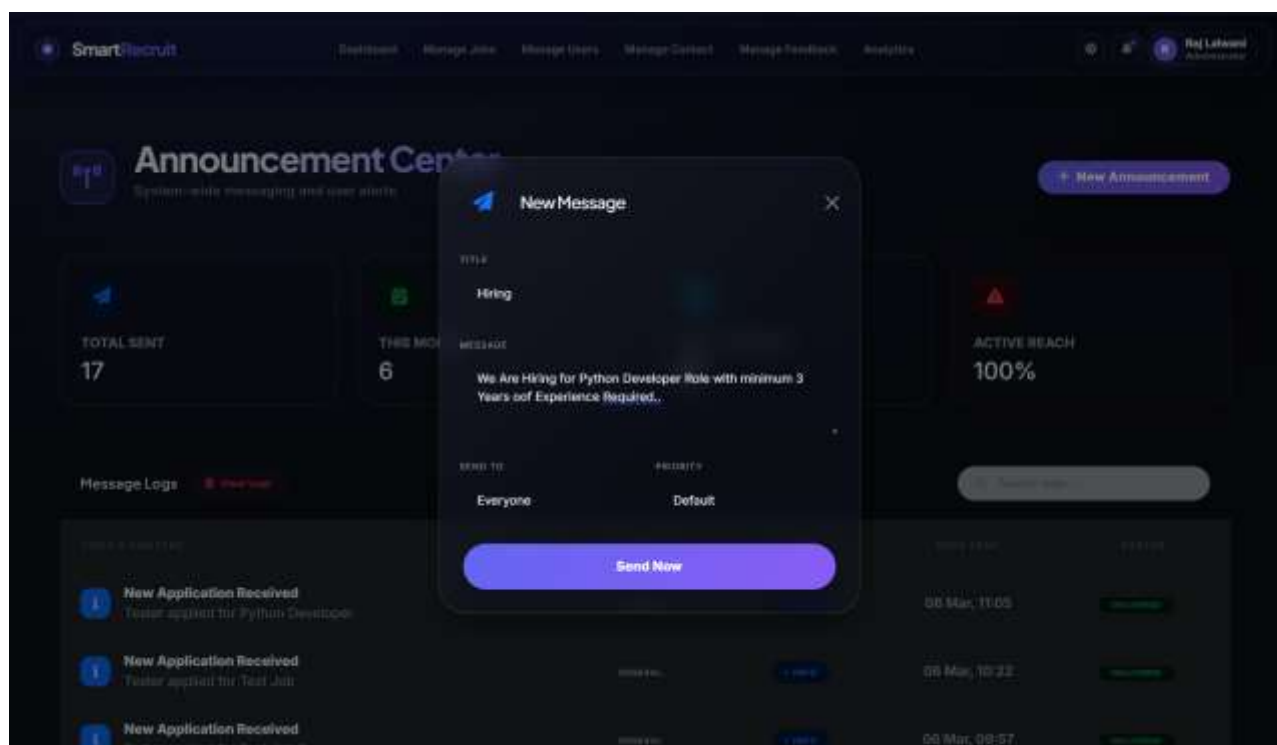
127.0.0.1:8000/jobs/analytics/export/pdf/?range=yearly&recruiter_id=&start_date=2025-04-14&end_date=2026-04-14

2/2

- **Message Broadcasting Page I:** Admin can view and Delete Broadcasting history, send Broadcast Messages to Candidate, Recruiter and All user at once via this Page.



- **Message Broadcasting Page II:** : Admin can view and Delete Broadcasting history, send Broadcast Messages to Candidate, Recruiter and All user at once via this Page.



TESTING

9.1 TESTING PLAN

- The Testing Plan outlines the rigorous strategy and steps that will be followed to ensure the Smart Recruit System meets all functional and non-functional requirements. The primary goal is to identify software bugs, assess AI latency issues, and ensure the absolute highest quality and reliability of the recruitment platform before final corporate deployment.

➤ Objectives of Testing

- To ensure the system handles application workloads exactly as expected in different hiring scenarios.
- To validate that all dedicated features (Candidate portal and Recruiter/Admin side) function flawlessly.
- To boldly identify and fix backend bugs before enterprise deployment.
- To ensure the system is completely secure, responsive on mobile devices, and user-friendly for all applicants.

➤ Types of Testing

1. Unit Testing:

- Individual operational modules (such as User Login, Resume Uploads, Application Submission, and AI Question Generation) will be tested separately.
- Ensures each independent function behaves exactly as expected in total isolation.

2. Integration Testing:

- Tests the direct interaction between different modules (e.g., linking the Job Postings inventory with the Candidate Application Pipeline, or connecting the AI Test evaluate APIs with the database).
- Verifies robust and error-free data flow between the Bootstrap-rendered frontend and the Python Django backend.

3. System Testing:

- Full end-to-end testing of the complete Applicant Tracking System (ATS).
- Ensures the platform as a whole flawlessly integrates and meets the defined organizational hiring requirements.

4. User Acceptance Testing (UAT):

- Performed directly by end-users (Sample HR Representatives and Test Candidates) to verify the system meets real-world, high-pressure hiring expectations.
- Helps actively identify UI/UX flow issues and deeply understand user expectations during the application process.

5. Regression Testing:

- Strictly ensures that ongoing code changes, future updates, or newly integrated AI features do not accidentally break existing functional job pipelines.

6. Performance Testing:

- Checks the system's backend performance under various computational loads.
- Ensures the application server can securely handle massive loads, such as hundreds of candidates attempting technical assessments simultaneously.

7. Security Testing:

- Ensures that profoundly sensitive proprietary recruiter data and candidate personal data (resumes, emails, scores) are strictly protected, guaranteeing the system is not vulnerable to severe external attacks like SQL injection, CSRF tampering, and XSS.

9.2 **TESTING STRATEGY**

The Testing Strategy is a crucial and highly formalized document typically designed by the Lead Quality Assurance (QA) Manager. It fundamentally helps in calculating the exact testing effort, timelines, and testing budget required for the project. Essentially, the testing strategy determines exactly which recruitment features will be heavily scrutinized and which architectural facilities will be bypassed. The testing space for the Smart Recruit platform is formally divided into two distinct parts:

- In Scope: The modules that are to be tested rigorously (e.g., Candidate Authentication, Resume Uploads, ATS Pipeline Routing, Gemini AI Question Generation, and Admin Action Analytics).
- Out Scope: The external systems that do not require internal structural testing (e.g., the internal server uptime of Google's Gemini API, external email SMTP servers, or local browser hardware).

Key Elements of the Smart Recruit Testing Strategy:

- A test strategy is a comprehensive outline that strategically describes the specific testing approach adopted throughout the software development lifecycle (SDLC) of the ATS.
- The deep purpose of this strategy is to provide a transparent, rational deduction mapping high-level organizational hiring objectives to actual, lower-level test scripts to meet those targets from a total Quality Assurance (QA) perspective.
- The creation and documentation of this test strategy was done in a highly systematic way to completely ensure that all recruitment workflows are fully covered and deeply understood by all HR stakeholders and developers. It will also be frequently reviewed, challenged, and dynamically updated as the organization scales and the platform evolves over time.
- This enterprise test strategy fundamentally aims to flawlessly align different stakeholders of quality assurance—synchronizing them in terms of uniform ATS terminology, precise test integration levels, specific roles and responsibilities (Admin vs Recruiter), defect traceability, and proper planning of computational resources.
- Test strategies heavily describe exactly how the critical product risks of the HR stakeholder—such as data breaches or faulty AI scoring—are defensively mitigated at the testing level. It outlines which specific types of testing are to be strictly performed (e.g., Regression, Security), and exactly what entry and exit criteria apply before pushing updates to the live server.
- These strategies reflect heavily on backend and frontend design documents. Core system design architectures, Data Flow Diagrams (DFDs), and Use-Case blueprints are constantly referred to. As the Smart Recruit platform expands to enable new feature sets, upcoming strategies are drafted to ensure no new vulnerability threatens the existing hiring ecosystem.

9.3 TESTING METHODS

Following are the common testing strategies:

[1] Black box testing

- Black box testing is a software testing methodology that examines the functional behavior of the Smart Recruit application without peering into its internal Django coding or backend architecture.
- The primary source for executing black box tests is the strict requirement specification document initially laid out for the Applicant Tracking System.
- In this process, the tester selects a specific UI function (e.g., submitting a job application or generating an AI question bank) and inputs dummy values to check if the candidate or recruiter dashboard produces the expected output.
- If the function perfectly updates the recruiter's pipeline view, it passes. Otherwise, the functional bug is logged and sent to the development team for UI/UX correction.

[2] White box testing

- The white box testing approach (also known as clear box, open box, or structural testing) deeply examines the internal routing, Django views, and Python logic of the software.
- It revolves around rigorously checking the internal syntax and database query structures. For instance, testing how efficiently the system parses a candidate's resume through the AI API before displaying the matched keyword score.
- Programming skills in Python are required to design these test cases. The primary goal is to verify the flow of inputs through complex logic gates (like moving a candidate from Round 1 to Round 2 automatically).
- The developer identifies unoptimized code algorithms, removes infinite loops within the search features, and performs one strict round of white box testing before handing it over to system testers.

[3] Unit testing

- Unit testing involves heavily isolating and testing each atomic component of the ATS. It serves as the absolute first level of functional testing.
- A "unit" in Smart Recruit could be a single Python function, such as the encryption script for user passwords or the specific controller handling PDF resume uploads.
- The purpose of unit testing is to definitively prove the correctness of isolated code. Normally, developers execute these tests actively during the programming phase using the unittest or pytest Python libraries.
- Whenever a single module (e.g., the User Profile Module) is fully coded, it is checked independently to ensure the basic building blocks of the recruitment portal are structurally sound.

[4] Integration testing

- Integration testing is the critical second tier of the software testing process, occurring directly after Unit Testing is finalized.
- Here, individual software components (like the Job Listing Module and the Applicant Tracking Module) are combined and tested as a unified group. The focus is to expose any severe data loss or crashes when two independent systems interact.
- Because SmartRecruit relies on multiple subsystems (such as the SQLite/MySQL database interacting with the Django Object-Relational Mapper (ORM), and the ORM talking to the Bootstrap Frontend), data flow must be impeccable.
- It ensures that when a candidate clicks "Apply", the data flows correctly from the frontend interface to the backend database, seamlessly alerting the associated recruiter.

[5] System testing

- System Testing involves a complete, end-to-end examination of the fully integrated Smart Recruit ATS platform.
- Software is initially developed in isolated units; however, no ATS can function properly unless all units confidently interface natively with web servers, hardware resources, and browser clients.
- This phase involves acting exactly like a real-world user. A tester navigates through the entire business pipeline: Creating an admin account, posting a job, logging in as a candidate, taking the AI assessment, and finally verifying if the recruiter sees the overall scorecard correctly.
- Overall, system testing proves that the final packaged product wholly satisfies the initial organizational use-case.

[6] Acceptance testing

- Acceptance testing is the highly formal testing tier conducted to gauge raw user satisfaction and real-world compliance. It operates predominantly as an advanced form of Black Box testing.
- Done fundamentally by real HR representatives, recruiters, and dummy candidates before the final software delivery, it answers the question: "Does this software actually make the hiring process easier?"
- During UAT, evaluators strictly concentrate on day-to-day business scenarios such as quickly filtering applicants by specific coding languages or communicating interview schedule times. Based on this, final UI tweaks are recommended before deployment.

[7] Regression testing

- Regression testing strictly authenticates that pushing a brand-new code update or a bug fix deeply within the application does not accidentally break existing core functionalities.
- For example, if a new "Video Interview" feature is added to the recruiter dashboard, Regression Testing heavily ensures that the standard "Resume Upload" function still works perfectly without bugs.
- Automated test scripts are usually run repeatedly because executing hundreds of repetitive pipeline checks manually every time a new patch is pushed is excessively time-consuming.
- In short, it guarantees the backward compatibility and inherent stability of the Smart Recruit system as it continuously scales across new version updates.

[8] Performance testing

- Performance Testing is a paramount technique executed to drastically evaluate the ATS's speed, reactivity, sensitivity, and systemic stability under heavy computational pressure.
- An essential goal for Smart Recruit is handling heavy web traffic. What happens if 500 candidates attempt the technical coding assessment at the exact same time on the exact same job listing?
- Stress tests and load tests are applied to identify database bottlenecks. This testing aggressively prevents server timeouts and ensures the Django backend can handle enterprise-scale transactional workloads.

[9] Security testing

- Deep security testing is an integral, non-negotiable part of modern recruitment software because an ATS handles highly protected PII (Personally Identifiable Information) such as candidate addresses, emails, and employment contracts.
- The primary objective is to aggressively hunt for fatal vulnerabilities, weaknesses, and structural loopholes that external hackers could exploit.
- Testers simulate malicious attacks to verify that the Smart Recruit portal successfully blocks vicious SQL Injections within the job search bars, intensely prevents Cross-Site Scripting (XSS), and actively stops unauthorized users from bypassing the role-based access control matrix.

9.4 TEST CASES

9.4.1 PURPOSE

The primary purpose of formalized test cases is to systematically evaluate and deeply scrutinize the overall functionality of the Smart Recruit software system by testing its backend and frontend responses to various inputs and user actions. Well-structured test cases are designed to ensure that the ATS completely aligns with the defined recruitment requirements. Below are the key strategic objectives that these test cases aim to accomplish:

1. **Verify Correctness of AI and System Results**

Test cases are specifically designed to calculate the correct algorithmic results based on the inputs provided (e.g., verifying if the Gemini API assigns accurate keyword scores to an uploaded resume). This confirms that Smart Recruit processes Applicant Data accurately, ensuring the final candidate outcome aligns strictly with the established hiring business logic.

2. **Ensure Safe and Secure Operations**

The entire ATS ecosystem must operate heavily in a safe, fail-proof manner, strictly free from catastrophic errors or malfunctions that could compromise sensitive PII (Personally Identifiable Information). Test cases critically check for deep system stability during operations, ensuring no unforeseen portal crashes, devastating memory leaks, or critical vulnerabilities arise during mass recruiter usage.

3. **Confirm Exact Software Task Execution**

Test cases validate that the recruitment application executes all required administrative and candidate-oriented tasks identically as defined in the architectural documentation. Whether it's processing an applicant's login, generating detailed Analytics Reports from the database, or sending automated SMTP emails, the test cases ensure the application behaves intelligently without skipping steps.

4. **Handle All Edge Cases and Inputs Properly**

Testing scenarios are designed to ensure the system flawlessly handles all spectrums of inputs, including valid parameters, edge cases, and completely invalid data. This specifically covers testing how the Smart Recruit system behaves when a user uploads unsupported video formats instead of PDF resumes, inputs special malicious characters in the job search bar, or leaves mandatory application fields blank.

5. **Check for Technical Defects and Continuously Measure Quality**

One of the most consequential objectives of deploying test cases is to hunt down and pinpoint software defects deeply within the code. By running rigorous test phases on the Django backend, developers can easily spotlight incorrect pipeline functionalities, AI

processing bottlenecks, or UI inconsistencies. This constant defect-checking significantly refines the final ATS product.

6. Deep Risk Assessment and Quality Measurement

Test cases provide transparent, empirical insights into the intrinsic risks associated with deploying an enterprise recruitment system. They help isolate application areas highly prone to failure (like concurrent server limits during live assessments) or architectural degradation. By performing these tests, developers can easily mitigate algorithmic risks and prioritize optimization areas.

7. Establish Absolute Confidence in the Platform

Consistent testing solidly builds immense stakeholder confidence inherently in the software's reliability, architectural security, and workload performance. A heavily comprehensive suite of test cases, if executed triumphantly under stressful conditions, profoundly reassures internal HR departments that the system is fully functional, vastly reducing the likelihood of catastrophic post-release emergencies.

8. Prevent Late-Stage Expensive Defects

Proactive and assertive testing fundamentally allows the core development team to identify and quarantine coding issues incredibly early in the software lifecycle, definitively preventing them from transforming into very costly, hard-to-patch defects later in production. This practice aggressively reduces future maintenance costs and solidifies the system's long-term enterprise sustainability.

9. Validation Against Original Requirements

Testing cases provide the perfect opportunity to aggressively validate that the delivered software faithfully meets all business, technical, and UX requirements originally outlined at the beginning of the SmartRecruit project. By heavily executing functional, non-functional, and cyber-security test loops, it ensures the core hiring objectives of the organization are conclusively achieved.

10. Documentation and Agile Continuous Improvement

Lastly, rigorous test cases organically play a critical role in thoroughly documenting the functionality of the digital infrastructure. The logs and historical results of test case execution provide a concrete record of the ATS's behavior over time. This dynamic documentation acts as a roadmap during future maintenance updates, ensuring massive upgrades don't mysteriously break existing features.

9.4.2 TEST CASES

➤ Module: Authentication and Authorization

No	Test Scenario	Expected Result	Actual Result	Status
1	Login with intentionally incorrect, invalid, or fake credentials	System triggers an authentication error message; user remains on the login page	As Expected	Pass
2	Login with deeply validated correct Candidate credentials	System accurately authenticates and securely redirects user to the Candidate Application Dashboard	As Expected	Pass
3	Login with correctly verified Admin/Recruiter credentials	System grants high-level role access and actively redirects to the Enterprise Recruiter Dashboard	As Expected	Pass
4	Attempt to click 'Login' while leaving mandatory fields entirely blank	System prevents execution and dynamically displays "This field is required" error prompts	As Expected	Pass
5	Click the 'Logout' securely encrypted session termination button	System successfully kills active session cookies and forces a redirect strictly to the main login portal	As Expected	Pass

➤ Module: Recruiter / Admin Dashboard

No	Test Scenario	Expected Result	Actual Result	Status
1	Click and sequentially navigate through all available options in the Sidebar/Navbar	Appropriate internal routing occurs, opening pages with zero HTTP 404/500 unexpected rendering errors	Appropriate pages render flawlessly without critical visual breaks	Pass
2	Load the primary Admin Performance Dashboard interface	Dashboard fetches live DB data and vividly renders all visual charts, analytics, and pipeline summary details	As Expected	Pass

3	Refresh the dashboard forcefully under simulated moderate system load	Dashboard safely reloads all statistical components without dropping database connections	As Expected	Pass
---	---	---	-------------	------

➤ **Module: Update Security Password**

No	Test Scenario	Expected Result	Actual Result	Status
1	User strongly attempts to update the password by providing an incorrect 'Current Password'	System securely blocks execution and throws a database warning: "Current password entered is deeply incorrect"	As Expected	Pass
2	User attempts to update the password where "New Password" and "Confirm Password" do not match	Password change is forcefully rejected with frontend validation alerting the mismatch	As Expected	Pass
3	User successfully updates password providing all parameters 100% correctly	System hashes the new password, posts a "Success" message, terminates the session, and triggers a forced re-login	As Expected	Pass

➤ **Module: User Profile & Resume Management**

No	Test Scenario	Expected Result	Actual Result	Status
1	Select the "View Profile" operational tab	User consistently views all fetched demographic and personal profile details accurately matching the database	As Expected	Pass
2	Rapidly update multiple text fields (Name, Location, GitHub URL) and click "Save Changes"	System correctly commits the updated strings to the active database and refreshes the UX interface	As Expected	Pass
3	Upload an invalid file format (e.g., .exe or .mp4) in the designated "Resume PDF Upload" sector	System automatically triggers error validation rejecting upload stating "Only .PDF file formats are supported"	As Expected	Pass

FUTURE ENHANCEMENT

To continuously improve the usability, performance, and overall effectiveness of the Smart Recruit System, several major future enhancements are proposed. These strategic upgrades heavily aim to provide a more seamless experience for users and profoundly expand the system's capabilities in line with cutting-edge HR industry standards.

1. Advanced Candidate Semantic Sourcing

- Implement deep NLP (Natural Language Processing) to scan through thousands of database resumes to find exact passive matches.
- Add map-integrated location searches to dynamically calculate candidate commute times or remote suitability.

2. Digital e-Signature API Integration

- Allow recruiters and candidates to securely generate and sign employment offer letters directly within the platform using APIs like DocuSign or Adobe Sign.
- Generate automatically authenticated and legally compliant digital PDF contracts.

3. Granular Enterprise Role Configurations

- Introduce profoundly dynamic role-based access control allowing custom combinations of user privileges (e.g., distinguishing between Hiring Managers, Technical Interviewers, and Super Admins).
- Provide a drag-and-drop dashboard to manage extreme data-level permissions.

4. Real-Time Chat & Video Interview System

- Integrate in-browser WebRTC capabilities for seamless real-time video and text messaging between recruiters and candidates right from the interface.
- Built-in communication records and interview recording functionalities for subsequent audit reviews.

5. Multi-Tenant SaaS Architecture Modeling

- Transform the ATS into a Subscription-based SaaS (Software as a Service) platform where multiple distinct corporate businesses can buy 'Workspaces' to manage their own recruiting independently.
- Feature restrictions based on subscription tiers (e.g., Free, Corporate, Enterprise).

6. Dedicated Mobile App Ecosystem

- Develop exceptionally responsive native Android/iOS applications to allow candidates to easily swipe and interact with job postings dynamically on the go.
- Utilize Flutter or React Native to offer real-time push notifications regarding application pipeline shifts.

7. Deep Learning Candidate Matchmaker

- Launch a machine-learning-driven intelligent 'Recommendation Engine' predicting exactly how successful a candidate will be in a target role.
- Suggest passive candidates directly to recruiters based on past 'Hired' traits and active candidate browsing behavior.

8. Predictive Hiring Analytics & ML Reporting

- Deploy predictive modeling on the Admin dashboard to intensely forecast organizational hiring hurdles, recruiter bottlenecks, and estimated time-to-hire parameters.
- Automatically generate and email weekly downloadable CSV/PDF performance packets to Senior HR Leadership.

9. Global Multi-Language & Compliance Support

- Introduce dynamic localization capabilities to instantly translate the ATS portal into multiple languages (e.g., Spanish, German, Hindi) for global corporate branches.
- Automated regional legal constraints handling, strictly managing specific GDPR (Europe) and CCPA (California) data privacy compliance acts.

10. Automated Candidate Verification APIs

- Allow the platform to interface deeply with 3rd-party Background Check APIs (like Checkr) for highly automated identity, criminal, and educational verification prior to onboarding.
- Automatically attach 'Verified Identity' badges to authenticated candidate profiles, significantly improving hiring speed and trust.

11. Smart Calendar Integration Protocols

- Enable candidates to dynamically choose open date/time interview blocks pulled directly from the Recruiter's live calendar.
- Provide frictionless two-way syncing with external Corporate Google Workspace Calendars and Microsoft Outlook to eliminate double booked instances.

12. Omni-Channel Automated Alert Matrix

- Expand the current email notification system into an Omni-channel platform capable of autonomously blasting updates via the WhatsApp Business API.
- Deliver live interview reminders, quick assessment links, and assessment outcome scores instantly to candidate mobility devices.

CONCLUSION

- The **Smart Recruit System** successfully stands as an extensively efficient, user-friendly, and technologically advanced Applicant Tracking System (ATS) specifically designed to automate the modern recruitment lifecycle. By entirely eliminating the outdated constraints of manual spreadsheet tracking and disorganized email threads, the platform introduces a secure, centralized digital ecosystem tailored perfectly for both hiring teams and candidates. Built upon the robust backbone of the **Python Django web framework**, the system provides a fast and secure environment for managing large-scale recruitment operations with data-driven precision.
- A core strength of the project is its structured **four-stage assessment pipeline** that ensures a standardized and rigorous evaluation process. The initial screening is efficiently handled through **two automated MCQ rounds**: Round 1 focuses on general Aptitude and logical reasoning, while Round 2 provides a deeper dive into Practical, domain-specific technical knowledge. These automated checkpoints allow the organization to effortlessly filter high volumes of applications, ensuring that only the most qualified candidates proceed to the more resource-intensive evaluation stages.
- Following the automated screening, the platform facilitates **two critical Manual evaluation rounds** that bridge the gap between digital tracking and human interaction. Round 3 is a comprehensive **Technical Interview** and Round 4 is the final **HR Round**, both of which are seamlessly coordinated via external professional platforms like **Google Meet or Zoom**. While these discussions occur live, the recruitment status, scheduling links, and interviewer feedback are entirely managed within the SmartRecruit dashboard, maintaining a single source of truth for every candidate's progress.
- The system's usability is further empowered by a professional **Glassmorphism-based UI architecture**, offering an intuitive experience for candidates to track their application journey in real-time. For administrators and recruiters, the platform provides advanced analytics and role-based access control, allowing for a top-down view of the hiring funnel. This design ensures that all stakeholders have immediate visibility into the recruitment pipeline, significantly reducing administrative delays and enhancing overall communication transparency between the applicant and the enterprise.
- Ultimately, the **Smart Recruit project** represents far more than a software upgrade; it acts as a major strategic shift in systematic talent acquisition. By providing a secure and scalable platform that intelligently balances automated testing with manual human judgment, it addresses the most persistent bottlenecks in modern hiring. With its inherently future-ready architecture, the system lays a solid technological foundation for the complete digital transformation of the recruitment industry, positioning it perfectly for real-world enterprise deployment.

BIBLIOGRAPHY

For the successful design, development, and thorough implementation of this project, I have extensively referred to various official technical documentations, online developer forums, and logic architectures. Several advanced backend concepts and artificial intelligence integrations were successfully studied and adapted from open-source references and global tech communities.

➤ Websites & Documentation Referred:

- **docs.djangoproject.com** (Official Django Framework concepts and backend security)
- **ai.google.dev** (Google Gemini AI API integration documentation)
- **getbootstrap.com** (Bootstrap 5 frontend UI structuring methodologies)
- www.freecodecamp.com
- www.youtube.com
- www.w3schools.com
- www.geeksforgeeks.org
- www.stackoverflow.com
- www.github.com